Occupation counts of rarely switching Markov chains: stratified local laws, support selection and the boundary layer

Arjun Pemmasani [1*]

[1*]Department of Mathematics, Harvey Mudd College, Claremont, CA, 91711, USA.

Corresponding author(s). E-mail(s): apemmasani@hmc.edu;

Abstract

A Markov chain with exit rates q_i runs for N steps with transition matrix $I + (T/N)Q$. On each irreducible face that carries start mass, we prove a local limit theorem for the occupation counts with an explicit constant and relative error $O(1/T)$, uniformly for $1 \le T \le \varepsilon N$ with $\varepsilon \max_i q_i < 1$ and on compact subsets of the open face. Up to $O(1)$, minus the log-probability of counts n with support F is $|F|-1$ times $\log(N/\sqrt{T})$ plus N times the Donsker and Varadhan functional of $I + (T/N)Q$ at n/N. Every admissible face has an explicit power of T in its largest probability for large $T \le N^{1/3}$, and a constant given by a finite-dimensional variational formula. For $T = \tau \log N$ the face of the mode minimizes $\tau\lambda_F + |F| - 1$ at first order, with λ_F the exit rate. If an admissible face has $\Delta \ge 1$ more states than another and an exit rate smaller by $\delta > 0$, we locate every crossing of their largest probabilities for large $T \le N^{1/3}$ up to $O(1/\log N)$. For two irreducible faces the crossing is unique for large N in $1 \le T \le \varepsilon N$, with coefficient $1/2$ of $\log\log N$. Near a facet, the continuous-time occupation density is the facet density times an explicit kernel, with relative error $O(1/T)$ at times $x \le cT$ off the facet, c small. The counts obey a lattice form with extra error T^2/N for $T \le \varepsilon_1\sqrt{N}$. The kernel is Bessel for $x = o(T)$ at facet profiles where the removed state has equal entry and exit laws, and at a profile where the rate has a first correction, of order $m \ge 3$, if and only if $x = o(T^{1-2/m})$.

Keywords: occupation times of Markov chains, mode, local limit theorem, principal Dirichlet eigenvalue, Donsker and Varadhan functional, Bessel kernel

MSC Classification: Primary 60J10; Secondary 60J27 , 60F10 , 60F05 , 15B48

1 Introduction

Let Q be a generator on d states, with exit rates $q_i = \sum_{j \neq i} q_{ij}$, and run the chain for N steps with transition matrix $I + (T/N)Q$. Its path follows the continuous-time chain with generator Q up to time T, while the transition matrix tends to the identity as N grows. Hence the chain degenerates. Its spectral gap is T/N times that of Q, and for fixed T the visit fractions converge in law to the occupation fractions of the continuous-time chain at time T, which charge the faces of the simplex. We let T range from 1 to $h_* N$, for a fixed h_* with $h_* \max_i q_i < 1$, and describe the law of the occupation counts $n_i = \#\{t \leq N : X_t = i\}$. The support $F = \{i : n_i > 0\}$ of the count vector n, which we also call a composition, is a face of the simplex.

Our main result is a local limit theorem for the count vector on each face that satisfies the condition (H) below, with an explicit constant and uniformly in T (Theorem A). On compact subsets of the open face, up to $O(1)$, $-\log P_N(n)$ is $(|F| - 1)\log(N/\sqrt{T})$ plus N times the Donsker and Varadhan functional of the discrete chain at n/N. Hence each dimension of the face costs a factor N up to a power of $\log N$ when T is of order $\log N$, and a factor of order $\sqrt{N}$ when T is of order N. The functional is one functional of the whole chain, and on each face it is that of the chain killed on leaving the face. Two consequences follow. First, the law selects the support of the most likely count vector, the mode, and it locates the values of T at which two faces have equal largest probabilities (Theorem B). Second, near a facet the occupation density of the continuous-time chain on a face is the density on the facet times an explicit kernel, which is Bessel in a range of the time off the facet that we determine, and the counts follow the same law with a lattice kernel. Section 1.3 summarizes the results on the complete graph and near a vertex, which are in the arXiv version of this paper [40].

To state the law, let J_V be the Donsker and Varadhan rate function of the empirical measure of the continuous-time chain with generator Q, given in (3). For $h = T/N$ let $hJ_V^{(h)}$ be the same rate for the chain with transition matrix $I + hQ$ (Section 3.3). It satisfies $J_V^{(h)} \geq J_V$ and tends to J_V as $h \to 0$. We say that a face F satisfies (H) if the chain can move between any two states of F without leaving F and the start gives F positive mass. Write λ_F for the exit rate of F and M_F for the largest probability of a count vector with support F (Definition 2.1).

Theorem A (the law on a face) *Fix the rates q_{ij}, the law μ of X_1 and h_* with $h_* \max_i q_i < 1$. Let F satisfy* (H)*, with $k = |F|$ states. If $k \geq 2$, let $\mathcal{K}$ be a compact subset of the open face. Uniformly for* $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ *and* $1 \leq T \leq h_* N$,

$$-\log P_N(n) = (k-1)\log N + T\, J_V^{(h)}(n/N) - \frac{k-1}{2}\log T - \log C_F^{(h)}(n/N) + O\Big(\frac{1}{T}\Big),$$

where $C_F^{(h)} > 0$ is an explicit function that tends to the function C_F of Definition 5.1 as $h \to 0$. Uniformly for $1 \leq T \leq h_* N$,

$$M_F = N^{-(k-1)}\, T^{(k-1)/2} K_F^{(h)}\, (1 - h\lambda_F)^N \big(1 + O(1/T)\big),$$

where $K_F^{(h)} = C_F^{(h)}(\alpha_F^*)$ and α_F^* is the quasi-ergodic law of F. For $k = 1$ the second display holds exactly, with $K_F^{(h)} = \mu_i/(1 - hq_i)$ for $F = \{i\}$. For $1 \leq T \leq N^{1/3}$ both displays

hold with J_V, C_F, $K_F = C_F(\alpha_F^*)$ *(with* $K_{\{i\}} = \mu_i$*) and* $e^{-T\lambda_F}$ *in place of* $J_V^{(h)}$, $C_F^{(h)}$, $K_F^{(h)}$ *and* $(1-h\lambda_F)^N$*, and with the error* $O(1/T + T^2/N)$*. The second display then holds for every admissible face* F*, one that a path of positive probability can fill, with* $k = |F|$*, for* $T_1 \le T \le N^{1/3}$*, with* T_1 *a large constant (Theorem 4.5), and with* $T^{(k-1)/2}$ *and* K_F *replaced by an explicit power* T^{p_F} *and a constant* $K_F^{\mathrm{red}} > 0$ *given by a finite-dimensional variational formula.*

Theorem A is Theorems 5.6, 5.7 and 4.5. Good's formula for the counts is an integral over a torus (Lemma A.2). In the variable of the torus the integrand is analytic in h up to $h = 0$, where it is the integrand of the continuum density of the chain killed on leaving F. Hence one Laplace argument with h as a parameter gives the law for $1 \le T \le h_* N$ (Theorem 5.4). Only for bounded T do we compare with the continuum, to keep the ratio bounded below. The rate of the killed chain is the restriction of $J_V^{(h)}$ to the face (Proposition 3.5). Its least value is $-\frac{1}{h}\log(1-h\lambda_F)$, attained only at α_F^* for every h (Proposition 3.6). With the continuum rate the error T^2/N cannot be avoided, since $TJ_V^{(h)} - TJ_V$ is T^2/N times a function of n/N that vanishes at most at one point, up to $O(T^3/N^2)$. On faces with (H) the continuum form follows from the discrete one for $T \le N^{1/2}$. We state it for $T \le N^{1/3}$, the range of a second proof through the continuum density, which needs $T^2/N \le 1/T$ to locate the most likely counts and which we make uniform over a class of chains. The factor $T^{(k-1)/2}$ appears because the visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $k-1$ free directions. For reversible faces K_F involves the square root of a weighted count of spanning trees (Proposition 5.2). The face maxima hold with a field v that weights a composition by $e^{Tv\cdot n/N}$, uniformly for v in a compact set, and uniformly over a compact class of reversible chains (Proposition 5.12). If the states of F do not all communicate, the time goes to the components of least exit rate and least size k^*, and to one of them when $k^* \ge 2$. Every other component C gets time of order one and costs $(T/N)^{|C|-1}$. Each switch from one component to the next costs T/N too. Hence the power is $p_F = (k^*-1)/2 + |F| - k^*$.

1.1 The mode and its crossings

When $T = \tau \log N$ the cost $e^{-T\lambda_F}$ of staying in F and the cost $N^{-(|F|-1)}$ of spreading over the count vectors of F are both powers of N. Put $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$. A face is admissible if a path of positive probability has exactly that support.

Theorem B (the support of the mode) *Fix the rates* q_{ij} *and the law* μ *of* X_1*.*

(i) Let $T = \tau \log N$ *with* $\tau > 0$ *fixed. Every admissible face* F *has* $M_F = N^{-\Phi_\tau(F)+o(1)}$*. For large* N *the support of every most likely composition is an admissible face that minimizes* Φ_τ *among the admissible faces. If each such minimizing face satisfies* (H)*, then the visit fractions of every most likely composition are within* $O(1/\log N)$ *of the quasi-ergodic law of its support.*

(ii) Let F *and* G *satisfy* (H)*, and suppose that* F *has* $\Delta \ge 1$ *more states than* G *and an exit rate smaller by* $\delta > 0$*. For large* N *the equation* $M_F = M_G$ *has exactly one root* T *in*

$[1, h_* N]$, *for any fixed* h_* *with* $h_* \max_i q_i < 1$, *and it satisfies*

$$\delta T + \frac{\Delta}{2} \log T = \Delta \log N - \log \frac{K_F}{K_G} + O\Big(\frac{1}{\log N}\Big).$$

If F *and* G *are only admissible, every root in* $[T_1, N^{1/3}]$ *satisfies this equation with* $(p_F - p_G) \log T$ *in place of* $\frac{\Delta}{2} \log T$ *and* K^{red} *in place of* K.

Part (i) is Theorem 4.2(ii)–(iv) and part (ii) is Theorem 5.8 and Corollary 4.6. The face maxima of Theorem A keep every root in a window of order $\log N$, and there the uniqueness comes from a derivative of the local law in T (Lemma A.5). Theorem 4.4 reads the winners in (i) off the lower-left boundary of the convex hull of the points $(\lambda_F, |F| - 1)$. Hence as τ grows the winning size can only grow. Solving the equation in (ii) gives $T = \frac{\Delta}{\delta}[\log N - \frac{1}{2} \log \log N + O(1)]$. The coefficient $\frac{1}{2}$ of $\log \log N$ holds for every such pair, whatever the graph, the rates, the sizes or the start, and without reversibility. For admissible faces it is $(p_F - p_G)/\Delta$, which is 1 and then 0 in Example 5.11. If Q has symmetric support or μ full support, and at each switch of (i) only two values of $(\lambda_F, |F|)$ compete, Corollary 5.10 locates every change of the mode for T of order $\log N$ up to $O(1/\log N)$. When more compete, Theorem 5.9 describes the window of T in which they do.

1.2 The boundary layer

Section 6 follows the occupation density of the continuous-time chain from the inside of a face F up to a facet $G = F \setminus \{i\}$. Let x be the time spent in i, $S = T - x$, and β the profile of the time on G, a point of the open facet. Let π_β^{in} be the law of the state from which the chain tilted to β enters i, and π_β^{out} the law of the state into which it leaves i. The Bessel kernel is $e^{-\gamma x} \sqrt{\omega S / x}\, I_1(2\sqrt{\omega S x})$, where ω and γ are explicit and $\gamma - q_i$ is $\omega/2$ times the effective resistance between π_β^{in} and π_β^{out} (Proposition 6.6). For every irreducible face whose facet satisfies (H), Theorem 6.1 writes the density on F as the density on G times an explicit generalized kernel, with relative error $O(1/T)$ for x up to a small multiple of T. Proposition 6.2 gives the range of the Bessel form at each β. At the β with $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$, which are all β when Q is reversible and some β for some chains that are not, the Bessel kernel holds with relative error $O(1/T + \sqrt{x/T})$, so for $x = o(T)$. At a β where the rate has a term beyond the Bessel exponent, in powers of $\sqrt{x/S}$, let $m \geq 3$ be the order of the first one. There the relative Bessel form holds if and only if $x = o(T^{1-2/m})$. We do not know whether such a term exists whenever $\pi_\beta^{\mathrm{in}} \neq \pi_\beta^{\mathrm{out}}$. For the cycle on three states with rates 2 in one direction and 1 in the other, at the uniform profile on a facet of 2 states, $m = 3$, and the logarithm of the ratio to the Bessel form is $\frac{49}{64\sqrt{\omega}}\, x^{3/2} S^{-1/2}$ to leading order. For a pair face the kernel is the facet form of Pedler's formula, which we prove with an explicit error for all x (Proposition 6.5). The proof rests on a local law for a weighted chain on the facet with a complex parameter (Theorem 6.8). We prove it on a torus through a complex critical point, with a Laplace lemma for a complex Hessian. The counts follow the same law. With n_i visits to i, $P_N(n)$ is the probability of the counts on G in $N - n_i$ steps times a lattice kernel, the generalized kernel with Laguerre polynomials in place of the Bessel function, with the extra relative error T^2/N for T up to a small multiple

of $N^{1/2}$ (Theorem 6.12). The lattice kernel is T/N times the generalized kernel up to $O(T^2/N)$, so the Bessel forms hold for the counts with T^2/N added to the error. For a pair face the discrete Pedler formula is exact, and we compare it with the lattice kernel at every T (Theorem 6.11). For three or more states the counts are open when T^2/N is of order one.

1.3 The complete graph and the layer near a vertex

On the complete graph with the uniform start, $Q = J - dI$ with J the all-ones matrix, every face ties at first order and the constants decide. Sections 7 and 8 of the arXiv version of this paper [40] treat this case, partly by the general theory and partly by exact arguments of our preprint [39], which this paper supersedes. A face of $k \geq 2$ states crosses a vertex exactly once, at a value $T_{N,k}$ with

$$T_{N,k} + \tfrac{1}{2} \log T_{N,k} = \log N + a_k + O\Big(\frac{1}{\log N}\Big), \qquad a_k = \frac{1}{2} \log(4\pi) - \frac{k + \frac{1}{2}}{k - 1} \log k.$$

For $T \leq N/d$ the most likely compositions with a given support are those whose positive coordinates differ by at most 1. Since $a_2 > a_3 > \cdots > a_d$, for large N and T up to a fixed multiple of $\log N$ the mode changes exactly once, from the vertices straight to the compositions with d positive coordinates. Near a vertex, for one large count b and small counts $a_1, \ldots, a_\ell$, the crossing with a vertex lies at the odds $\sqrt{y_a/b}\,(1 + O(\sqrt{(|a| + 1)/b}))$ of a switch, where y_a is the positive root of $\mathcal{B}(y) = 1$ for an explicit polynomial $\mathcal{B}$ built from Laguerre polynomials. Then T is of order $\sqrt{N}$, and the lattice kernel of Theorem 6.12 cannot be replaced by the Bessel kernel. This version leaves out those two sections and keeps the numbers of the arXiv version, so Section 6 is followed by Section 9.

1.4 Related work

The exact law of the count vector goes back to Whittle [43], Dawson and Good [16], Goodman [24] and Good [23]. For two states it is the Markov binomial law, whose exact form Gabriel [22] gave. Dobrushin [19] found the limit laws of the number of visits of a two-state chain whose switch probabilities scale with the length, and for switch probabilities of order $1/N$ the limit is not Gaussian. This is the case $d = 2$ of our regime. Čekanavičius and Vellaisamy [11] give compound Poisson approximations to the Markov binomial law and cite the work on it since Dobrushin. Stadje [42] finds the joint laws of the numbers of visits and their limits when one state becomes nearly absorbing, with the transition probabilities out of it of order $1/N$. Here every row of the transition matrix degenerates at the same rate. Good proposed a saddle-point analysis of his formula (10) and did not carry it out [23, pp. 42–43], and Theorem 5.4(a) is that analysis, uniform in T and with its constant. Part (b) does the same for his discrete formula (3), uniformly in $T/N \leq h_*$. Polettini [41] and Carugno, Vivo and Coghi [10] derive the exact law from the BEST and matrix-tree theorems.

For a fixed chain the occupation counts form a Markov additive process, and its large deviations are classical. Miller [33] proves a local large deviation theorem from

the Perron root of the transform matrix, Ney and Nummelin [34, 35] obtain the rates from its convergence parameter on general state spaces, Kontoyiannis and Meyn [30, Theorem 6.3] give exact large deviation asymptotics for a real functional, and Hervé and Ledoux [26] prove a local limit theorem for densities of the additive component. These results concern a fixed transition matrix, and their constants depend on it. For instance, Theorems 6.3 and 6.5 of [30] give the exact asymptotics of $\mathbb{P}\{S_n \geq nc\}$ as $n \to \infty$, where S_n is the partial sum of a real functional of one geometrically ergodic chain, in the non-lattice and the lattice case, with no rate in n and no uniformity over kernels. Here the matrix $I + hQ$ tends to the identity and its spectral gap T/N tends to 0, so these results give no bound uniform in N. Theorem 5.4(b) is an exact local large deviation estimate for the whole count vector of the chain killed on leaving F, with an explicit constant, and the new point is that it is uniform as $I + hQ \to I$. At $h = 0$ it joins Theorem 5.4(a), the same estimate for the occupation times of the continuous-time chain. We found no statement in these sources that covers either, and none that is uniform as the chain degenerates.

In continuous time, Brydges, van der Hofstad and König [9] give an explicit formula for the joint density of the local times on the event that the range is a given set, and use it for large deviation upper bounds beyond the exponential scale. On the event that the range up to time T is F, Lemma 2.4 gives this density, mixed over the start and written as Good's run expansion, and Theorem 5.4(a) gives its exact asymptotics on compact subsets of the open face. Huang, Kious, Sidoravicius and Tarrès [27] write the joint density of the local times, the last exit tree and the cycling numbers of a chain with symmetric rates as a product of Bessel functions, one for each edge. Pedler's occupation-time density for two-state chains [37] and the Bessel forms of Kovchegov, Meredith and Nir [31] for the occupation time of one state are exact formulas for a fixed chain too. In the boundary layer the Bessel kernel arises differently. It is the large-T form of the contribution of the visits to one state, after the facet is integrated out by a local law, and without $\pi^{\mathrm{in}} = \pi^{\mathrm{out}}$ it can fail beyond x of order $T^{1/3}$.

The rate inside a face is the Donsker and Varadhan functional of the killed chain [12, 20, 21]. Guillin, Nectoux and Wu [25] prove a large deviation principle for the occupation measure of a process conditioned to stay in a set. The quasi-ergodic law goes back to Darroch and Seneta [14, 15], and Collet, Martínez and San Martín [13] is the standard reference. Dietz and Sethuraman [17, 18], Bouguet and Cloez [8] and Benaïm, Bravo and Faure [2] study chains whose switching decays along the trajectory, while here the rate is fixed during each path and changes with N. Bertini, Gabrielli and Landim [6], Landim [32] and Kim and Landim [29] expand the rate functions of a fixed chain in a parameter, the last for the joint rate of the empirical measure and the flow [4, 5]. Here the functional enters through its restriction to each face, and faces of different dimension compete.

The two-state walk is treated at every N in [38]. Our preprint [39] on the complete graph is superseded by this paper. Table 1 lists the results of the preprint that we reprove, where, which part, and how. Sections 7 and 8 of the arXiv version of this paper [40] reprove in addition Proposition 2.2, Theorems 3.1, 3.2 and 3.8, Lemma 3.5, Proposition 3.6, Corollary 3.9, Example 3.10, Lemma 6.1 and Theorem 6.4 of [39], and Table 1 there says how. The other results of [39] are not reproved here,

Table 1 Results of the preprint [39] reproved here. "General" means that the proof is a case of the general theory of Sections 2 to 5

Item of [39]	Result here	Part covered	Method
Thm. 3.3	Thm. 5.6	the local law (7) up to the closed form of its constant, not the root (8)	general
Lemma 3.4	Prop. 5.2(b)	all	general

namely Proposition 2.1, Proposition 3.7, Sections 4 and 5 on a start and a field, Theorem 6.2, Remark 6.3 and Appendix C. A companion paper, in preparation, asks what the Donsker and Varadhan functional and the modal supports determine about the generator.

Organization. Section 2 sets up the exact law and Section 3 the killed functional, in continuous and in discrete time. Section 4 proves first-order face selection and gives the face maxima of reducible faces. It uses the local law and the face maxima of Section 5, whose proofs do not use Section 4. Section 6 treats the boundary layer, and Section 9 the checks and the open questions. The arXiv version of this paper [40] has in addition Sections 7 and 8, on the complete graph and near a vertex, with Appendices C and D. Its other sections and statements have the same numbers as here.

2 Setting, notation and the exact law

2.1 Setting and notation

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$. Its rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. We also write q for the vector (q_i). The start μ is a probability law on $[d]$, and $\mu(F) = \sum_{i \in F} \mu_i$. We write $V = [d]$ for the face of all states. For $N \geq 2$ and $T > 0$ put

$$h = \frac{T}{N}, \qquad L = \log N, \qquad \tau = \frac{T}{L}.$$

The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + hQ$. So at each step it jumps from i to j with probability hq_{ij}, and N steps look like the continuous-time chain with generator Q run for time T. We assume $hq_{\max} \leq \frac{1}{2}$ unless we say otherwise. The *composition* is the vector of visit counts $n = (n_1, \dots, n_d)$, with $n_i = \#\{t \leq N : X_t = i\}$, so that $n_1 + \cdots + n_d = N$. We write $P_N(n)$ for the probability that the chain has composition n, $\operatorname{supp} n = \{i : n_i > 0\}$ for its support, and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$, of size $k = k_F = |F|$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F, and Δ_F° the set of those that are positive on F. We number the states of F as $1, \dots, k$ and use the coordinates $(\alpha_2, \dots, \alpha_k)$ on Δ_F and $(t_2, \dots, t_k)$ on $\{t \in \mathbb{R}^F : t_1 + \cdots + t_k = T\}$. We write $x \cdot y = \langle x, y \rangle = \sum_{i \in F} x_i y_i$ and $x \circ y = (x_i y_i)_{i \in F}$. Let $A_F = (q_{ij})_{i,j \in F}$ with zero diagonal, and $Q_F = A_F - \operatorname{diag}(q_i)_{i \in F}$.

Since q_i includes the rates of leaving F, Q_F generates the chain killed when it leaves F. The switching graph G_F has an edge $i \to j$ whenever $i, j \in F$, $i \neq j$ and $q_{ij} > 0$.

Perron roots. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and then its Perron root $\varrho(B)$ is its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [3].

Definition 2.1 (faces) Let F be a face. We call F *irreducible* if G_F is strongly connected. In particular every singleton is irreducible. We call F *admissible* if for some N some path $x_1 \cdots x_N$ of positive probability has support exactly F (for $0 < hq_{\max} < 1$ this does not depend on h). We say that F satisfies (H) if F is irreducible and $\mu(F) > 0$. The *exit rate* of F is $\lambda_F = -\varrho(Q_F)$, so $\lambda_{\{i\}} = q_i$. If F is irreducible, r_F and l_F are right and left eigenvectors of Q_F for $-\lambda_F$ with positive entries and $\langle l_F, r_F \rangle = 1$, and $\alpha_F^* = l_F \circ r_F$ is the *quasi-ergodic law* of F [14, 15]. The *face maximum* is $M_F = \max\{P_N(n) : \operatorname{supp} n = F\}$, with $\max \emptyset = 0$.

Note that λ_F is a true exit rate. If F is irreducible, then $e^{TQ_F} r_F = e^{-T\lambda_F} r_F$ with r_F positive. Hence the chance that the continuous-time chain started in F stays in F up to time T lies between $e^{-T\lambda_F} \min r_F / \max r_F$ and $e^{-T\lambda_F} \max r_F / \min r_F$. If F is reducible, we order the strongly connected components $C_1, \ldots, C_r$ of G_F so that every edge between components goes forward. Then Q_F is block triangular with diagonal blocks Q_{C_j}, and $\lambda_F = \min_j \lambda_{C_j}$. In all cases $0 \le \lambda_F \le \min_{i \in F} q_i$, since the rows of Q_F have sums at most 0 and a Perron root is at least every diagonal entry. For a face F satisfying (H), α_F^* is the limit as $T \to \infty$ of the law of the visit fractions of the continuous-time chain conditioned to stay in F up to time T.

Conventions. We fix Q, μ and the faces we discuss. Constants, implied constants in $O(\cdot)$ and thresholds N_0 depend only on these and on the compact set $\mathcal{K}$, if there is one, unless we say otherwise. By a bound "uniformly for $1 \le T \le N^{1/3}$" we mean one implied constant for all $N \ge N_0$ and all T in that range. Some results fix h_* with $h_* q_{\max} < 1$ and hold uniformly for $1 \le T \le h_* N$. There $h \le h_*$ replaces $hq_{\max} \le \frac{1}{2}$, and the constants also depend on h_*. Subscripts as in $O_{\mathcal{K}}$ stress a dependence.

2.2 Admissible faces

Lemma 2.2 (admissible faces) *Let F be a face and $C_1, \ldots, C_r$ the strongly connected components of G_F.* (a) *F is admissible if and only if the components can be numbered so that $\mu(C_1) > 0$ and, for each $j < r$, $q_{xy} > 0$ for some $x \in C_j$ and $y \in C_{j+1}$.* (b) *An irreducible face is admissible if and only if $\mu(F) > 0$. So* (H) *holds exactly for the irreducible admissible faces.* (c) *If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.*

Proof We prove this by translating paths into walks in G_F. Take a path of positive probability with support F and list the states of its maximal runs. This list is a walk in G_F that starts at a state of $\operatorname{supp} \mu$ and visits every state of F. Conversely, every such walk $v_1, \ldots, v_J$ is the list of runs of a path of positive probability with support F, as soon as $N \ge J$. The path follows the walk and then stays in its last state, and a stay has positive probability. Hence F is admissible exactly when G_F has a walk that starts in $\operatorname{supp} \mu$ and visits every state of F.

(a) Suppose that there is such a walk. Once it has left a component C for a state $y \notin C$, it cannot come back to C. Otherwise y could be reached from C and C from y, and y would belong to C. Hence the walk visits each component in one block of consecutive times, and it passes from one block to the next along a single edge. We number the components in the order of these blocks. The walk starts in C_1, so $\mu(C_1) > 0$, and the edge from block j to block $j+1$ is an edge $x \to y$ with $x \in C_j$ and $y \in C_{j+1}$. This is the condition in (a).

Conversely, suppose that the components are numbered as in (a), with edges $x_j \to y_{j+1}$ from C_j to C_{j+1} for $j < r$. Take $y_1 \in C_1$ with $\mu_{y_1} > 0$ and any $x_r \in C_r$. Since C_j is strongly connected, there is a walk inside C_j from y_j to x_j that visits every state of C_j. We join these r walks by the edges $x_j \to y_{j+1}$. The result is a walk that starts in $\operatorname{supp}\mu$ and visits every state of F.

(b) This is (a) with $r = 1$.

(c) If Q has symmetric support, an edge from one component to another would come with an edge back, and the two components would be the same. Hence no edge joins two components, and the condition in (a) forces $r = 1$. □

Hence $M_F > 0$ for large N if F is admissible, and $M_F = 0$ otherwise.

2.3 The exact law

For $m \in \mathbb{Z}_{\geq 1}^F$ put $M = \sum_{i \in F} m_i$ and $W(m) = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}$, the sum over the words $c = c_1 \cdots c_M$ in F with $c_j \neq c_{j+1}$ in which each $i \in F$ occurs m_i times. Such a word lists the states of the runs of a path, and m_i is the number of runs in state i.

Lemma 2.3 (run expansion) *Let $hq_{\max} \leq 1$ and let n have support F. Then, with $\binom{n_i-1}{m_i-1} = 0$ for $m_i > n_i$ and $0^0 = 1$,*

$$P_N(n) = \sum_{m \in \mathbb{Z}_{\geq 1}^F} W(m)\, h^{M-1} \prod_{i \in F} \binom{n_i - 1}{m_i - 1} (1 - hq_i)^{n_i - m_i}.$$

This is Good's coefficient identity [23, (3)], with the exponents $n_i - 1$, written for the chain killed on leaving F.

Proof We prove this by counting words. The states $c_1 \cdots c_M$ of the maximal runs of a path with composition n form a word as in $W(m)$. The path starts at c_1, of probability μ_{c_1}, it has $M - 1$ switches, of probabilities $hq_{c_j c_{j+1}}$, and $n_i - m_i$ stays in state i, of probability $1 - hq_i$ each. The number of paths with the word c and composition n is $\prod_i \binom{n_i-1}{m_i-1}$, the number of ways to write each n_i as an ordered sum of m_i positive run lengths. Summing over the words c with given m gives $W(m)$, and summing over m gives the claim. □

For $F = \{i\}$ the only path is i^N, so uniformly for $1 \leq T \leq N^{1/2}$

$$M_{\{i\}} = \mu_i (1 - hq_i)^{N-1} = \mu_i e^{-q_i T} \big(1 + O(T^2/N)\big). \tag{1}$$

For $t \in (0, \infty)^F$ let $H(t) = \sum_m W(m) \prod_{i \in F} t_i^{m_i - 1}/(m_i - 1)!$, which is finite since $W(m) \leq k^M q_{\max}^{M-1}$. In the continuum limit the binomial coefficients of Lemma 2.3 become the powers $t_i^{m_i-1}/(m_i-1)!$, and Lemma 5.5 below shows that the discrete law is close to the following density for $T \leq N^{1/2}$.

Lemma 2.4 (the continuum density) *Let $(Y_s)_{s\ge 0}$ be the continuous-time chain with generator Q and $Y_0 \sim \mu$, let $\mathcal{O}_i = \int_0^T \mathbf{1}\{Y_s = i\}\,ds$, and let E_F be the event that Y stays in F during $[0,T]$ and visits every state of F. Then $\mathbb{P}(E_F,\ \mathcal{O} \in \cdot\,)$ has the density $e^{-\langle q,t\rangle} H(t)$ in $(t_2,\dots,t_k)$ on $\{t \in (0,\infty)^F : \sum_i t_i = T\}$, and $\mathbb{P}(E_F,\ \mathcal{O}/T \in \cdot\,)$ has the density*

$$f_T(\alpha) = T^{k-1} e^{-T\langle q,\alpha\rangle} H(T\alpha)$$

in $(\alpha_2,\dots,\alpha_k)$. For n with support F and $t = hn$, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t)$. For $F = \{i\}$, $f_T = \mathbb{P}(E_F) = \mu_i e^{-q_i T}$.

The density is Good's formula [23, (10)] for the killed chain, whose diagonal keeps the full exit rates q_i. The proof checks the exponents $m_i - 1$.

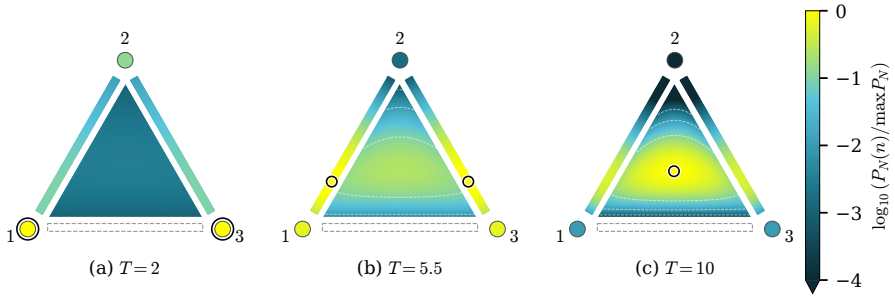
Proof Up to a null set, E_F is the disjoint union of the events E_c that Y visits exactly $c_1,\dots,c_M$ in this order during $[0,T]$, over the words c in the sums $W(m)$. Fix c, and let s_j be the time that Y spends in its j-th run, so that $s_1 + \cdots + s_M = T$. For $j < M$ the run has the density $q_{c_j} e^{-q_{c_j} s_j}$ and ends with a jump to c_{j+1} of probability $q_{c_j c_{j+1}}/q_{c_j}$, and the last run survives with probability $e^{-q_{c_M} s_M}$. So on $\{s_j > 0\ \forall j\}$ the event E_c has the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,\mathcal{O}\rangle}$ in $(s_1,\dots,s_{M-1})$, where $\mathcal{O}_i$ is the sum of the s_j with $c_j = i$. For each $i \ne c_M$ we replace the last holding time in state i by $\mathcal{O}_i$, a triangular change of variables with unit diagonal. With $\mathcal{O} = t$ fixed, the $m_i - 1$ other holding times in state i are positive with sum less than t_i, so they range over a simplex of volume $t_i^{m_i-1}/(m_i-1)!$. Integrating them out gives the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,t\rangle} \prod_i t_i^{m_i-1}/(m_i-1)!$ in the coordinates t_i with $i \ne c_M$, and any $k-1$ of the coordinates give the same Lebesgue measure on $\{\sum_i t_i = T\}$. Summing over c and m gives $e^{-\langle q,t\rangle} H(t)$, and $t = T\alpha$ has Jacobian T^{k-1}. For $F = \{i\}$ the event E_F says that Y starts at i and does not jump before time T. □

Example 2.5 (two models) (i) For $d = 2$, $q_{12} = q_{21} = 1$ and uniform μ the chain gives the directions of the persistent random walk on $\mathbb{Z}$ of [38], which repeats its previous step with probability $p = 1 - h$. So $T = N(1-p)$, and the faces $\{1\}$ and $\{2\}$ are the endpoints. (ii) For $Q = J - dI$, with J the all-ones matrix, the chain switches to each other state with probability h and keeps its state with probability $p = 1 - (d-1)h$. With uniform μ it is the model of [39], whose switch probability r is our h. A word $x_1 \cdots x_N$ then has probability $\frac{1}{d} h^{N-1} (p/h)^{\#\{s<N:\, x_s = x_{s+1}\}}$, so the path law is the one-dimensional d-state Potts chain with free boundaries [44]. Here $q_i = d - 1$ for every i, so Lemma 2.3 reads

$$P_N(n) = p^{N-1} \sum_{a \in F} \mu_a S_n^{(a)}(u), \qquad u = \frac{h}{p}, \tag{2}$$

for $p > 0$ and $\operatorname{supp} n = F$, where $S_n^{(a)}(u)$ is the sum of u^j over the words with the counts n that start with a and have j switches. Indeed such a word has probability $\mu_a p^{N-1-j} h^j = \mu_a p^{N-1} u^j$. In both models Q has symmetric support and μ full support, so all admissible faces satisfy (H) by Lemma 2.2.

Figure 1 shows the exact support competition for the three-state path. There λ_F is 1 for an end, $(3-\sqrt{5})/2$ for a pair of neighbors and 0 for all three states, so to first order the mode moves at $\tau = (1+\sqrt{5})/2$ and at $\tau = (3+\sqrt{5})/2$, that is, at $T = 7.3$ and $T = 11.8$ for $N = 90$. At $N = 90$ the mode actually moves near $T = 4.9$ and $T = 8.5$. Corrections of this size are the subject of Section 5.



(a) $T=2$ (b) $T=5.5$ (c) $T=10$

Fig. 1 The exact law of the composition for the path 1–2–3 (rate 1 on each edge, uniform start, $N = 90$) on the simplex of visit fractions, with the faces drawn apart. The triangle holds the compositions with three states, each band those with two, and each disk a vertex. The color is the probability relative to the mode, the thin lines are spaced by a factor of ten, and the rings mark the modes. The mode is (a) at an end of the path, (b) on a pair of neighbors and (c) inside. The dashed band $\{1, 3\}$ has probability zero

3 The killed functional

Every rate in this paper is the restriction of one functional, the Donsker and Varadhan functional of the chain, or of its discrete version of Section 3.3.

3.1 The restriction identity

For a probability vector α on $[d]$ let

$$J_V(\alpha) = \sup_{u\in(0,\infty)^d} \sum_{i\in[d]} \alpha_i \Big(-\frac{(Qu)_i}{u_i}\Big) = \sup_{u\in(0,\infty)^d} \sum_{i\in[d]} \alpha_i \Big(q_i - \sum_{j\neq i} q_{ij}\frac{u_j}{u_i}\Big), \tag{3}$$

the Donsker and Varadhan rate of the empirical measure of the continuous-time chain with generator Q [20]. For a face F and $\alpha \in \Delta_F$ let

$$J_F(\alpha) = \sup_{u\in(0,\infty)^F} \sum_{i\in F} \alpha_i \Big(-\frac{(Q_F u)_i}{u_i}\Big) = \sup_{u\in(0,\infty)^F} \sum_{i\in F} \alpha_i \Big(q_i - \sum_{j\in F\setminus\{i\}} q_{ij}\frac{u_j}{u_i}\Big),$$

the same functional for the chain killed on leaving F. Let $q_i^{\mathrm{out}}(F) = \sum_{j\notin F} q_{ij}$. Taking $u = \mathbf{1}$ gives $J_F(\alpha) \geq \sum_i \alpha_i q_i^{\mathrm{out}}(F)$, and dropping the terms with q_{ij} gives $J_F(\alpha) \leq \sum_i \alpha_i q_i$.

Proposition 3.1 (the restriction identity) *For every face F and every $\alpha \in \Delta_F$, $J_F(\alpha) = J_V(\alpha)$. Hence $J_F = J_{F'}$ on $\Delta_{F'}$ for every nonempty $F' \subseteq F$.*

Proof Fix $\alpha \in \Delta_F$ and $u \in (0,\infty)^d$. The terms of (3) with $i \notin F$ vanish, since $\alpha_i = 0$ there. For $i \in F$ the terms with $j \notin F$ are nonnegative and are subtracted, so

$$\sum_{i\in F} \alpha_i \Big(q_i - \sum_{j\neq i} q_{ij}\frac{u_j}{u_i}\Big) \leq \sum_{i\in F} \alpha_i \Big(q_i - \sum_{j\in F\setminus\{i\}} q_{ij}\frac{u_j}{u_i}\Big) \leq J_F(\alpha).$$

Hence $J_V(\alpha) \le J_F(\alpha)$. Conversely, let $w \in (0,\infty)^F$ and let u equal w on F and ε off F. The terms with $j \notin F$ are then $-q_{ij}\varepsilon/w_i$, which tend to zero as $\varepsilon \downarrow 0$, and the supremum over w gives $J_V(\alpha) \ge J_F(\alpha)$. The second claim is the first for F and for F'. □

3.2 The variational formula

For a face F and a field $v \in \mathbb{R}^F$ let

$$\lambda_F(v) = -\varrho\big(Q_F + \operatorname{diag} v\big), \tag{4}$$

so that $\lambda_F(0) = \lambda_F$. If F is irreducible, let r and l be positive right and left Perron vectors of $Q_F + \operatorname{diag} v$ with $l \cdot r = 1$, and put $\alpha_F^*(v) = l \circ r$, so that $\alpha_F^*(0) = \alpha_F^*$. Since $\sum_i \alpha_i = 1$,

$$J_F(\alpha) - v \cdot \alpha = \sup_{u \in (0,\infty)^F} \sum_{i \in F} \alpha_i \Big(- \frac{((Q_F + \operatorname{diag} v)u)_i}{u_i} \Big). \tag{5}$$

Hence a field replaces the diagonal q_i by $q_i - v_i$ and changes nothing else.

Lemma 3.2 (monotonicity in the face) *If $F \subsetneq G$ are faces and Q_G is irreducible, then $\lambda_G(v_G) < \lambda_F(v_F)$ for every field $v \in \mathbb{R}^G$. Now let Q be irreducible. Then $\lambda_V(v) < \lambda_F(v_F)$ for every proper face F, and for every fixed $v \in \mathbb{R}^d$ there is $\gamma(v) > 0$ with $\lambda_F(v_F) \ge \lambda_V(v) + \gamma(v)$ for all proper F. So the score $\Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v_F)$ is least at V for $\tau > (d-1)/\gamma(v)$.*

Note that the last two claims need Q irreducible. For $d = 2$, $q_{12} = 1$ and $q_{21} = 0$ we have $\lambda_V = 0 = \lambda_{\{2\}}$.

Proof Choose c with $B = Q_G + \operatorname{diag} v_G + cI \ge 0$. Its principal submatrix on F is $Q_F + \operatorname{diag} v_F + cI$, since the diagonal of a killed generator keeps the full exit rates. Let $\widetilde{B}$ be this submatrix padded with zeros to $G \times G$. Then $0 \le \widetilde{B} \le B$, and $\widetilde{B} \ne B$ because the irreducible matrix B has a positive entry in a row of F and a column off F. So $\varrho(\widetilde{B}) < \varrho(B)$ [3, Ch. 2], and $\varrho(\widetilde{B})$ is the Perron root of the submatrix on F, since the padding adds only the eigenvalue 0. This holds whether or not F is irreducible, and subtracting c gives the first claim. If Q is irreducible, the first claim with $G = V$ gives the second. For fixed v there are finitely many proper faces, so $\gamma(v)$, the least of the numbers $\lambda_F(v_F) - \lambda_V(v)$, is positive. Finally $\Phi_{\tau,v}(F) - \Phi_{\tau,v}(V) = |F| - d + \tau(\lambda_F(v_F) - \lambda_V(v)) \ge -(d-1) + \tau\gamma(v)$. □

Proposition 3.3 (the rate on a face) *Let F be irreducible with $k = |F| \ge 2$, and let $v \in \mathbb{R}^F$.*

(a) On Δ_F, $J_F(\alpha) - v \cdot \alpha \ge \lambda_F(v)$, with equality exactly at $\alpha = \alpha_F^*(v)$. In particular $J_F \ge \lambda_F$, with equality exactly at $\alpha_F^* = l_F \circ r_F$.

(b) The rate J_F is convex on Δ_F and strictly convex on Δ_F°. For $\alpha \in \Delta_F^\circ$ the supremum defining $J_F(\alpha)$ is attained at a vector $u = r_\alpha$, which is unique up to a positive factor, and α is the stationary law of the generator on F with off-diagonal entries $q_{ij} r_{\alpha,j}/r_{\alpha,i}$.

(c) If there is $\nu \in (0,\infty)^F$ with $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i, j \in F$, then for every $\alpha \in \Delta_F$

$$J_F(\alpha) = \sum_{\{i,j\} \subseteq F,\ i \ne j} \big(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}}\big)^2 + \sum_{i \in F} \alpha_i q_i^{\mathrm{out}}(F).$$

(d) *For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $J_F(\alpha) - v\cdot\alpha = J_{F'}(\alpha) - v\cdot\alpha \geq \lambda_{F'}(v_{F'}) > \lambda_F(v)$.*

For symmetric Q_F, $l_F = r_F$ and α_F^* is the entrywise square of the Perron vector normalized in ℓ^2. Without reversibility the proof of (c) still gives the formula as an upper bound.

Proof We prove this through the matrix B. Put $B = Q_F + \operatorname{diag} v$, an irreducible Metzler matrix with the switching graph G_F and $\lambda_F(v) = -\varrho(B)$.

Setup. With $u = e^{\psi}$, (5) reads $J_F(\alpha) - v\cdot\alpha = \sup_\psi \Psi_\alpha(\psi)$ with

$$\Psi_\alpha(\psi) = \sum_i \alpha_i(q_i - v_i) - \sum_{i\neq j} \alpha_i q_{ij} e^{\psi_j - \psi_i},$$

where i, j run over F. It is concave in ψ, affine in α, and unchanged when a constant is added to ψ. Let G^ψ be the conservative generator on F with off-diagonal entries $q_{ij}e^{\psi_j-\psi_i}$. It has the graph G_F, so it has exactly one stationary law, and it does not involve v. Since $\nabla_\psi \Psi_\alpha = -\alpha^\top G^\psi$ and a differentiable concave function attains its maximum exactly at its critical points, ψ attains the supremum if and only if α is the stationary law of G^ψ. We call this the criterion.

A lower bound on every face. Let F' be any face, irreducible or not, and $B' = Q_{F'} + \operatorname{diag} v_{F'}$. We show $J_{F'}(\alpha) - v\cdot\alpha \geq -\varrho(B')$ on $\Delta_{F'}$. Let $s > \varrho(B')$ and write $B' = B_0 - cI$ with $B_0 \geq 0$. Then $s + c > \varrho(B_0)$, so $u = (sI - B')^{-1}\mathbf{1} = \sum_{\ell\geq 0}(s+c)^{-\ell-1}B_0^\ell \mathbf{1}$ converges and is positive. Since $(sI - B')u = \mathbf{1}$, we get $-(B'u)_i/u_i = -s + 1/u_i > -s$ at every state, and letting $s \downarrow \varrho(B')$ gives the claim.

The value at $\alpha_F^*(v)$. Take $\psi = \log r$ and $R = \operatorname{diag} r$. Then $G^\psi = R^{-1}(B - \varrho(B)I)R$, since the off-diagonal entries agree and $Br = \varrho(B)r$. So $(l\circ r)^\top G^\psi = l^\top(B - \varrho(B)I)R = 0$, and by the criterion ψ attains the supremum at $\alpha_F^*(v)$. Every ratio $-(Br)_i/r_i$ equals $\lambda_F(v)$, so the value is $\lambda_F(v)$.

(b) Convexity holds because J_F is a supremum of affine functions. Let $\alpha \in \Delta_F^\circ$ and take $v = 0$.

The supremum is attained. Normalize $\min_i \psi_i = 0$ and put $\bar\psi = \max_i \psi_i$. Let a be the least positive rate q_{ij} in F. A shortest path in G_F from a state where $\psi = 0$ to a state where $\psi = \bar\psi$ has at most $k-1$ edges, so one of its edges $i \to j$ has $\psi_j - \psi_i \geq \bar\psi/(k-1)$. Keeping only this edge gives $\Psi_\alpha(\psi) \leq \sum_i \alpha_i q_i - a(\min_i \alpha_i)e^{\bar\psi/(k-1)}$, which tends to $-\infty$ as $\bar\psi \to \infty$. Hence the continuous function Ψ_α attains its supremum on a bounded set. Write $r_\alpha = e^\psi$ for a maximizer. The last claim of (b) is the criterion.

The maximizer is unique up to a factor. If ψ and $\psi + \delta$ both attain the supremum, Ψ_α is constant on the segment between them. There it is a constant minus a sum of the convex functions $\alpha_i q_{ij} e^{\psi_j - \psi_i + s(\delta_j - \delta_i)}$, so each of them is affine in s, and $\delta_j = \delta_i$ on every edge of G_F. As G_F is strongly connected, δ is constant.

Strict convexity. Let $\alpha^0 \neq \alpha^1$ lie in Δ_F°, put $\alpha^s = (1-s)\alpha^0 + s\alpha^1$ with $0 < s < 1$, and let ψ attain $J_F(\alpha^s)$. Since $\Psi_\alpha(\psi)$ is affine in α, $J_F(\alpha^s) = (1-s)\Psi_{\alpha^0}(\psi) + s\Psi_{\alpha^1}(\psi) \leq (1-s)J_F(\alpha^0) + sJ_F(\alpha^1)$. Equality would make α^0 and α^1 both stationary for G^ψ by the criterion, which is impossible.

(d) The equality is Proposition 3.1. The lower bound on F' gives $J_{F'}(\alpha) - v\cdot\alpha \geq \lambda_{F'}(v_{F'})$, and Lemma 3.2 with $G = F$ gives $\lambda_{F'}(v_{F'}) > \lambda_F(v)$.

(a) The lower bound and the value at $\alpha_F^*(v)$ are proved above. Let $\alpha \neq \alpha_F^*(v)$. If α lies on the boundary of Δ_F, then (d) with $F' = \operatorname{supp}\alpha$ gives $J_F(\alpha) - v\cdot\alpha > \lambda_F(v)$. If $\alpha \in \Delta_F^\circ$ and equality held, strict convexity would give a value below $\lambda_F(v)$ at $\frac{1}{2}(\alpha + \alpha_F^*(v))$.

(c) Since $q_i = \sum_{j\in F\setminus\{i\}} q_{ij} + q_i^{\mathrm{out}}(F)$, the function inside the supremum is $\sum_i \alpha_i q_i^{\mathrm{out}}(F)$ plus, for each pair $\{i,j\}$, the term $\alpha_i q_{ij}(1-u_j/u_i) + \alpha_j q_{ji}(1-u_i/u_j)$. By the AM–GM inequality this term is at most $(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2$, which gives the upper bound. For $\alpha \in \Delta_F^\circ$ the vector $u_i = \sqrt{\alpha_i/\nu_i}$ gives $\alpha_i q_{ij} u_j/u_i = \alpha_j q_{ji} u_i/u_j$ by $\nu_i q_{ij} = \nu_j q_{ji}$, so there is equality for every pair at once. If some α_i vanish, take $u_i = \varepsilon$ for these i and let $\varepsilon \to 0$. A pair with $\alpha_i = 0 < \alpha_j$ then has the term $\alpha_j q_{ji}(1-\varepsilon/u_j) \to \alpha_j q_{ji}$, which is its bound, and a pair with $\alpha_i = \alpha_j = 0$ has the term 0. □

Proposition 3.4 (the variational formula) *Let F be irreducible. For every $v \in \mathbb{R}^F$,*

$$\lambda_F(v) = \min_{\alpha\in\Delta_F} \big(J_F(\alpha) - v\cdot\alpha\big),$$

and the minimum is attained only at $\alpha_F^*(v)$. Equivalently, $-\lambda_F(v)$ is the Legendre transform of J_F at v, and J_F is the Legendre transform of $v \mapsto -\lambda_F(v)$ on Δ_F°. The map $v \mapsto \lambda_F(v)$ is concave and real-analytic, and $\lambda_F(v + c\mathbf{1}) = \lambda_F(v) - c$. If Q_F is reversible with respect to ν, let $S_F = D_{\sqrt{\nu}}(-Q_F)D_{\sqrt{\nu}}^{-1}$, where $D_x = \operatorname{diag} x$. Then with $\psi_i = \sqrt{\alpha_i}$,

$$J_F(\alpha) - v\cdot\alpha = \sum_{i\in F}(q_i - v_i)\alpha_i - \sum_{i\neq j\in F}\sqrt{q_{ij}q_{ji}}\,\psi_i\psi_j$$

is the Rayleigh quotient of the symmetric matrix $S_F - \operatorname{diag} v$ at the unit vector ψ, and $\lambda_F(v) = \lambda_{\min}(S_F - \operatorname{diag} v)$.

The formula is the finite form of the Donsker and Varadhan formula for the principal eigenvalue [20], close to the characterization of Collatz and Wielandt [3, Ch. 2].

Proof The formula and the unique minimizer are Proposition 3.3(a). As a minimum of affine functions of v, $\lambda_F(v)$ is concave. It is real-analytic because the Perron root of an irreducible Metzler matrix is a simple eigenvalue, and the shift is clear.

Legendre duality. The formula says $-\lambda_F(v) = \max_\alpha(v\cdot\alpha - J_F(\alpha))$. Conversely $J_F(\alpha) \geq v\cdot\alpha + \lambda_F(v)$ for all α and v. Let $\alpha \in \Delta_F^\circ$, take r_α from Proposition 3.3(b), and put $v = q - A_F r_\alpha/r_\alpha$, with the quotient taken entrywise. Then $Q_F + \operatorname{diag} v$ has the off-diagonal entries of Q_F and the diagonal $-A_F r_\alpha/r_\alpha$, so $r_\alpha > 0$ is a null vector of it. Only the Perron root has a positive eigenvector, so $\lambda_F(v) = 0$. Also $J_F(\alpha) = \langle q - A_F r_\alpha/r_\alpha, \alpha\rangle = v\cdot\alpha$, so equality holds at this v.

The reversible form. Reversibility gives $\sqrt{\nu_i}\,q_{ij}/\sqrt{\nu_j} = \sqrt{q_{ij}q_{ji}}$, so S_F is symmetric with diagonal q_i and off-diagonal entries $-\sqrt{q_{ij}q_{ji}}$. Expanding the squares in Proposition 3.3(c) gives $J_F(\alpha) = \langle\psi, S_F\psi\rangle$, and subtracting $v\cdot\alpha = \langle\psi, \operatorname{diag}(v)\psi\rangle$ gives the display. Every unit vector $\psi \geq 0$ arises from $\alpha = \psi\circ\psi$, and the off-diagonal entries of $S_F - \operatorname{diag} v$ are nonpositive, so passing to the entrywise modulus does not increase the quadratic form. Hence the minimum over Δ_F is $\lambda_{\min}(S_F - \operatorname{diag} v)$. □

3.3 The functional of the discrete chain

When T is a fixed fraction of N the counts follow the functional of the discrete chain itself. Let $0 < h < 1/q_{\max}$ and $p_i = 1 - hq_i$, so that $I + hQ$ is a stochastic matrix with

positive diagonal. For a face F and $\alpha \in \Delta_F$ let

$$J_F^{(h)}(\alpha) = \frac{1}{h} \sup_{u \in (0,\infty)^F} \sum_{i \in F} \alpha_i \log \frac{u_i}{((I + hQ_F)u)_i},$$

and put $J_F^{(0)} = J_F$. For $F = V$, $hJ_V^{(h)}$ is the Donsker and Varadhan rate of the empirical measure of the chain with transition matrix $I + hQ$ [20]. For $\psi \in \mathbb{R}^F$ put $\theta_i(\psi) = \sum_{j \in F \setminus \{i\}} q_{ij} e^{\psi_j - \psi_i}$, and let $L_h(y) = \int_0^1 y\, ds/(1 - shy)$, which is $-\log(1 - hy)/h$ for $h > 0$ and y for $h = 0$. With $u = e^\psi$ we have $((I + hQ_F)u)_i/u_i = p_i + h\theta_i(\psi)$, so

$$J_F^{(h)}(\alpha) = \sup_{\psi \in \mathbb{R}^F} G_h(\alpha, \psi), \qquad G_h(\alpha, \psi) = \sum_{i \in F} \alpha_i L_h\big(q_i - \theta_i(\psi)\big),$$

and at $h = 0$ this is the supremum that defines J_F. Since $L_h(y) \geq y$, we have $J_F^{(h)} \geq J_F$. We say that a function on $[0, h_*] \times U$ is real-analytic up to $h = 0$ if near each point it is the restriction of a real-analytic function on an open set.

Proposition 3.5 (the discrete functional) *Fix h_* with $h_* q_{\max} < 1$.*

(a) For every face F, every $0 < h < 1/q_{\max}$ and every $\alpha \in \Delta_F$, $J_F^{(h)}(\alpha) = J_V^{(h)}(\alpha)$. Hence $J_F^{(h)} = J_{F'}^{(h)}$ on $\Delta_{F'}$ for every nonempty $F' \subseteq F$.

(b) Let F be irreducible with $k \geq 2$, and let $(h, \alpha) \in [0, h_*] \times \Delta_F^\circ$. The supremum defining $J_F^{(h)}(\alpha)$ is attained at exactly one $\psi = \psi^{(h)}(\alpha)$ with $\psi_1 = 0$, and there the Hessian of $G_h(\alpha, \cdot)$ in $(\psi_2, \dots, \psi_k)$ is negative definite. The maps $\psi^{(h)}(\alpha)$, $\theta^{(h)}(\alpha) = \theta(\psi^{(h)}(\alpha))$ and $J_F^{(h)}(\alpha)$ are real-analytic on $[0, h_*] \times \Delta_F^\circ$ up to $h = 0$, and $\theta^{(0)}(\alpha) = A_F r_\alpha / r_\alpha$ with r_α from Proposition 3.3(b). The function $J_F^{(h)}$ is convex on Δ_F and strictly convex on Δ_F°.

(c) Let F be irreducible with $k \geq 2$, $0 \leq h \leq h_*$, $\alpha \in \Delta_F^\circ$ and $\psi \in \mathbb{R}^F$, and put $\theta = \theta(\psi)$. Then ψ attains the supremum if and only if the vector $(\alpha_i/(p_i + h\theta_i))_{i \in F}$ is proportional to $l \circ e^\psi$, where l is a positive left null vector of $A_F - \operatorname{diag} \theta$.

Proof We prove (a), then (b), then (c). (a) As in the proof of Proposition 3.1, the terms $hq_{ij}u_j$ with $j \notin F$ only increase $((I + hQ)u)_i$, which gives $J_V^{(h)} \leq J_F^{(h)}$ on Δ_F, and $u = \varepsilon$ off F with $\varepsilon \downarrow 0$ gives the converse, since the diagonal of $I + hQ_F$ keeps the full exit rates. The second claim is the first for F and for F'.

(c) We have $\partial\theta_i/\partial\psi_m = q_{im} e^{\psi_m - \psi_i}$ for $m \neq i$ and $\partial\theta_i/\partial\psi_i = -\theta_i$, and $L_h'(y) = 1/(1 - hy)$. So with $w_i = \alpha_i/(p_i + h\theta_i)$ the gradient of $G_h(\alpha, \cdot)$ is $-w^\top G^\psi$, where G^ψ is the conservative generator of the proof of Proposition 3.3, with off-diagonal entries $q_{ij} e^{\psi_j - \psi_i}$. The function $G_h(\alpha, \cdot)$ is concave by the proof of (b) below. Hence ψ attains the supremum if and only if w is proportional to the stationary law of G^ψ. With $U = \operatorname{diag}(e^\psi)$ we have $G^\psi = U^{-1}(A_F - \operatorname{diag} \theta)U$, whose stationary law is proportional to $l \circ e^\psi$.

(b) *Concavity.* At $h = 0$ the Hessian of $-G_0$ in a direction δ is $\sum_i \alpha_i \sum_{j \neq i} q_{ij} e^{\psi_j - \psi_i} (\delta_j - \delta_i)^2$. For $h > 0$,

$$-hG_h(\alpha, \psi) = \sum_{i \in F} \alpha_i \log\Big(p_i + \sum_{j \in F \setminus \{i\}} hq_{ij} e^{\psi_j - \psi_i}\Big).$$

The i-th term is α_i times the logarithm of a sum of exponentials of the affine functions 0 and $\psi_j - \psi_i$, with the positive weights p_i and hq_{ij}. Its second derivative in a direction δ is α_i times the variance of the values 0 and $\delta_j - \delta_i$ under the weights proportional to p_i and $hq_{ij}e^{\psi_j-\psi_i}$. As $p_i > 0$, it vanishes only if $\alpha_i = 0$ or $\delta_j = \delta_i$ on every edge $i \to j$ of G_F. So for every $h \in [0, h_*]$ the function $G_h(\alpha, \cdot)$ is concave, and for $\alpha \in \Delta_F^\circ$ its Hessian is negative definite on $\{\delta_1 = 0\}$, since G_F is strongly connected. This is where $h_* q_{\max} < 1$ is used.

The supremum is attained. Let $\psi_1 = 0$ and let $\bar\psi$ be the largest difference of two entries of ψ. As in the proof of Proposition 3.3(b), some edge $i \to j$ has $\psi_j - \psi_i \ge \bar\psi/(k-1)$, so $\theta_i(\psi) \ge ae^{\bar\psi/(k-1)}$ with a the least positive rate in F. Since L_h is increasing and tends to $-\infty$ at $-\infty$, the term of i tends to $-\infty$ as $\bar\psi \to \infty$, while every term is at most $\alpha_i L_h(q_i) \le q_{\max}/p_{\min}$, with $p_{\min} = 1 - h_* q_{\max}$. Hence the supremum is attained, and by strict concavity at exactly one ψ with $\psi_1 = 0$.

Analyticity. The function $L_h(y)$ is real-analytic in (h, y) where $1 - hy > 0$, including $h = 0$ and small negative h, and $1 - h(q_i - \theta_i) \ge p_i > 0$ for $h \in [0, h_*]$. So G is real-analytic near every point (h, α, ψ) with $h \in [0, h_*]$. At the maximizer the gradient in $(\psi_2, \dots, \psi_k)$ vanishes and the Hessian is invertible. The implicit function theorem gives a real-analytic solution near (h, α), which is $\psi^{(h)}(\alpha)$ for $h \ge 0$ by uniqueness. Hence $\theta^{(h)}(\alpha)$ and $J_F^{(h)}(\alpha) = G_h(\alpha, \psi^{(h)}(\alpha))$ are real-analytic up to $h = 0$. At $h = 0$ the maximizer is $\log r_\alpha$, normalized.

Convexity. As a supremum of affine functions, $J_F^{(h)}$ is convex. Let $\alpha^0 \ne \alpha^1$ lie in Δ_F°, let $0 < s < 1$, and let ψ attain $J_F^{(h)}(\alpha^s)$ at $\alpha^s = (1-s)\alpha^0 + s\alpha^1$. Equality in $J_F^{(h)}(\alpha^s) \le (1-s)J_F^{(h)}(\alpha^0) + sJ_F^{(h)}(\alpha^1)$ would make ψ attain the supremum at α^0 and at α^1. By (c) both would then be proportional to $((p_i + h\theta_i)l_i e^{\psi_i})_{i\in F}$, and as probability vectors they would be equal. □

For a field $v \in \mathbb{R}^F$, $J_F^{(h)}(\alpha) - v\cdot\alpha$ is $1/h$ times the supremum of $\sum_i \alpha_i \log(u_i/(Mu)_i)$ with $M = e^{h \operatorname{diag} v}(I + hQ_F)$, since $-hv_i\alpha_i = \alpha_i \log e^{-hv_i}$. For any face F', any $v \in \mathbb{R}^{F'}$ and $0 < h < 1/q_{\max}$ we put $\lambda_{F'}^{(h)}(v) = -\frac{1}{h}\log \varrho\big(e^{h \operatorname{diag} v}(I + hQ_{F'})\big)$, which is defined since the matrix is nonnegative with a positive diagonal.

Proposition 3.6 (the discrete exit rate) *Let F be irreducible with $k \ge 2$, let $0 < h < 1/q_{\max}$ and $v \in \mathbb{R}^F$, and put $M = e^{h \operatorname{diag} v}(I + hQ_F)$. Let r and l be positive right and left Perron vectors of M with $l \cdot r = 1$, and put $\alpha_F^{*(h)}(v) = l \circ r$.*

(a) On Δ_F, $J_F^{(h)}(\alpha) - v\cdot\alpha \ge \lambda_F^{(h)}(v)$, with equality only at $\alpha_F^{*(h)}(v)$, which lies in Δ_F°. For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $J_F^{(h)}(\alpha) - v\cdot\alpha \ge \lambda_{F'}^{(h)}(v_{F'}) > \lambda_F^{(h)}(v)$.

(b) At $v = 0$, $\varrho(M) = 1 - h\lambda_F$ and $\alpha_F^{*(h)}(0) = \alpha_F^*$, so $\lambda_F^{(h)}(0) = -\frac{1}{h}\log(1 - h\lambda_F)$ for every h.

(c) Fix h_* with $h_* q_{\max} < 1$, and put $\lambda_F^{(0)}(v) = \lambda_F(v)$ and $\alpha_F^{*(0)}(v) = \alpha_F^*(v)$. Then $\lambda_F^{(h)}(v)$ and $\alpha_F^{*(h)}(v)$ are real-analytic on $[0, h_*] \times \mathbb{R}^F$ up to $h = 0$.

Proof We prove this through the matrix B. Put $B = (M - I)/h$. Its off-diagonal entries $e^{hv_i}q_{ij}$ are nonnegative and positive exactly on the edges of G_F, so B is an irreducible Metzler matrix. Its entries are real-analytic in (h, v) up to $h = 0$, and $B = Q_F + \operatorname{diag} v$ at $h = 0$. Let $\kappa = \varrho(B)$. Then $\varrho(M) = 1 + h\kappa$, and M and B have the same Perron vectors. As κ is a

simple eigenvalue of B, it and the normalized Perron vectors are real-analytic in (h, v) up to $h = 0$, and so is $\lambda_F^{(h)}(v) = -\kappa \int_0^1 ds/(1 + sh\kappa)$. This gives (c). At $v = 0$ we have $B = Q_F$ for every h, which gives (b).

(a) *A lower bound on every face.* Let $F' \subseteq F$ be nonempty, irreducible or not, and let M' be the principal submatrix of M on F', which is $e^{h \operatorname{diag} v_{F'}}(I + hQ_{F'})$ since the diagonal keeps the full exit rates. For $\alpha \in \Delta_{F'}$, part (a) of Proposition 3.5 gives $J_F^{(h)}(\alpha) = J_{F'}^{(h)}(\alpha)$, so $J_F^{(h)}(\alpha) - v \cdot \alpha$ is $1/h$ times the supremum of $\sum_{i \in F'} \alpha_i \log(u_i/(M'u)_i)$ over $u \in (0, \infty)^{F'}$. Let $s > \varrho(M')$. As $M' \geq 0$, the vector $u = (sI - M')^{-1}\mathbf{1} = \sum_{j \geq 0} s^{-j-1} M'^j \mathbf{1}$ is positive, and $M'u = su - \mathbf{1}$. Also $M'u > 0$, since M' has a positive diagonal. So $u_i/(M'u)_i > 1/s$ for every i, and letting $s \downarrow \varrho(M')$ gives $J_F^{(h)}(\alpha) - v \cdot \alpha \geq \lambda_{F'}^{(h)}(v_{F'})$. If $F' \neq F$, then M' padded with zeros is at most M and differs from it, since the irreducible M has a positive entry in a row of F' and a column off F'. So $\varrho(M') < \varrho(M)$ [3, Ch. 2], which gives the second claim. The case $F' = F$ is the lower bound.

Equality. With $u = r$ every ratio $r_i/(Mr)_i$ is $1/\varrho(M)$. The function $\psi \mapsto \sum_i \alpha_i(\psi_i - \log(Me^{\psi})_i)$ is concave, and at $\psi = \log r$ its derivative in ψ_m is $\alpha_m - \sum_i \alpha_i M_{im} r_m/(\varrho(M) r_i)$. For $\alpha = l \circ r$ this is $\alpha_m - (l^{\top} M)_m r_m/\varrho(M) = 0$. So $u = r$ attains the supremum at $\alpha_F^{*(h)}(v)$, and equality holds there. This point lies in Δ_F° since $r, l > 0$.

Uniqueness. On the boundary of Δ_F the inequality is strict by the second claim. In Δ_F°, $J_F^{(h)}(\alpha) - v \cdot \alpha$ is strictly convex by Proposition 3.5(b), so it has only one minimizer. □

4 First-order face selection

4.1 An example with four states on a cycle

Take $d = 4$ states on a cycle, with rate 1 between neighbors and the uniform start. Up to symmetry the admissible faces are a vertex, an adjacent pair, an arc of three states and the whole cycle, with exit rates 2, 1, $2 - \sqrt{2}$ and 0. For the arc, $-Q_F = 2I - A$ with A the adjacency matrix of the path 1–2–3, whose largest eigenvalue is $\sqrt{2}$. The chain stays in a face F up to time T with probability about $e^{-T\lambda_F}$, and this mass is shared by about $(N/\sqrt{T})^{|F|-1}$ compositions. Hence $M_F \approx N^{-(|F|-1)} T^{(|F|-1)/2} e^{-T\lambda_F}$, and with $T = \tau \log N$ this is $N^{-\Phi_\tau(F)}$ up to powers of $\log N$. Figure 2(a) plots the four exponents. The vertices win for $\tau < 1$, the pairs for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The arcs never win, since they beat the pairs only for $\tau > 1 + \sqrt{2}$, where the cycle beats them. The whole cycle carries almost all the mass for every $\tau > 0$, and it carries the mode only for $\tau > 2$, because it spreads its mass over about N^3 compositions. Theorems 4.2 and 5.7 make this rule exact, and for a pair Corollary 5.3 gives the constant $1/\sqrt{2\pi}$ in front.

4.2 Face selection

We also need an upper bound on a whole face. This includes the region near its boundary, where the local law does not apply. For an irreducible face F put $V_F = \{\theta \in \mathbb{R}^F : \varrho(A_F - \operatorname{diag}\theta) = 0\}$. Each $\theta \in V_F$ gives an exponential bound with the linear rate $\alpha \mapsto \langle q - \theta, \alpha \rangle$, and J_F is the supremum of these linear rates. Indeed, for $u \in (0, \infty)^F$ the entrywise quotient $\theta = A_F u/u$ lies in V_F, since u is a positive null vector of the irreducible Metzler matrix $A_F - \operatorname{diag}\theta$ and only its Perron root has a

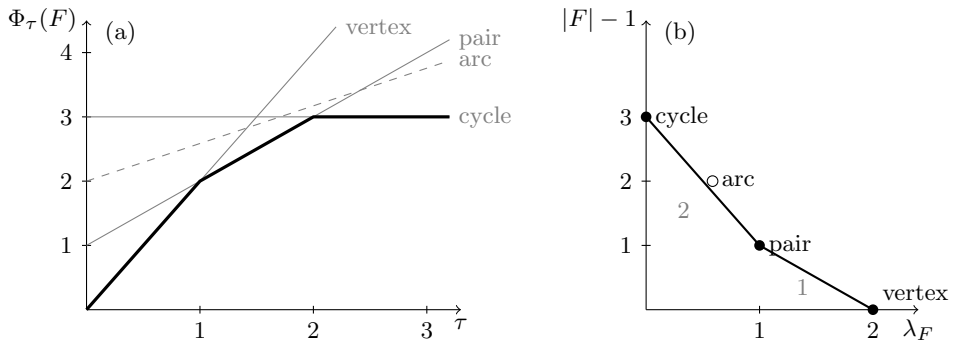


Fig. 2 The 4-cycle. (a) The exponents $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ of the four kinds of faces against τ, and their lower envelope (bold). The mode sits on the face whose line is lowest, namely a vertex for $\tau < 1$, an adjacent pair for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The dashed line of the arc of three states is never lowest. (b) The same faces as points $(\lambda_F, |F| - 1)$, with the lower-left boundary of their convex hull. The gray numbers are minus the slopes of its edges, the values of τ where the winner changes (Theorem 4.4). The arc (open) lies above the boundary

positive eigenvector. Then $\langle q - \theta, \alpha\rangle$ is the function of u inside the supremum that defines $J_F(\alpha)$. Conversely every $\theta \in V_F$ arises in this way from a positive Perron vector of $A_F - \operatorname{diag}\theta$. For a field $v \in \mathbb{R}^F$ the vector $\theta = q - v - \lambda_F(v)\mathbf{1}$ lies in V_F, since $A_F - \operatorname{diag}\theta = Q_F + \operatorname{diag} v + \lambda_F(v)I$.

Lemma 4.1 (a uniform upper bound) *Let F be irreducible with $k = |F| \geq 2$, and let $\theta \in V_F$. There is C_θ, depending on θ, Q and μ and bounded for θ in compact subsets of V_F, such that for every $n \in \mathbb{Z}^F_{\geq 0}$ with $\operatorname{supp} n = F$ and every h with $hq_{\max} \leq \frac{1}{2}$, the probability that $X_1, \dots, X_{|n|}$ have composition n is at most $C_\theta h^{k-1} \exp\bigl(-h\langle n, q - \theta\rangle\bigr)$.*

The proof is in Appendix A. It writes the probability as an integral over a torus and bounds the integrand by its value at the real point. The lemma holds for a path of any length $|n|$, so it also applies to the blocks of a path. With $\theta = q - v - \lambda_F(v)\mathbf{1}$ and $|n| = N$ it gives

$$P_N(n)e^{Tv\cdot n/N} \leq C_\theta h^{k-1} e^{-T\lambda_F(v)} \tag{6}$$

for every n with support F. The next theorem uses the local law and the face maxima of Section 5, which are proved there without it. Its part (i) is a weak form of Theorem 5.6(b), and its part (ii) is the only part that bounds the face maxima of reducible faces.

Theorem 4.2 (first-order face selection) *Fix Q and μ.*

(i) Let F satisfy (H)*, let $k = |F|$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There is N_0 such that, uniformly for $N \geq N_0$, $1 \leq T \leq N^{1/3}$ and $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$,*

$$\log P_N(n) = -(k-1)L - T\,J_F(n/N) + O_{\mathcal{K}}(1 + \log T).$$

(ii) Let F be admissible with components $C_1, \dots, C_r$, and let k^* be the least size of a component C_j with $\lambda_{C_j} = \lambda_F$. There are κ, T_0 and N_0 such that for $N \geq N_0$ and

$T_0 \le T \le N^{1/3}$

$$\log M_F \ge -(k_F-1)L + (k_F - k^*)\log T - T\lambda_F - \kappa,$$
$$\log M_F \le -(k_F-1)L + (k_F - 1)\log T - T\lambda_F + \kappa.$$

If F is not admissible, $M_F = 0$.

(iii) Let $T = \tau L$ with $\tau > 0$ fixed, and let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ. Then $\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L)$, with the minimum over admissible faces, and for large N every global mode $\hat{n}$ has $\operatorname{supp}\hat{n} \in \mathcal{F}_\tau$. This holds for every start μ.

(iv) If in (iii) every face in $\mathcal{F}_\tau$ is irreducible, then every global mode $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/L)$ with $F = \operatorname{supp}\hat{n}$. This is the case when Q has symmetric support or μ has full support (Theorem 4.4(c)).

Proof (i) This is Theorem 5.6(b), since $J_V = J_F$ on Δ_F, $\log C_F$ is bounded on $\mathcal{K}$ and $J_{\{i\}} = q_i$.

(ii) This follows from Theorem 4.5 below, whose proof in Appendix E does not use the present theorem. After T_1 and N_1 are enlarged, that theorem gives $\log M_F = -(k_F-1)L + p_F \log T - T\lambda_F + O(1)$ for $N \ge N_1$ and $T_1 \le T \le N^{1/3}$. Since $T \ge 1$ and $k_F - k^* \le p_F \le k_F - 1$, both bounds hold with $T_0 = T_1$ and $N_0 = N_1$. If F is not admissible, no path has support F, so $M_F = 0$. Appendix A.7 gives a direct proof by explicit paths, which does not use the local law.

(iii) For large N we have $T_0 \le T = \tau L \le N^{1/3}$, and both bounds in (ii) read $\log M_F = -L\,\Phi_\tau(F) + O(\log L)$ for every admissible face, while $M_F = 0$ for the others. Since $\max_n P_N(n) = \max_F M_F$ over finitely many faces, the first claim follows. Let $m_\tau = \min_F \Phi_\tau(F)$. An admissible $F \notin \mathcal{F}_\tau$ has $\Phi_\tau(F) \ge m_\tau + \varepsilon_0$ for a fixed $\varepsilon_0 > 0$, so $M_F \le N^{-m_\tau - \varepsilon_0 + o(1)}$, which is smaller than $\max_n P_N(n) = N^{-m_\tau + o(1)}$ for large N. A global mode has positive probability, so its support is admissible, and hence it lies in $\mathcal{F}_\tau$.

(iv) By (iii), for large N a global mode $\hat{n}$ maximizes P_N on its face $F \in \mathcal{F}_\tau$, which satisfies (H) by Lemma 2.2(b). So Theorem 5.7(b) with $T = \tau L \le N^{1/3}$ gives $\hat{n}/N = \alpha_F^* + O(1/L)$. □

On a reducible face the power of T can differ from $(k_F - 1)/2$, as in Example 5.11. Theorem 4.5 below gives the power and the constant for every admissible face.

Remark 4.3 (a field) Let $v \in \mathbb{R}^d$, put $M_F(T, v) = \max\{P_N(n) e^{Tv\cdot n/N} : \operatorname{supp} n = F\}$ and $\Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v_F)$. On an admissible face, $Q_F + \operatorname{diag} v_F$ is block triangular in the order of the components, so $\lambda_F(v_F) = \min_j \lambda_{C_j}(v_{C_j})$. The upper bound in (ii) holds for $M_F(T, v)$ with $\lambda_F(v_F)$ in place of λ_F. The proof in Appendix A.7 applies with (6) on each block and the exact weight $e^{hv_i N_j}$ on a block $\{i\}$ of length N_j. For F satisfying (H), Theorem 5.7 gives the matching lower bound. If μ has full support, a reducible admissible face F has a component C with $\lambda_C(v_C) = \lambda_F(v_F)$, which is admissible with fewer states, so $\Phi_{\tau,v}(C) \le \Phi_{\tau,v}(F) - 1$. Hence for fixed τ and v, (iii) and (iv) hold for the weighted law with $\Phi_{\tau,v}$ and $\alpha_F^*(v)$, every face that minimizes $\Phi_{\tau,v}$ is irreducible, and a strict minimizer carries every maximizer of the weighted law for large N.

4.3 Reading the winners off a convex hull

For an admissible face F put $P_F = (\lambda_F, |F| - 1)$, and let $\mathcal{S}$ be the set of these points. Then $\Phi_\tau(F) = \tau x + y$ at $(x, y) = P_F$. Picture a line of slope $-\tau$ moving up from far

below. The first points of $\mathcal{S}$ that it touches are the winners at τ. For each $\tau > 0$ the points of $\operatorname{conv}\mathcal{S}$ where $\tau x + y$ is least form a vertex or an edge of $\operatorname{conv}\mathcal{S}$, and we call the union of these over all $\tau > 0$ the lower-left boundary of $\operatorname{conv}\mathcal{S}$. Figure 2(b) shows it for the 4-cycle.

Theorem 4.4 (geometry of the winners) *Let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ.*

(a) The lower-left boundary of $\operatorname{conv}\mathcal{S}$ is a path whose vertices lie in $\mathcal{S}$. Read from right to left, they are $v_i = (\lambda^{(i)}, y^{(i)})$, $0 \le i \le m$, with $\lambda^{(0)} > \cdots > \lambda^{(m)}$ and $y^{(0)} < \cdots < y^{(m)}$. Put $\tau_i = (y^{(i)} - y^{(i-1)})/(\lambda^{(i-1)} - \lambda^{(i)})$ for $1 \le i \le m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \cdots < \tau_m$. For $\tau_i < \tau < \tau_{i+1}$, $\mathcal{F}_\tau$ is the set of faces F with $P_F = v_i$. At $\tau = \tau_i$ it is the set of faces whose point lies on the closed edge $[v_{i-1}, v_i]$.

(b) If $0 < \tau < \tau'$, $F \in \mathcal{F}_\tau$ and $G \in \mathcal{F}_{\tau'}$, then $\lambda_F \ge \lambda_G$ and $|F| \le |G|$. Both inequalities are strict when $\mathcal{F}_\tau \cap \mathcal{F}_{\tau'} = \emptyset$.

(c) If Q has symmetric support, or μ has full support, then every face in $\mathcal{F}_\tau$ is irreducible, for every $\tau > 0$.

Proof (a) We prove this through the least value $g(\tau) = \min_{(x,y)\in\mathcal{S}}(\tau x + y)$ for $\tau > 0$. Each point (x, y) of $\mathcal{S}$ gives the affine function $\tau \mapsto \tau x + y$, whose slope is x, and different points give different functions. Hence g is a minimum of finitely many affine functions, so it is concave and piecewise affine. Let $\tau_1 < \cdots < \tau_m$ be the points of $(0, \infty)$ where its slope changes, and put $\tau_0 = 0$ and $\tau_{m+1} = \infty$.

Between two kinks. On (τ_i, τ_{i+1}) the function g is affine. The affine function of a point of $\mathcal{S}$ is nowhere below g. So if it equals g at one τ inside this interval, it equals g on the whole interval. Hence exactly one point of $\mathcal{S}$ attains g there. We call it $v_i = (\lambda^{(i)}, y^{(i)})$, and for $\tau_i < \tau < \tau_{i+1}$ the set $\mathcal{F}_\tau$ consists of the faces F with $P_F = v_i$. The slope of g on this interval is $\lambda^{(i)}$. The slope of a concave function decreases, and it changes at each τ_i, so $\lambda^{(0)} > \cdots > \lambda^{(m)}$. By continuity both v_{i-1} and v_i attain $g(\tau_i)$, namely

$$\tau_i \lambda^{(i-1)} + y^{(i-1)} = \tau_i \lambda^{(i)} + y^{(i)}.$$

This is the formula for τ_i in the statement, and since $\tau_i > 0$ it gives $y^{(i)} > y^{(i-1)}$.

At a kink. The points (x, y) of $\mathcal{S}$ that attain $g(\tau_i)$ are those on the line $\tau_i x + y = g(\tau_i)$, which passes through v_{i-1} and v_i. Let (x, y) be one of them. For τ slightly below τ_i we have $\tau x + y \ge g(\tau) = \tau\lambda^{(i-1)} + y^{(i-1)}$, with equality at $\tau = \tau_i$. Subtracting gives $(\tau - \tau_i)(x - \lambda^{(i-1)}) \ge 0$, so $x \le \lambda^{(i-1)}$. In the same way, τ slightly above τ_i gives $x \ge \lambda^{(i)}$. So (x, y) lies on the closed edge $[v_{i-1}, v_i]$. Conversely every point of $\mathcal{S}$ on this edge lies on the line and so attains $g(\tau_i)$.

The boundary. On $\operatorname{conv}\mathcal{S}$ the function $\tau x + y$ is least exactly on the convex hull of the points of $\mathcal{S}$ where it is least. By the two steps above this hull is the vertex v_i for $\tau_i < \tau < \tau_{i+1}$ and the edge $[v_{i-1}, v_i]$ for $\tau = \tau_i$. So the lower-left boundary is the path $v_0 \cdots v_m$, and its vertices lie in $\mathcal{S}$.

(b) We prove this by adding two inequalities. Since F wins at τ and G wins at τ', we have $\Phi_\tau(F) \le \Phi_\tau(G)$ and $\Phi_{\tau'}(G) \le \Phi_{\tau'}(F)$. Adding them gives $(\tau' - \tau)(\lambda_F - \lambda_G) \ge 0$, so $\lambda_F \ge \lambda_G$. Then the first inequality gives $|F| - |G| \le \tau(\lambda_G - \lambda_F) \le 0$. If $\lambda_F = \lambda_G$, the two inequalities give $|F| = |G|$, so $\Phi_{\tau'}(F) = \Phi_{\tau'}(G)$ and $F \in \mathcal{F}_{\tau'}$. So if $\mathcal{F}_\tau$ and $\mathcal{F}_{\tau'}$ are disjoint, then $\lambda_F > \lambda_G$, and the first inequality gives $|F| < |G|$.

(c) With symmetric support every admissible face is irreducible by Lemma 2.2(c). Now let μ have full support, and let F be admissible and reducible. Take a component C of F with $\lambda_C = \lambda_F$. It is irreducible with $\mu(C) > 0$, so it is admissible by Lemma 2.2(b). It has the same exit rate as F and fewer states, so $\Phi_\tau(C) \le \Phi_\tau(F) - 1$. Hence $F \notin \mathcal{F}_\tau$. □

Note that a face with small λ_F for its size is a set of states that the chain rarely leaves. As τ grows the exit rate counts for more, so the mode moves to larger such sets. It need not grow around its first vertex and it can move to a disjoint set. Faces of one size share the term $|F| - 1$. Hence only the faces with the least exit rate of each size can win.

4.4 Reducible faces

Let F be admissible, with components $C_1, \dots, C_r$ numbered as in Lemma 2.2(a). A path of positive probability with support F visits them in this order, each in one block of consecutive times. The order is unique, since consecutive blocks are joined by an edge and the components form an acyclic graph, in which at most one path passes through every vertex. Put $k_j = |C_j|$ and $\lambda_j = \lambda_{C_j}$, so $\lambda_F = \min_j \lambda_j$. Let k^* be the least k_j with $\lambda_j = \lambda_F$, let S^* be the set of j with $\lambda_j = \lambda_F$ and $k_j = k^*$, and put

$$a_F = \frac{k^* - 1}{2}, \qquad p_F = a_F + k_F - k^*, \qquad R_j = (q_{xy})_{x \in C_j,\, y \in C_{j+1}}.$$

For an irreducible C with $|C| \ge 2$, states $y, x \in C$ and $t \in (0,\infty)^C$ let $\Phi_C(t)_{yx} = e^{-\langle q,t\rangle} H_{A_C, e_y, e_x}(t)$, in the notation before Lemma A.3. As in Lemma 2.4, this is the density of the occupation times of the chain started at y on the event that it stays in C up to time $|t|$, visits every state of C and ends at x. For $C = \{i\}$ put $\Phi_C(t) = e^{-q_i t}$. For $j \in S^*$ let r and l be the vectors r_{C_j} and l_{C_j} of Definition 2.1, with $l \cdot r = 1$, and let

$$A_j = \sup\, e^{\lambda_F |t^<|}\, \mu_{C_1}^\top \Phi_{C_1}(t^{(1)}) R_1 \cdots \Phi_{C_{j-1}}(t^{(j-1)}) R_{j-1}\, r,$$
$$B_j = \sup\, e^{\lambda_F |t^>|}\, l^\top R_j \Phi_{C_{j+1}}(t^{(j+1)}) R_{j+1} \cdots \Phi_{C_r}(t^{(r)})\, \mathbf{1},$$

over all $t^{(m)} \in (0,\infty)^{C_m}$ for $m < j$, respectively $m > j$, where $|t^<|$ and $|t^>|$ are their total times, $A_1 = \mu_{C_1} \cdot r$ and $B_r = l \cdot \mathbf{1}$. Finally let

$$K_F^{\mathrm{red}} = \max_{j \in S^*} \frac{A_j B_j}{(2\pi)^{a_F} \sqrt{\det \Sigma_{C_j}}},$$

where Σ_{C_j} is Σ_α of Definition 5.1 for the face C_j at $\alpha = \alpha^*_{C_j}$, and the denominator is 1 if $k^* = 1$. If F is irreducible, then $p_F = (k_F - 1)/2$ and $K_F^{\mathrm{red}} = K_F$.

Theorem 4.5 (face maxima on reducible faces) *Let F be admissible. Then $0 < K_F^{\mathrm{red}} < \infty$, and there are T_1, N_1 and C such that for $N \ge N_1$ and $T_1 \le T \le N^{1/3}$*

$$\Big| \frac{M_F}{N^{-(k_F-1)} T^{p_F} e^{-T\lambda_F}} - K_F^{\mathrm{red}} \Big| \le C\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

In particular the ratio lies between $\frac{1}{2}K_F^{\mathrm{red}}$ *and* $2K_F^{\mathrm{red}}$ *after* T_1 *and* N_1 *are enlarged.*

Hence the power of T is p_F, between the two powers of Theorem 4.2(ii). The time goes to the components of least exit rate and least size k^*, and to one of them when $k^* \geq 2$. Such a component contributes $h^{k^*-1}T^{-(k^*-1)/2}$ through its local law, and each other component C gets time of order one and contributes $h^{|C|-1}$. Each switch from one component to the next contributes one more factor h. The constants A_j and B_j are maxima over these short blocks. For irreducible F the theorem is Theorem 5.7(b) at $v = 0$. The proof is in Appendix E.

Corollary 4.6 (crossings of admissible faces) *Let* F *and* G *be admissible, with* $\Delta = k_F - k_G \geq 1$ *and* $\delta = \lambda_G - \lambda_F > 0$. *Put* $\pi = p_F - p_G$, $\kappa = \log(K_F^{\mathrm{red}}/K_G^{\mathrm{red}})$ *and* $D_N = \log M_F - \log M_G$, *and let* T_1 *serve for both faces in Theorem 4.5.*

(i) For large N, D_N *has a zero in* $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.

(ii) For large N, *every zero* $T_N \in [T_1, N^{1/3}]$ *of* D_N *satisfies* $\delta T_N + \pi \log T_N = \Delta L - \kappa + O(1/L)$.

(iii) All such zeros lie in one interval of length $O(1/L)$, *and*

$$T_N = \frac{\Delta}{\delta}\Big[L - \frac{\pi}{\Delta}\log L + b_{FG}\Big] + O\Big(\frac{\log L}{L}\Big), \qquad b_{FG} = -\frac{\pi}{\Delta}\log\frac{\Delta}{\delta} - \frac{\kappa}{\Delta}.$$

The coefficient of $\log\log N$ is π/Δ. It is $\frac{1}{2}$ when F and G are irreducible, which gives Theorem 5.8(i) to (iii) for zeros in $[T_1, N^{1/3}]$. In Example 5.11, where $q_{12} = b$, we have $p_{\{1\}} = 0$, $p_{\{1,2\}} = 1$ and $p_{\{1,2,3\}} = 1$, and $K^{\mathrm{red}}_{\{1,2\}} = b$ from $A_2 = \sup_t be^{(1-b)t} = b$ and $B_2 = 1$. So the corollary gives the coefficients 1 and 0 of parts (ii) and (iii) there, with their constants. We do not claim that the zero is unique.

5 The second-order theory

To locate a crossing up to $o(1)$ we need the power of T and the constant in front of $N^{-(k-1)}e^{-T\lambda_F}$ for every face that satisfies (H). Throughout this section gradients and Hessians in α are taken in the coordinates $\alpha' = (\alpha_2, \dots, \alpha_k)$, $|\cdot|$ is the Euclidean norm, and $q = (q_i)_{i\in F}$.

5.1 The tilted chain and the constant

To find the probability of a composition with fractions α we change the rates so that α becomes typical. The optimal vector r_α in the definition of $J_F(\alpha)$ gives a new generator $\widehat{Q}_\alpha$ on F, with the same switching graph, whose stationary law is exactly α. Under it the visit fractions satisfy a central limit theorem with covariance Γ_α/T, and the price of the change of rates is $e^{-TJ_F(\alpha)}$ times boundary factors for the start and the end of the path. The constant $C_F(\alpha)$ collects these factors and the Gaussian normalization.

Definition 5.1 (tilt and constants) Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ let r_α attain the supremum defining $J_F(\alpha)$ (Proposition 3.3(b)), and put $\theta(\alpha) = A_F r_\alpha / r_\alpha$ and

$l_\alpha = \alpha / r_\alpha$ (entrywise). The *tilted generator* is $\widehat{Q}_\alpha = R^{-1}(A_F - \operatorname{diag}\theta(\alpha))R$ with $R = \operatorname{diag} r_\alpha$. With $D = \operatorname{diag}\alpha$ let

$$Z_\alpha = (\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top, \qquad \Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D,$$

and let Σ_α be Γ_α with the row and column of state 1 deleted. Finally let

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}\sqrt{\det \Sigma_\alpha}}, \qquad K_F = C_F(\alpha_F^*).$$

For a single state we put $K_{\{i\}} = \mu_i$. Nothing in Definition 5.1 changes when r_α is scaled, and nothing in it involves the diagonal of Q_F. Hence the objects depend on A_F and μ only, and a field does not change them. For $v \in \mathbb{R}^F$ we put $K_F(v) = C_F(\alpha_F^*(v))$, so that $K_F(0) = K_F$, and $K_{\{i\}}(v) = \mu_i$. Also $l_\alpha \cdot r_\alpha = 1$ and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$. Since r_α minimizes $u \mapsto \sum_{i\neq j} \alpha_i q_{ij} u_j / u_i$, its derivative in u_j, which is $\sum_i l_{\alpha,i} q_{ij} - l_{\alpha,j}\theta_j(\alpha)$, vanishes at r_α. Hence

$$l_\alpha^\top \big(A_F - \operatorname{diag}\theta(\alpha)\big) = 0. \tag{7}$$

Also $(A_F - \operatorname{diag}\theta(\alpha))r_\alpha = 0$ with $r_\alpha > 0$, so $\theta(\alpha) \in V_F$ [3]. Hence $\widehat{Q}_\alpha$ is an irreducible generator with the same off-diagonal zero pattern as A_F, and by (7) its stationary law is α. If $(\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)v = 0$, multiplying by $\alpha^\top$ gives $\alpha^\top v = 0$, so $\widehat{Q}_\alpha v = 0$. Hence v is a multiple of $\mathbf{1}$, and $\alpha^\top v = 0$ gives $v = 0$. Thus Z_α exists, and

$$Z_\alpha \mathbf{1} = 0, \qquad \alpha^\top Z_\alpha = 0, \qquad -\widehat{Q}_\alpha Z_\alpha = I - \mathbf{1}\alpha^\top. \tag{8}$$

Hence $\Gamma_\alpha \mathbf{1} = 0$. As Γ_α is symmetric, its adjugate is then a multiple of $\mathbf{1}\mathbf{1}^\top$, so deleting another state instead of state 1 gives the same $\det \Sigma_\alpha$. At α_F^* the vector $u = r_F$ gives the value $\lambda_F = J_F(\alpha_F^*)$, so $r_{\alpha_F^*}$ is a multiple of r_F, $\theta(\alpha_F^*) = q - \lambda_F \mathbf{1}$, and $l_{\alpha_F^*}$ is the matching multiple of l_F.

For conductances $c_{ij} = c_{ji}$ on F let $\operatorname{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\}\in\mathcal{T}} c_{ij}$, the sum over the spanning trees $\mathcal{T}$ of the complete graph on F.

Proposition 5.2 (the constant) *Let F be irreducible with $k \geq 2$.*

(a) The maps $\alpha \mapsto \theta(\alpha)$, $J_F(\alpha)$, Σ_α, r_α and l_α (with $\sum_i r_{\alpha,i} = 1$) are real-analytic on Δ_F°, Σ_α is positive definite, and $\nabla^2 J_F(\alpha) = \Sigma_\alpha^{-1}$. Hence $\det \Sigma_\alpha \cdot \det \nabla^2 J_F(\alpha) = 1$ and $C_F(\alpha) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})\sqrt{\det \nabla^2 J_F(\alpha)/(2\pi)^{k-1}}$.

(b) Suppose that $\nu_i q_{ij} = \nu_j q_{ji}$ for $i, j \in F$ and some $\nu \in (0,\infty)^F$. Then r_α is a multiple of $\sqrt{\alpha/\nu}$ and l_α of $\sqrt{\alpha\nu}$, and $\widehat{Q}_\alpha$ is reversible with conductances $c_{ij}(\alpha) = \sqrt{\alpha_i\alpha_j}\, q_{ij}\sqrt{\nu_i/\nu_j}$. Also $J_F(\alpha) = \sum_i q_i\alpha_i - \sum_{i\neq j} c_{ij}(\alpha)$, and

$$\det \Sigma_\alpha = \frac{2^{k-1}(\prod_i \alpha_i)^2}{\operatorname{tree}(c(\alpha))}, \qquad C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(4\pi)^{(k-1)/2}} \frac{\sqrt{\operatorname{tree}(c(\alpha))}}{\prod_i \alpha_i}.$$

Proof We prove (a) in Appendix A, using Lemma A.1.

(b) We guess the vector r_α and check that it is a critical point. Put $u = \sqrt{\alpha/\nu}$ and $l = \alpha/u = \sqrt{\alpha\nu}$ entrywise. As we saw before (7), the derivative of $\sum_{i\neq j} \alpha_i q_{ij} u_j/u_i$ in u_j is $\sum_i l_i q_{ij} - l_j (A_F u)_j / u_j$. Here

$$\sum_i l_i q_{ij} = \sum_i \sqrt{\alpha_i \nu_i}\, q_{ij} \qquad \text{and} \qquad \frac{l_j (A_F u)_j}{u_j} = \nu_j \sum_i q_{ji} \sqrt{\alpha_i/\nu_i} = \sum_i \sqrt{\alpha_i \nu_i}\, q_{ij},$$

where the last step uses $\nu_j q_{ji} = \nu_i q_{ij}$. Hence the derivative vanishes at u. The function is convex in $\log u$, so its critical point u is a minimizer, and the minimizer is unique up to a factor (Proposition 3.3(b)). Hence r_α is a multiple of u and l_α is the matching multiple of l. Then $\alpha_i (\widehat{Q}_\alpha)_{ij} = l_i q_{ij} u_j = c_{ij}(\alpha)$ for $i \neq j$, and $c_{ij}(\alpha) = c_{ji}(\alpha)$ because $q_{ij}\sqrt{\nu_i/\nu_j} = q_{ji}\sqrt{\nu_j/\nu_i}$. Hence $\widehat{Q}_\alpha$ is reversible with these conductances. Also $\alpha_i \theta_i(\alpha) = l_i (A_F u)_i = \sum_{j\neq i} c_{ij}(\alpha)$, and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$ gives the formula for J_F.

For the determinant we express the quadratic form of Σ_α through the Laplacian of the conductances, namely $\mathcal{L} = -D\widehat{Q}_\alpha$. It is symmetric, its off-diagonal entries are $-c_{ij}(\alpha)$, and $\mathcal{L}\mathbf{1} = 0$. Fix $\eta \in \mathbb{R}^F$ with $\eta_1 = 0$, and put $w = Z_\alpha \eta$ and $\zeta = D\eta - (\alpha\cdot\eta)\alpha$. By (8), $\mathcal{L} w = D(I - \mathbf{1}\alpha^\top)\eta = \zeta$ and $\alpha \cdot w = 0$. So $\eta^\top \Gamma_\alpha \eta = 2(D\eta)^\top w = 2\zeta^\top w$, where the second step uses $\alpha \cdot w = 0$. Next we remove state 1. Let $v = w - w_1 \mathbf{1}$, so that $v_1 = 0$. Since $\zeta \cdot \mathbf{1} = 0$ and $\mathcal{L}\mathbf{1} = 0$, we have $\zeta^\top w = \zeta^\top v$ and $\mathcal{L} v = \zeta$. We write primes for the coordinates $2, \dots, k$, $D' = \operatorname{diag}\alpha'$, and $\mathcal{L}_{(1)}$ for $\mathcal{L}$ without row and column 1. As $v_1 = 0$, the rows $2, \dots, k$ of $\mathcal{L} v = \zeta$ say that $\mathcal{L}_{(1)} v' = \zeta'$, and as $\eta_1 = 0$ we have $\zeta' = (D' - \alpha'\alpha'^\top)\eta'$. The matrix-tree theorem [7] gives $\det \mathcal{L}_{(1)} = \operatorname{tree}(c(\alpha))$, which is positive because F is irreducible. Hence $\mathcal{L}_{(1)}$ is invertible, and

$$\eta'^\top \Sigma_\alpha \eta' = 2\zeta'^\top v' = 2\eta'^\top (D' - \alpha'\alpha'^\top)\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^\top)\eta'$$

for every $\eta' \in \mathbb{R}^{k-1}$. Both matrices are symmetric, hence $\Sigma_\alpha = 2(D' - \alpha'\alpha'^\top)\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^\top)$. We take determinants. The matrix determinant lemma gives $\det(D' - \alpha'\alpha'^\top) = \prod_{i\geq 2} \alpha_i \,(1 - \sum_{i\geq 2} \alpha_i) = \prod_i \alpha_i$. Hence $\det \Sigma_\alpha = 2^{k-1} (\prod_i \alpha_i)^2 / \operatorname{tree}(c(\alpha))$, and putting this into Definition 5.1 gives the formula for $C_F(\alpha)$. □

Corollary 5.3 (symmetric faces) *Let F satisfy* (H) *with $k \geq 2$, and let Q_F be symmetric with constant row sums. Then $\lambda_F = q_i - \sum_{j\in F\setminus\{i\}} q_{ij}$ for every $i \in F$, α_F^* is uniform on F, and*

$$K_F = \mu(F)\, k^{(k+1)/2} \sqrt{\operatorname{tree}(q_F)}\, (4\pi)^{-(k-1)/2},$$

where $\operatorname{tree}(q_F)$ *is* $\operatorname{tree}(c)$ *for the conductances $c_{ij} = q_{ij}$, $i, j \in F$.*

Proof The positive vector $\mathbf{1}$ is a right and left eigenvector of Q_F, so it belongs to the Perron root. Hence λ_F is minus the row sum, r_F and l_F are multiples of $\mathbf{1}$, and $\alpha_F^* = \mathbf{1}/k$. At $\alpha = \mathbf{1}/k$, Proposition 5.2(b) with $\nu = \mathbf{1}$ gives $r_\alpha = \mathbf{1}$, $l_\alpha = \alpha$ and $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \mu(F)$. Also $c_{ij}(\alpha) = q_{ij}/k$, so $\operatorname{tree}(c(\alpha)) = k^{-(k-1)} \operatorname{tree}(q_F)$, and $\prod_i \alpha_i = k^{-k}$. □

5.2 The local law and the face maxima

A field $v \in \mathbb{R}^F$ enters the continuum density as an exact factor. By Lemma 2.4, $f_T(\alpha) e^{Tv\cdot\alpha}$ is the density f_T of the killed matrix $Q_F + \operatorname{diag} v$, which has the same A_F and μ and the diagonal $q - v$ in place of q. On the lattice the factor is $e^{Tv\cdot n/N} = e^{h\langle v, n\rangle}$.

The discrete constant. Let F be irreducible with $k \geq 2$, fix h_* with $h_* q_{\max} < 1$, and let $(h, \alpha) \in [0, h_*] \times \Delta_F^\circ$. Let $\theta = \theta^{(h)}(\alpha)$ be the vector of Proposition 3.5(b), let r and l be positive right and left null vectors of $A_F - \operatorname{diag}\theta$ with $l \cdot r = 1$, put $c_h(\alpha) = \sum_i \alpha_i/(1 - hq_i + h\theta_i)$, and let

$$C_F^{(h)}(\alpha) = c_h(\alpha)\,(\mu \cdot r)(l \cdot \mathbf{1})\sqrt{\det \nabla^2 J_F^{(h)}(\alpha)/(2\pi)^{k-1}}.$$

Lemma A.6 shows that $\nabla^2 J_F^{(h)}$ is positive definite and that $C_F^{(h)}$ is positive and real-analytic up to $h = 0$. At $h = 0$ we have $c_0 = 1$ and $\theta^{(0)}(\alpha) = \theta(\alpha)$, so $C_F^{(0)} = C_F$ by Proposition 5.2(a). For $v \in \mathbb{R}^F$ put $K_F^{(h)}(v) = C_F^{(h)}(\alpha_F^{*(h)}(v))$ and $K_F^{(h)} = K_F^{(h)}(0)$, and for a single state $K_{\{i\}}^{(h)}(v) = \mu_i/(1 - hq_i)$. At α_F^* the maximizer of Proposition 3.5(b) is $\log r_F$ up to a constant, for every h, by part (c) of that proposition. Hence $\theta^{(h)}(\alpha_F^*) = q - \lambda_F \mathbf{1}$ and $c_h(\alpha_F^*) = 1/(1 - h\lambda_F)$.

Theorem 5.4 (local law) *Let F satisfy* (H) *with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact.*

(a) Uniformly for $\alpha \in \mathcal{K}$, $T \geq 1$ and $v \in \mathbb{R}^F$,

$$f_T(\alpha) e^{Tv\cdot\alpha} = T^{(k-1)/2} C_F(\alpha) e^{-T(J_F(\alpha) - v\cdot\alpha)}\big(1 + O_{\mathcal{K}}(1/T)\big).$$

(b) Fix h_* with $h_* q_{\max} < 1$. Uniformly for $1 \leq T \leq h_* N$, $\operatorname{supp} n = F$ with $\alpha = n/N \in \mathcal{K}$, and $v \in \mathbb{R}^F$,

$$P_N(n) e^{Tv\cdot\alpha} = N^{-(k-1)} T^{(k-1)/2} C_F^{(h)}(\alpha) e^{-T(J_F^{(h)}(\alpha) - v\cdot\alpha)}\big(1 + O_{\mathcal{K}}(1/T)\big).$$

For $F = \{i\}$, $f_T = \mu_i e^{-q_i T}$ and $P_N(Ne_i) = \mu_i (1 - hq_i)^{N-1}$.

The field cancels from the two sides, so the case $v = 0$ is the whole content, and the error does not depend on v. In (b) the ratio of the two sides lies between two positive constants for all T in the range, so the error is relative.

Idea of the proof. We give the proof in Appendix A. The generating function of the words in H is a matrix resolvent, so Cauchy's formula writes H as an integral over a torus with the radii $\theta_2(\alpha), \dots, \theta_k(\alpha)$ (Lemma A.2). All the coefficients are nonnegative, so the integrand is largest at the real point, and a Laplace lemma with a parameter (Lemma B.1) gives the expansion. The Hessian at the real point comes from the Perron root of a perturbed generator (Lemma A.1), which brings in Σ_α. Note that $f_T(\alpha)$ is continuous and positive in (T, α), as H is a convergent power series with some $W(m) > 0$ by (H). For (b) the same torus carries Good's discrete formula, with $e^{t_i z_i}$ replaced by $(1 - hq_i + hz_i)^{n_i - 1}$ (Lemma A.2). In the variable z the integrand is analytic in h up to $h = 0$, where it is the integrand of (a), so one Laplace argument with (h, α) in the compact set $[0, h_*] \times \mathcal{K}$ covers $1 \leq T \leq h_* N$. Only for bounded T does the proof use the comparison of Lemma 5.5, to keep the ratio bounded below.

Lemma 5.5 (discrete and continuum) *Let F satisfy* (H) *with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$, $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O_{\mathcal{K}}(T^2/N))$.*

We prove this in Appendix A. A field multiplies both sides by the same factor $e^{Tv\cdot n/N}$. Theorem 5.4(b) makes this transfer unnecessary for the local law, but the proofs of Theorem 5.4(b) for bounded T and of Lemma A.5 use it. Hence, together with the restriction identities, the local law gives one display for all faces.

Theorem 5.6 (the stratified local law) *Fix Q, a start μ and h_* with $h_* q_{\max} < 1$. Let F satisfy* (H) *with $|F| = k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact.*

(a) Uniformly for $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ and $1 \leq T \leq h_* N$,

$$-\log P_N(n) = (k-1)L + T\, J_V^{(h)}(n/N) - \frac{k-1}{2}\log T - \log C_F^{(h)}(n/N) + O_{\mathcal{K}}\Big(\frac{1}{T}\Big).$$

(b) Uniformly for $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/3}$,

$$-\log P_N(n) = (k-1)L + T\, J_V(n/N) - \frac{k-1}{2}\log T - \log C_F(n/N) + O_{\mathcal{K}}\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

For a singleton, $-\log P_N(Ne_i) = -\log \mu_i - (N-1)\log(1-hq_i)$, which is $Tq_i - \log \mu_i + O(T^2/N)$. With $T = \tau L$ the coefficient of L in (b) is $(k-1) + \tau J_V(n/N)$, the lattice entropy of the face plus τ times the Donsker and Varadhan cost of the point.

Proof (a) This is Theorem 5.4(b) with $v = 0$ and Proposition 3.5(a), which replaces $J_F^{(h)}$ by $J_V^{(h)}$ on Δ_F. (b) By Proposition 3.5(b) and Lemma A.6, $J_F^{(h)}$ and $\log C_F^{(h)}$ are real-analytic in (h, α) up to $h = 0$, so $TJ_F^{(h)} = TJ_F + O(Th) = TJ_F + O(T^2/N)$ and $\log C_F^{(h)} = \log C_F + O(T/N)$ uniformly on $\mathcal{K}$. Put this into (a) and replace J_F by J_V on Δ_F by Proposition 3.1. Part (b) also follows from Theorem 5.4(a) and Lemma 5.5. The singleton case is the path i^N and (1). □

For a field $v \in \mathbb{R}^d$ we write $\lambda_F(v)$, $\alpha_F^*(v)$, $K_F(v)$, $\lambda_F^{(h)}(v)$, $\alpha_F^{*(h)}(v)$ and $K_F^{(h)}(v)$ for the quantities of v_F, and

$$M_F(T, v) = \max\{P_N(n) e^{Tv\cdot n/N} : \operatorname{supp} n = F\},$$

so that $M_F(T, 0) = M_F$.

Theorem 5.7 (face maxima) *Let F satisfy* (H)*, let $\mathcal{V} \subset \mathbb{R}^d$ be compact, and fix h_* with $h_* q_{\max} < 1$.*

(a) If $k \geq 2$, then uniformly for $v \in \mathcal{V}$ and $1 \leq T \leq h_* N$,

$$M_F(T, v) = N^{-(k-1)} T^{(k-1)/2} K_F^{(h)}(v) e^{-T\lambda_F^{(h)}(v)}\big(1 + O_{\mathcal{V}}(1/T)\big),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^{*(h)}(v) + O_{\mathcal{V}}(1/T)$. At $v = 0$ this reads $M_F = N^{-(k-1)} T^{(k-1)/2} K_F^{(h)} (1 - h\lambda_F)^N (1 + O(1/T))$ with $\hat{n}/N = \alpha_F^* + O(1/T)$. For $k = 1$, $M_{\{i\}}(T, v) = e^{Tv_i}\mu_i (1 - hq_i)^{N-1} = K_{\{i\}}^{(h)}(v) e^{-T\lambda_{\{i\}}^{(h)}(v)}$ exactly.

(b) If $k \ge 2$, then uniformly for $v \in \mathcal{V}$ and $1 \le T \le N^{1/3}$,

$$M_F(T,v) = N^{-(k-1)} T^{(k-1)/2} K_F(v) e^{-T\lambda_F(v)} \big(1 + O_{\mathcal{V}}(1/T + T^2/N)\big),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^*(v) + O_{\mathcal{V}}(1/T)$. For $k = 1$, $M_{\{i\}}(T,v) = e^{Tv_i} M_{\{i\}}$ with $M_{\{i\}}$ given by (1). In both cases, uniformly for $v \in \mathcal{V}$ and $1 \le T \le N^{1/3}$,

$$\log M_F(T,v) = -(k-1)L + \tfrac{k-1}{2} \log T + \log K_F(v) - T\lambda_F(v) + O_{\mathcal{V}}(1/T + T^2/N). \quad (9)$$

The constants depend on $\mathcal{V}$ only through the compact set $\{\alpha_F^{*(h)}(v) : 0 \le h \le h_*,\ v \in \mathcal{V}\}$, the bounds of $\lambda_F^{(h)}$ and $K_F^{(h)}$ on $[0,h_*] \times \mathcal{V}$, and finitely many vectors of V_F.

Proof (a) For $k = 1$ the only composition is Ne_i. Let $k \ge 2$. For $(h,v) \in [0,h_*] \times \mathcal{V}$ put $J^{h,v}(\alpha) = J_F^{(h)}(\alpha) - v \cdot \alpha - \lambda_F^{(h)}(v)$, which is nonnegative on Δ_F and vanishes only at $\alpha_F^{*(h)}(v)$, by Proposition 3.6(a) for $h > 0$ and Proposition 3.3(a) for $h = 0$. Put $X = N^{-(k-1)} T^{(k-1)/2} K_F^{(h)}(v) e^{-T\lambda_F^{(h)}(v)}$, the value that (a) claims, and write $x = \alpha' - \alpha_F^{*(h)}(v)'$ for the displacement of α from $\alpha_F^{*(h)}(v)$. Below, constants depend only on Q, μ, h_* and $\mathcal{V}$.

The balls. By Proposition 3.6(c) the set $\mathcal{A} = \{\alpha_F^{*(h)}(v) : 0 \le h \le h_*,\ v \in \mathcal{V}\}$ is a compact subset of Δ_F°, and $\lambda_F^{(h)}(v)$ and $\log K_F^{(h)}(v)$ are bounded. By Proposition 3.5(b) and Lemma A.6, $J_F^{(h)}$ and $\log C_F^{(h)}$ are real-analytic in (h,α) up to $h = 0$, with $\nabla^2 J_F^{(h)}$ positive definite, and $J^{h,v}$ has a critical point at $\alpha_F^{*(h)}(v)$. Taylor's theorem with bounds on the derivatives of order up to three on $[0,h_*]$ times a compact neighborhood of $\mathcal{A}$ gives $\rho_0 > 0$ and positive constants c_U, C_U, C_1 and C_3 with the following properties. The set $\mathcal{K}_{\mathcal{V}}$ of points within ρ_0 of $\mathcal{A}$ lies in Δ_F°, and for $(h,v) \in [0,h_*] \times \mathcal{V}$ the ball $U(h,v) = \{|x| \le \rho_0\}$ satisfies

$$c_U |x|^2 \le J^{h,v} \le C_U |x|^2, \qquad \big|\log(C_F^{(h)}/K_F^{(h)}(v)) - g \cdot x\big| \le C_1 |x|^2$$

on $U(h,v)$, where $g = \nabla \log C_F^{(h)}(\alpha_F^{*(h)}(v))$ has $|g| \le C_1$. Also $\log C_F^{(h)}$ is C_3-Lipschitz on $\mathcal{K}_{\mathcal{V}}$.

Outside $U(h,v)$. For $\beta \in \Delta_F$ the definition of $J_F^{(h)}$ and $((I + hQ_F)u)_i/u_i = 1 - hq_i + h\theta_i$ with $\theta = A_F u/u$ give $J_F^{(h)}(\beta) = \sup_{\theta \in V_F} \sum_i \beta_i L_h(q_i - \theta_i)$, and for fixed $\theta \in V_F$ the sum is continuous in $(h,\beta) \in [0,h_*] \times \Delta_F$, since $1 - h(q_i - \theta_i) \ge 1 - h_* q_{\max}$. The set $\mathcal{E}$ of triples $(h,v,\beta) \in [0,h_*] \times \mathcal{V} \times \Delta_F$ with $\beta \notin U(h,v)^\circ$ is compact, and $J^{h,v}(\beta)$ is positive and lower semicontinuous on it. So $J^{h,v} \ge 3\eta$ on $\mathcal{E}$ for some $\eta > 0$. For each point of $\mathcal{E}$ the supremum gives $\theta \in V_F$ with $\sum_i \beta_i L_h(q_i - \theta_i) - v \cdot \beta > \lambda_F^{(h)}(v) + 2\eta$, and the same θ works near that point. By compactness finitely many vectors $\theta_j \in V_F$ suffice. Now let $\operatorname{supp} n = F$ with $(h,v,n/N) \in \mathcal{E}$, and take j that works there. The bound (A5) with $\theta = \theta_j$, multiplied by $e^{h\langle v,n\rangle}$, gives, since $(1 - hq_i + h\theta_{j,i})^{n_i} = e^{-hn_i L_h(q_i - \theta_{j,i})}$ with $hn_i = T\beta_i$ for $\beta = n/N$,

$$P_N(n) e^{Tv \cdot n/N} \le C(T/N)^{k-1} e^{-T(\lambda_F^{(h)}(v) + 2\eta)} = X \cdot O\big(T^{(k-1)/2} e^{-2\eta T}\big).$$

Inside $U(h,v)$. By Theorem 5.4(b) with $\mathcal{K} = \mathcal{K}_{\mathcal{V}}$, for $\alpha = n/N \in U(h,v)$,

$$P_N(n) e^{Tv \cdot n/N} = X \, \frac{C_F^{(h)}(\alpha)}{K_F^{(h)}(v)} \, e^{-TJ^{h,v}(\alpha)} (1 + E), \qquad |E| \le C_2/T, \quad (10)$$

and $1 + E$ lies between two positive constants. For $T \ge 2C_1/c_U$ the logarithm of the two middle factors is at most

$$g \cdot x + C_1 |x|^2 - c_U T |x|^2 \le g \cdot x - \tfrac{c_U T}{2} |x|^2 \le \frac{|g|^2}{2c_U T},$$

the maximum of a concave quadratic. Hence $P_N(n)e^{Tv\cdot n/N} \le X(1+O(1/T))$ on $U(h,v)$. The bound outside $U(h,v)$ is smaller for large T, and hence $M_F(T,v) \le X(1+O(1/T))$. For the lower bound let n^0 have support F and $|n_i^0 - N\alpha_F^{*(h)}(v)_i| < 1$ for all i, which exists for $N \ge N_0$. Then $|x| \le k/N$, $C_F^{(h)}(n^0/N) \ge K_F^{(h)}(v)e^{-kC_3/N}$ and $TJ^{h,v}(n^0/N) \le C_U k^2 T/N^2 \le C_U k^2 h_*/N$. Since $1/N \le h_*/T$, (10) gives $P_N(n^0)e^{Tv\cdot n^0/N} \ge X(1-O(1/T))$.

Bounded T. Fix T_2 such that the two bounds give $M_F(T,v)/X \in [\frac12, \frac32]$ for $T \ge T_2$. For $1 \le T \le T_2$ the same three steps, with $1+E$ between two positive constants, give $M_F(T,v)/X \in [c,C]$ with $0 < c < C$. This proves the first display, and Proposition 3.6(b) gives the form at $v = 0$.

The maximizer. Let $\hat n$ be a maximizer with displacement $\hat x$. For $T \ge T_3$ the bound outside $U(h,v)$ is below $X/2 \le P_N(n^0)e^{Tv\cdot n^0/N}$, so $\hat n/N \in U(h,v)$, and $|E| \le \frac12$ in (10). Taking logarithms in the comparison of $\hat n$ with n^0, and using $\log(C_F^{(h)}/K_F^{(h)}(v)) \le C_3|\hat x|$ and $TJ^{h,v} \ge c_U T|\hat x|^2$, we get

$$c_U T|\hat x|^2 \le C_3|\hat x| + \frac{C_4}{T}$$

for all $T_3 \le T \le h_*N$. Solving the quadratic inequality gives $|\hat x| \le \bigl(C_3 + \sqrt{C_3^2 + 4c_U C_4}\bigr)/(2c_U T)$. For $1 \le T < T_3$ the claim holds since $|\hat x|$ is bounded.

(b) By Proposition 3.6(c) and Lemma A.6, uniformly for $v \in \mathcal{V}$, $K_F^{(h)}(v) = K_F(v)(1+O(h))$, $\lambda_F^{(h)}(v) = \lambda_F(v) + O(h)$ and $\alpha_F^{*(h)}(v) = \alpha_F^*(v) + O(h)$. For $T \le N^{1/3}$ we have $h \le T^2/N \le 1/T$ and $Th = T^2/N \le 1$. Hence the value X in (a) is $N^{-(k-1)}T^{(k-1)/2}K_F(v)e^{-T\lambda_F(v)}(1+O(T^2/N))$, and every maximizer is within $O(1/T)$ of $\alpha_F^*(v)$. For $k = 1$ the formula is the path i^N, and (9) is the logarithm of the two displays.

The continuum route to (b). Part (b) also follows from the proof of (a) with J_F, C_F, $K_F(v)$, $\lambda_F(v)$, Theorem 5.4(a) together with Lemma 5.5, and Lemma 4.1 in place of $J_F^{(h)}$, $C_F^{(h)}$, $K_F^{(h)}(v)$, $\lambda_F^{(h)}(v)$, Theorem 5.4(b) and (A5). In (10) the error is then $|E| \le C_2(1/T + T^2/N)$, which is at most $\frac12$ for $T_2 \le T \le N^{1/3}$ and large N. For bounded T we use that f_T is continuous and positive. In the step that locates the maximizer the comparison gives $c_U T|\hat x|^2 \le C_3|\hat x| + C_4(1/T + T^2/N)$, and $T^2/N \le 1/T$ gives $|\hat x| = O(1/T)$. Proposition 5.12 makes this route uniform over a class of chains. □

The range of part (b). The proof of Theorem 5.7(b) from part (a) uses only $Th \le 1$ and $h \le 1/T$, so it gives the display and the location of the maximizer for $1 \le T \le N^{1/2}$. The same lines give Theorem 5.6(b) for $T \le N^{1/2}$. We state both for $T \le N^{1/3}$, the range of the continuum route, which needs $T^2/N \le 1/T$ to locate the maximizer and is the route that Proposition 5.12 makes uniform. The error T^2/N is not an artifact of the proof. By the envelope theorem and $L_h(y) = y + hy^2/2 + O(h^2)$ for bounded y, uniformly on $\mathcal{K}$,

$$TJ_F^{(h)}(\alpha) = TJ_F(\alpha) + \frac{T^2}{2N}\sum_{i\in F}\alpha_i\bigl(q_i - \theta_i(\alpha)\bigr)^2 + O\Bigl(\frac{T^3}{N^2}\Bigr),$$

and $-N\log(1-h\lambda_F) = T\lambda_F + T^2\lambda_F^2/(2N) + O(T^3/N^2)$. The sum vanishes only where $\theta(\alpha) = q$. Since θ determines α, as the stationary law of $R^{-1}(A_F - \operatorname{diag}\theta)R$ with R the diagonal of the Perron vector, this happens only when $\lambda_F = 0$ and $\alpha = \alpha_F^*$. Hence the continuum exponent TJ_V fails when T is of order $N^{1/2}$, while the discrete functional

absorbs every power of T/N and has no threshold below h_*N. Theorems 4.2(ii) and 4.5, which cover reducible faces, and the crossing on the complete graph in the arXiv version of this paper [40] are stated for $T \le N^{1/3}$, and the tie windows of Theorem 5.9 for T of order L. We have not extended them.

5.3 Crossings and tie windows

Theorem 5.8 (the crossing equation) *Let F and G satisfy* (H)*, with $\Delta = k_F - k_G \ge 1$ and $\delta = \lambda_G - \lambda_F > 0$, and let $D_N(T) = \log M_F - \log M_G$.*

(i) For large N, D_N has a zero in $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.

(ii) Fix h_* with $h_* q_{\max} < 1$. For large N, every zero $T_N \in [1, h_*N]$ of D_N satisfies

$$\delta T_N + \frac{\Delta}{2}\log T_N = \Delta L - \log\frac{K_F}{K_G} + O\Big(\frac{1}{L}\Big). \tag{11}$$

(iii) All such zeros lie in one interval of length $O(1/L)$, and

$$T_N = \frac{\Delta}{\delta}\Big[L - \frac{1}{2}\log L + b_{FG}\Big] + O\Big(\frac{\log L}{L}\Big), \qquad b_{FG} = -\frac{1}{2}\log\frac{\Delta}{\delta} - \frac{1}{\Delta}\log\frac{K_F}{K_G}.$$

(iv) For large N, D_N has exactly one zero in $[1, h_*N]$.

Proof Where the zeros lie. We show that for large N every zero of D_N in $[1, h_*N]$ lies in $W = [\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$. By Theorem 5.7(a) with $v = 0$ for F, and for G too if $k_G \ge 2$, uniformly for $1 \le T \le h_*N$,

$$D_N(T) = -\Delta L + \Xi_N(T) + \log\frac{K_F^{(h)}}{K_G^{(h)}} + O\Big(\frac{1}{T}\Big), \qquad \Xi_N(T) = N\log\frac{1-h\lambda_F}{1-h\lambda_G} + \frac{\Delta}{2}\log T.$$

If $G = \{j\}$ this holds by the exact formula $M_G = K_G^{(h)}(1-h\lambda_G)^N$ of Theorem 5.7(a). Since $\lambda_F < \lambda_G \le q_{\max} < 1/h$,

$$\Xi_N'(T) = \frac{\lambda_G}{1-h\lambda_G} - \frac{\lambda_F}{1-h\lambda_F} + \frac{\Delta}{2T} = \frac{\delta}{(1-h\lambda_F)(1-h\lambda_G)} + \frac{\Delta}{2T} \ge \delta,$$

and the first term of Ξ_N vanishes at $T = 0$, so $\Xi_N(T) \ge \delta T$ for $T \ge 1$. The constants $K_F^{(h)}$ and $K_G^{(h)}$ are continuous and positive in $h \in [0, h_*]$, so the logarithm of their ratio is bounded. Hence a zero T_N has $\Xi_N(T_N) = \Delta L + O(1)$, hence $\delta T_N \le \Delta L + O(1)$ and $T_N < \frac{2\Delta}{\delta}L$. For $T \le \frac{2\Delta}{\delta}L$ we have $Th = O(L^2/N)$, so $\Xi_N(T) = \delta T + \frac{\Delta}{2}\log T + O(L^2/N)$, and $\Xi_N(\frac{\Delta}{2\delta}L) = \frac{\Delta}{2}L + O(\log L)$. As Ξ_N is increasing, $T_N > \frac{\Delta}{2\delta}L$ for large N. Also $W \subset [1, N^{1/3}]$ for large N.

On W. We subtract (9) for G from (9) for F. Uniformly on $[1, N^{1/3}]$,

$$D_N(T) = -\Delta L + \Xi(T) + \log\frac{K_F}{K_G} + O\Big(\frac{1}{T} + \frac{T^2}{N}\Big), \qquad \Xi(T) = \delta T + \frac{\Delta}{2}\log T.$$

The function Ξ is increasing, with $\Xi' \ge \delta$.

(i) At $T = \frac{\Delta}{2\delta}L$ the display gives $D_N = -\frac{\Delta}{2}L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta}L$ it gives $D_N = \Delta L + O(\log L) > 0$. For fixed N, M_F is a maximum of finitely many polynomials in h, and it is positive because F is admissible. The same holds for G. Hence D_N is continuous and has a zero in between.

(ii) Let $T_N \in [1, h_*N]$ be a zero. By the first step $T_N \in W$, so T_N is of order L, the error term of the display is $O(1/T_N + T_N^2/N) = O(1/L)$, and $D_N(T_N) = 0$ is (11).

(iii) Let T° be the solution of $\Xi(T^\circ) = \Delta L - \log(K_F/K_G)$. By (ii) every zero has $\Xi(T_N) = \Xi(T^\circ) + O(1/L)$, and $\Xi' \geq \delta$, so every zero is within $O(1/L)$ of T°. To solve for T°, we use $T^\circ = \frac{\Delta}{\delta} L + O(\log L)$, so $\log T^\circ = \log L + \log \frac{\Delta}{\delta} + O(\log L/L)$. Putting this into $\delta T^\circ = \Delta L - \frac{\Delta}{2} \log T^\circ - \log(K_F/K_G)$ and dividing by δ gives the explicit form.

(iv) By the first step, for large N every zero in $[1, h_* N]$ lies in W. We show that D_N is strictly increasing on W. For fixed N, M_F is the maximum of the finitely many polynomials $T \mapsto P_N(n)$ with $\operatorname{supp} n = F$ that are not identically zero. By Lemma 2.3 each term of such a polynomial is either zero for all h or positive for all $0 < h < 1/q_{\max}$, so these polynomials are positive on W. Hence $\log M_F$ has at each $T \in W$ a right derivative, the largest $\partial_T \log P_N(n)$ over the maximizers n at T, and a left derivative, the least of them. By Theorem 5.7 every maximizer has $n/N = \alpha_F^* + O(1/L)$ on W. Hence for large N it lies in a closed ball $\mathcal{K} \subset \Delta_F^\circ$ around α_F^*, and $J_F(n/N) = \lambda_F + O(1/L)$, since J_F is smooth on $\mathcal{K}$ with $J_F(\alpha_F^*) = \lambda_F$. Lemma A.5 then gives both one-sided derivatives of $\log M_F$ as $-\lambda_F + O(1/L)$ on W. The same holds for G if $k_G \geq 2$. If $G = \{j\}$, then $\log M_G = \log \mu_j + (N-1)\log(1 - hq_j)$ has the derivative $-(N-1)q_j/(N(1-hq_j)) = -\lambda_G + O(L/N)$. So both one-sided derivatives of D_N are $\delta + O(1/L) \geq \delta/2$ on W for large N. A continuous function whose right derivative is positive at every point of an interval is strictly increasing there. Hence D_N has at most one zero in W, and by (i) it has one. □

On the 4-cycle at $N = 10^{14}$ the explicit form in (iii) differs from the root of (11) without its error term by 0.035 for a vertex against a pair and by 0.087 for a pair against the cycle, the two crossings that carry the mode. Hence computations should solve (11).

At a switch value $\tau^* = \tau_i$ of Theorem 4.4(a) several faces minimize Φ_{τ^*}, sometimes more than two, as all faces do at $\tau^* = 1$ for the complete graph $Q = J - dI$, where $\lambda_F = d - |F|$. Then the competition plays out in a window $T = \tau^*(L - \frac{1}{2}\log L + t)$ with t of order 1. In this window each tied face has a line $\ell_F(t)$, and the mode follows the highest line.

Theorem 5.9 (tie windows) *Fix $\tau^* > 0$. Let m^* be the least value of Φ_{τ^*} over the admissible faces and $\mathcal{F}^*$ the set of minimizers. Assume that every face in $\mathcal{F}^*$ is irreducible. By Theorem 4.4(c) this holds if Q has symmetric support or μ has full support. Put $T = \tau^*(L - \frac{1}{2}\log L + t)$, and for $F \in \mathcal{F}^*$ let $\ell_F(t) = \log K_F + \frac{k_F - 1}{2} \log \tau^* - \tau^* \lambda_F t$ and $E(t) = \max_{G \in \mathcal{F}^*} \ell_G(t)$. The function E is convex and piecewise linear. We call the finitely many points where its slope changes the* kinks *of E, and we write t_j for them.*

(i) Uniformly for t in compact sets, for $F \in \mathcal{F}^*$, $\log M_F = -m^* L + \frac{m^*}{2} \log L + \ell_F(t) + O(\log L/L)$.

(ii) There are $\varepsilon_1, \varepsilon_0 > 0$ such that for large N and $|T/L - \tau^*| \leq \varepsilon_1$, every admissible $F \notin \mathcal{F}^*$ has $M_F \leq N^{-\varepsilon_0} \max_{G \in \mathcal{F}^*} M_G$.

(iii) There are $\varepsilon, t_0 > 0$ such that the following holds for large N. If $t_0 \leq t \leq \varepsilon L$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $[t_0, \infty)$. If $-\varepsilon L \leq t \leq -t_0$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $(-\infty, -t_0]$. A face of the first kind has the least λ_F in $\mathcal{F}^*$, and a face of the second kind has the largest.

The numbers ε_0, ε_1, ε and t_0 depend only on Q, μ and τ^*. Consequently the following holds for every $\eta > 0$ and large N. If $|t| \leq \varepsilon L$ and t is at distance at least η from every kink of E, then every global mode lies on a face whose line equals E near t. So for $|t| \leq \varepsilon L$ the line of the face that carries the global mode changes only at $T = \tau^*(L - \frac{1}{2}\log L + t_j) + o(1)$, where t_j

runs over the kinks of E, and it changes near each kink. Coincident lines do not distinguish their faces at this order.

Proof Every $F \in \mathcal{F}^*$ is admissible and irreducible, so it satisfies (H) by Lemma 2.2(b). We first put T into (9). Write $T = \tau^* L(1+y)$ with $y = (t - \frac{1}{2}\log L)/L$, so that $|y| \le \frac{1}{2}$ for $|t| \le L/4$ and large N. Then $\log T = \log\tau^* + \log L + \log(1+y)$ and $T\lambda_F = \tau^*\lambda_F(L - \frac{1}{2}\log L + t)$, and the error term of (9) is $O(1/L)$. Since $\tau^*\lambda_F + k_F - 1 = m^*$, we get, uniformly for $|t| \le L/4$,

$$\log M_F = -m^* L + \tfrac{m^*}{2}\log L + \ell_F(t) + \tfrac{k_F - 1}{2}\log(1+y) + O(1/L) \qquad (F \in \mathcal{F}^*). \tag{12}$$

(i) For t in a compact set, $y = O(\log L/L)$, so $\log(1+y) = O(\log L/L)$ in (12).

(ii) Let $g > 0$ be the least value of $\Phi_{\tau^*}(F) - m^*$ over the admissible $F \notin \mathcal{F}^*$. If there is no such face, (ii) says nothing, and we put $g = 1$. Put $\varepsilon_1 = \min\bigl(\tau^*/2,\, g/(4q_{\max} + 4)\bigr)$. If $|T/L - \tau^*| \le \varepsilon_1$, then $\frac{1}{2}\tau^* L \le T \le \frac{3}{2}\tau^* L$, so for large N we are in the range of Theorem 4.2(ii) and $\log T = O(\log L)$. Take an admissible $F \notin \mathcal{F}^*$ and any $G \in \mathcal{F}^*$. Since $0 \le \lambda_F, \lambda_G \le q_{\max}$ (Section 2.1), $\Phi_\tau(F) - \Phi_\tau(G)$ has slope at most $q_{\max}$ in absolute value, and it is at least g at τ^*. Hence it is at least $g/2$ for $|\tau - \tau^*| \le \varepsilon_1$. With $\tau = T/L$ the upper bound of Theorem 4.2(ii) for F and the lower bound for G give $\log M_F - \log M_G \le -(\Phi_\tau(F) - \Phi_\tau(G))L + O(\log L) \le -\frac{g}{4}L$ for large N. Hence (ii) holds with $\varepsilon_0 = g/4$.

(iii) Take $\varepsilon = \min(\frac{1}{4}, \varepsilon_1/(2\tau^*))$. Then for $|t| \le \varepsilon L$ and large N we have $|T/L - \tau^*| = \tau^*|y| \le \varepsilon_1$, so (ii) applies and no global mode lies on a face outside $\mathcal{F}^*$. It remains to compare the tied faces. Let $t_0 \ge 1$ be larger than every $|t_j|$, so that E is a single line on $[t_0, \infty)$. Let $F \in \mathcal{F}^*$ have this line, and let $G \in \mathcal{F}^*$ have another line. We show that $M_F > M_G$ for $t_0 \le t \le \varepsilon L$ if t_0 and N are large. Since $\ell_F \ge \ell_G$ on $[t_0, \infty)$, the slopes satisfy $\lambda_F \le \lambda_G$. As both faces are tied,

$$\Delta = k_F - k_G = \tau^*(\lambda_G - \lambda_F) \ge 0.$$

There are two cases.

First let $\Delta = 0$. Then the two lines are parallel, and $\ell_F - \ell_G = \log(K_F/K_G) > 0$. The terms with $\log(1+y)$ in (12) are the same for F and G, so $\log(M_F/M_G) = \log(K_F/K_G) + O(1/L) > 0$ for large N.

Now let $\Delta > 0$. Then $\ell_F(t) - \ell_G(t) = \Delta t + \ell_F(0) - \ell_G(0)$, and (12) gives

$$\log\frac{M_F}{M_G} = \Delta t + \ell_F(0) - \ell_G(0) + \frac{\Delta}{2}\log(1+y) + O\Bigl(\frac{1}{L}\Bigr).$$

For large N and $t \ge 1$ we have $|\log(1+y)| \le 2|y| \le (2t + \log L)/L \le t/2$. Hence the right side is at least $\frac{3}{4}\Delta t - C$, which is positive for $t \ge t_0$ once t_0 is large.

Hence for $t_0 \le t \le \varepsilon L$ every global mode lies on a face whose line is E on $[t_0, \infty)$. The slope of ℓ_F is $-\tau^*\lambda_F$, and E follows the line of largest slope on the far right, so such a face has the least λ_F in $\mathcal{F}^*$. The case $t \le -t_0$ is the same with the signs reversed.

The consequence. Fix $\eta > 0$ and let $S_\eta = [-t_0, t_0] \setminus \bigcup_j (t_j - \eta, t_j + \eta)$. For $G \in \mathcal{F}^*$ the function $E - \ell_G$ is convex, piecewise linear and nonnegative, with kinks only at kinks of E. So on each interval of S_η either $\ell_G = E$ or $E - \ell_G \ge g_\eta$ for some $g_\eta > 0$. Let $t \in S_\eta$ and $\ell_F(t) = E(t)$. By (i), every $G \in \mathcal{F}^*$ whose line is not E near t has $\log(M_F/M_G) \ge g_\eta - O(\log L/L) > 0$, and by (ii) the faces outside $\mathcal{F}^*$ are smaller still. With (iii) for $t_0 \le |t| \le \varepsilon L$ this gives the claim, and the line does change near each kink, since E is a different line on its two sides. □

Corollary 5.10 (where the mode changes) *Let Q have symmetric support or μ full support. Let $v_0, \dots, v_m$ and $\tau_1 < \cdots < \tau_m$ be the vertices and switch values of Theorem 4.4(a), and assume that no point of $\mathcal{S}$ other than v_{i-1} and v_i lies on the edge $[v_{i-1}, v_i]$, for $1 \le i \le m$.*

Put $\Delta_i = y^{(i)} - y^{(i-1)}$ and $\delta_i = \lambda^{(i-1)} - \lambda^{(i)}$, let $K^{(i)}$ be the largest K_F over the faces F with $P_F = v_i$, and let $T_{N,i}$ be the unique positive root of $\delta_i T + \frac{\Delta_i}{2}\log T = \Delta_i L - \log(K^{(i)}/K^{(i-1)})$. Fix $0 < a < b$. There are C and N_0 such that for $N \ge N_0$ and every $T \in [aL, bL]$ with $|T - T_{N,j}| \ge C/L$ for all j, every global mode $\hat{n}$ has $P_{\operatorname{supp}\hat{n}} = v_i$, where i is the number of j with $T_{N,j} < T$.

Proof By Theorem 4.4(a) each face whose point lies on the lower-left boundary belongs to some $\mathcal{F}_\tau$, so it satisfies (H) by Theorem 4.4(c) and Lemma 2.2(b). Let $\mathbb{F}_i$ be the set of faces F with $P_F = v_i$, and $M^{(i)} = \max_{F\in\mathbb{F}_i} M_F$. These faces share $|F| - 1 = y^{(i)}$ and $\lambda_F = \lambda^{(i)}$. So for T of order L, (9) gives $\log(M^{(i)}/M^{(i-1)}) = \Xi_i(T) - \Xi_i(T_{N,i}) + O(1/L)$ with $\Xi_i(T) = \delta_i T + \frac{\Delta_i}{2}\log T$, and $\Xi_i' \ge \delta_i$. Also $T_{N,i} = \tau_i L + O(\log L)$. Put $\tau_0 = 0$ and $\tau_{m+1} = \infty$.

Near a switch value. By Theorem 4.4(a) and the assumption, $\mathcal{F}_{\tau_i} = \mathbb{F}_{i-1} \cup \mathbb{F}_i$. Theorem 5.9(ii) with $\tau^* = \tau_i$ gives $\varepsilon_1 > 0$ such that for large N and $|T/L - \tau_i| \le \varepsilon_1$ no face outside $\mathbb{F}_{i-1} \cup \mathbb{F}_i$ carries a global mode. We shrink ε_1 below a third of every gap $\tau_{j+1} - \tau_j$. Then this window contains $T_{N,i}$ and no other $T_{N,j}$, and by the display there is C such that the global modes lie in $\mathbb{F}_i$ for $T \ge T_{N,i} + C/L$ and in $\mathbb{F}_{i-1}$ for $T \le T_{N,i} - C/L$.

Between switch values. Let $J \subset (\tau_i, \tau_{i+1})$ be compact. For $\tau \in J$ only v_i attains the least value of Φ_τ on $\mathcal{S}$. So $\Phi_\tau(F) - \Phi_\tau(G) \ge g_J > 0$ for $\tau \in J$, $G \in \mathbb{F}_i$ and every admissible F with $P_F \ne v_i$. The bounds of Theorem 4.2(ii), with $k^* = k_G$ for the irreducible face G, give $\log M_F - \log M_G \le -g_J L + (d-1)\log T + 2\kappa < 0$ for $T/L \in J$ and large N. Hence every global mode lies in $\mathbb{F}_i$, and no $T_{N,j}$ lies in LJ. Finitely many windows and intervals J cover $[a, b]$. □

The equation for $T_{N,i}$ is (11) for a face of largest constant on each side, so Theorem 5.8(iii) gives $T_{N,i}$ explicitly. If three or more points of $\mathcal{S}$ lie on one edge, Theorem 5.9 locates the changes only up to $o(1)$. At a kink of E where the line of G gives way to that of F, $\Delta = \tau^*\delta$ and $t_j = b_{FG}$, so the kink is where Theorem 5.8(iii) puts the crossing of F and G. The coefficient $\frac{1}{2}$ of $\log\log N$ comes from the power $T^{(k-1)/2}$ in Theorem 5.7, so it needs only (H).

5.4 A reducible face

The best composition of a reducible face can visit a state only once, with no Gaussian window in that direction. Hence the coefficient $\frac{1}{2}$ of Theorem 5.8 can fail, and in the next example it is 1 and then 0, as Corollary 4.6 predicts.

Example 5.11 (a directed 3-cycle) Let $d = 3$, $q_{12} = b > 2$, $q_{23} = q_{31} = 1$, all other rates 0, and $\mu_1 = 1$. From state 1 the chain moves only along $1 \to 2 \to 3 \to 1$, so the admissible faces are $\{1\}$, $\{1,2\}$ (reducible, $\lambda = 1$) and $\{1,2,3\}$ (irreducible, $\lambda = 0$).

(i) Exactly $M_{\{1\}} = (1-bh)^{N-1}$ and $M_{\{1,2\}} = bh(1-h)^{N-2}$, and the latter is attained only at $n = (1, N-1, 0)$. For the full face, $\alpha^* = (1, b, b)/(1+2b)$ and $K_{\{1,2,3\}} = (1+2b)^2/(2\sqrt{3}\,\pi b)$.

(ii) The equation $M_{\{1\}} = M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh < 1$, and $T_N^{(1)} = \frac{1}{b-1}[L - \log L + \log\frac{b-1}{b}] + O(\log L/L)$. The coefficient of $\log\log N$ in the bracket is 1.

(iii) Every root $T \in [L/2, 2L]$ of $M_{\{1,2\}} = M_{\{1,2,3\}}$ satisfies $T = L + \log \frac{2\sqrt{3}\,\pi b^2}{(1+2b)^2} + O(1/L)$, so the coefficient of $\log\log N$ is 0.

Proof (i) A path with support $\{1,2\}$ is $1^{n_1}2^{n_2}$, of probability $(1-bh)^{n_1-1}\,bh\,(1-h)^{n_2-1}$. Moving one visit from state 2 to state 1 multiplies it by $(1-bh)/(1-h) < 1$, so the maximum is at $n_1 = 1$. On the full face Q is a generator, so $\lambda = 0$, $r = \mathbf{1}$, $l = \alpha^*$ and α^* is the stationary law $(1, b, b)/(1+2b)$. So $(\mu\cdot r)(l\cdot\mathbf{1}) = 1$ and $K_{\{1,2,3\}} = (2\pi)^{-1}(\det\Sigma_{\alpha^*})^{-1/2}$. By the inequality of the arithmetic and geometric means, the infimum of $b\alpha_1u_2/u_1 + \alpha_2u_3/u_2 + \alpha_3u_1/u_3$ over $u > 0$ is $3\phi(\alpha)$ with $\phi(\alpha) = (b\alpha_1\alpha_2\alpha_3)^{1/3}$. So $J_V(\alpha) = b\alpha_1 + \alpha_2 + \alpha_3 - 3\phi(\alpha)$ by (3), and by Proposition 5.2(a), $\det\Sigma_{\alpha^*}$ is the inverse of the determinant of the Hessian of J_V in (α_2, α_3) at α^*. In all three coordinates the Hessian of -3ϕ is $\phi D^{-1}(I - \frac{1}{3}\mathbf{1}\mathbf{1}^\top)D^{-1}$ with $D = \operatorname{diag}\alpha$. At $\alpha^* = (c, bc, bc)$ with $c = 1/(1+2b)$ we have $\phi = bc$, and the substitution $\alpha_1 = 1 - \alpha_2 - \alpha_3$ gives the Hessian

$$\frac{1+2b}{3b}\begin{pmatrix} 2b^2+2b+2 & 2b^2+2b-1 \\ 2b^2+2b-1 & 2b^2+2b+2 \end{pmatrix},$$

whose determinant is $(1+2b)^4/(3b^2)$. Hence $\det\Sigma_{\alpha^*} = 3b^2/(1+2b)^4$, which gives the value of $K_{\{1,2,3\}}$.

(ii) The function $\log(M_{\{1,2\}}/M_{\{1\}}) = \log(bh) + (N-2)\log(1-h) - (N-1)\log(1-bh)$ has the derivative $1/h - (N-2)/(1-h) + (N-1)b/(1-bh) > 0$ in h, since $b > 1$ and $1 - bh < 1 - h$. It tends to $-\infty$ as $h \to 0$ and to $+\infty$ as $bh \to 1$, so it has exactly one zero. Its expansion $\log b + \log T - L + (b-1)T + O(T^2/N)$ gives $(b-1)T + \log T = L - \log b + O(T^2/N)$ at the zero, and with $\log T = \log L - \log(b-1) + O(\log L/L)$ this gives the formula.

(iii) For the full face $k = 3$, so Theorem 5.7 gives $\log M_{\{1,2,3\}} = -2L + \log T + \log K_{\{1,2,3\}} + O(1/L)$ on $[L/2, 2L]$, with the same power T^1 as $\log M_{\{1,2\}} = \log b + \log T - L - T + O(T^2/N)$. Hence $\log T$ cancels, and equating the two gives $T = L + \log b - \log K_{\{1,2,3\}} + O(1/L)$. □

5.5 Face maxima uniformly over a class

The constants of Theorem 5.7 depend on Q and μ. We make them uniform over a compact class of reversible chains on a fixed graph. The companion paper uses this statement to make its finite-horizon recovery uniform over such a class.

Fix a connected graph $\mathcal{G} = ([d], \mathcal{E})$, numbers $0 < w_* \le R$ and $0 < \mu_* \le 1/d$, and a compact set $\mathcal{V} \subset \mathbb{R}^d$ of fields. A generator Q is *on* $\mathcal{G}$ if $q_{ij} > 0$ exactly when $\{i,j\} \in \mathcal{E}$. For a reversible irreducible Q with stationary law π put $w_{ij} = q_{ij}q_{ji}$ and $S_V = D_{\sqrt{\pi}}(-Q)D_{\sqrt{\pi}}^{-1}$, with $D_x = \operatorname{diag} x$, the matrix S_F of Proposition 3.4 for $F = V$. It is symmetric, with diagonal q_i and the entries $-\sqrt{w_{ij}}$ on edges. Let $\mathcal{C}$ be the set of reversible irreducible generators on $\mathcal{G}$ with $\|S_V\|_2 \le R$ and $\sqrt{w_{ij}} \ge w_*$ on every edge, and let $\mathcal{M}$ be the set of starts with $\min_i \mu_i \ge \mu_*$. We use $M_F(T, v)$, $\alpha_F^*(v)$ and $K_F(v)$ as defined before Theorem 5.7.

Proposition 5.12 (uniform face maxima on the class) *Let $\mathcal{G}$ induce a connected graph on the face F, with $k = |F| \ge 2$. There are N_0 and C, depending only on $\mathcal{G}$, R, w_*, μ_* and $\mathcal{V}$, such that for all $Q \in \mathcal{C}$, $\mu \in \mathcal{M}$ and $v \in \mathcal{V}$ the following hold.*

(a) For all $N \ge 2$ and $T > 0$ with $TR \le N/2$, $M_F(T, v) \le C(T/N)^{k-1}e^{-T\lambda_F(v) + CT^2/N}$.

(b) For $N \ge N_0$ and $1 \le T \le N^{1/3}$, we have $C^{-1} \le K_F(v) \le C$ and

$$M_F(T,v) = K_F(v)\, N^{-(k-1)} T^{(k-1)/2} e^{-T\lambda_F(v)}(1+E), \qquad |E| \le C\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

For a singleton, $M_{\{i\}}(T,v) = \mu_i(1 - Tq_i/N)^{N-1} e^{Tv_i}$ exactly.

Proof For fixed (Q,μ), part (b) is Theorem 5.7(b) with the compact set $\mathcal{V}$ of fields, whose constants already do not depend on $v \in \mathcal{V}$. We use its continuum route, given at the end of the proof of Theorem 5.7. Part (a) is (6), that is, Lemma 4.1 with the Perron tilt $\theta = q - v - \lambda_F(v)\mathbf{1} \in V_F$, and the singleton is the path i^N. It remains to show that no constant depends on (Q,μ) either.

The parameter set is compact. Let $\mathcal{P} = \mathcal{C} \times \mathcal{M} \times \mathcal{V}$. Since S_V is symmetric and similar to $-Q$, it is positive semidefinite, so $0 \le q_l \le R$ for every l and $\sqrt{w_{ij}} \le R$ on every edge. On an edge $q_{ij} \ge w_*^2/R$, since $q_{ij}q_{ji} \ge w_*^2$ and $q_{ji} \le R$, and off the edges $q_{ij} = 0$. Reversibility is the closed condition of Kolmogorov on cycle products, and $\|S_V\|_2 \le R$ is closed since π is continuous. Hence $\mathcal{P}$ is compact, and every connected face is irreducible for every $Q \in \mathcal{C}$.

Joint regularity. The Perron root is simple, so $\lambda_F(v)$, the normalized Perron vectors and $\alpha_F^*(v)$ are real-analytic in (Q,v). The proof of Proposition 5.2(a), run with Q as an extra variable, makes $\theta(\alpha)$, r_α, l_α, Σ_α, J_F and C_F real-analytic in (α, Q), and C_F is linear in μ. Hence $K_F(v)$ is continuous and positive on $\mathcal{P}$, which gives its bounds, and there is $\varepsilon_0 > 0$ with $\alpha_{F,i}^*(v) \ge 2k\varepsilon_0$ on $\mathcal{P}$. Put $\mathcal{K}_0 = \{\alpha \in \Delta_F : \alpha_i \ge \varepsilon_0 \ \forall i\}$.

The constants. They are the following, and each is continuous on $\mathcal{P}$ or on $\mathcal{P}\times\mathcal{K}_0$, or comes from a compactness argument run there. (i) The constant C_θ of Lemma 4.1 is built from the factor $b(z')$ of Lemma A.2. This factor is continuous where the radius condition (A3) holds strictly, as it does on V_F. So it is continuous along the Perron tilt. (ii) In Theorem 5.4(a) the Perron root in (A3) at $\theta'(\alpha)$ is continuous and negative on $\mathcal{P}\times\mathcal{K}_0$, so one radius margin serves, and Lemma B.1 applies on the compact space $\mathcal{P} \times \mathcal{K}_0$, since the matrix $\mathcal{J}$ of Appendix A is continuous and positive definite and $\chi(\alpha,0) = (\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1})\prod_{i\ge2}\theta_i(\alpha)/\alpha_1$ is continuous and positive. (iii) Step 3 of the proof of Lemma 5.5 needs only $\Upsilon(\overline{1}) \le CT$, with C uniform on $\mathcal{P} \times \mathcal{K}_0$, and we prove this directly. The upper bound $H_e(T\alpha) \le e^{k-1} b(\theta') \prod_{i\ge2}\theta_i\, e^{eT\langle\alpha,\theta\rangle}$ of that step, with $\theta = \theta(\alpha)$, has a constant that is continuous on $\mathcal{P}\times\mathcal{K}_0$ by (i). For the lower bound take a walk c^0 in $\mathcal{G}$ through every state of F with at most $2k-1$ letters, for instance along a depth-first search of a spanning tree of the graph that $\mathcal{G}$ induces on F, and let m^0 count its letters. Every start has $\mu_{c_1^0} \ge \mu_*$ and every edge has $q_{ij} \ge w_*^2/R$, so for $T \ge 1$ and $\alpha \in \mathcal{K}_0$ the term of m^0 in $H(T\alpha)$ is at least

$$\mu_* \min\Big(1, \frac{w_*^2}{R}\Big)^{2k-2} \prod_{i\in F} \frac{(T\alpha_i)^{m_i^0-1}}{(m_i^0-1)!} \ge \mu_* \min\Big(1, \frac{w_*^2}{R}\Big)^{2k-2} \frac{\min(1,\varepsilon_0)^{k-1}}{(k-1)!} = c_0 > 0,$$

since $\sum_i (m_i^0 - 1) \le k-1$ and $T\alpha_i \ge \varepsilon_0$. Hence $\Upsilon(1) = \log H_e(T\alpha) - \log H(T\alpha) \le C + eT\max\theta_i - \log c_0 \le C'T$, uniformly on $\mathcal{P}\times\mathcal{K}_0$. Step 4 uses only $q_{\max} \le R$ and $n_{\min} \ge \varepsilon_0 N$. (iv) In the continuum route to Theorem 5.7(b) the bounds for Σ_α^{-1}, for the third derivatives of J_F and for two derivatives of $\log C_F$ are continuous on $\mathcal{P} \times \mathcal{K}_0$, which gives one radius $\rho_0 \le \varepsilon_0$ and uniform c_U, C_U, C_1 and C_3. Away from the ball, $J_F(\beta) - v\cdot\beta - \lambda_F(v)$ is lower semicontinuous and positive on a compact subset of $\mathcal{P} \times \Delta_F$, hence at least 3η, and finitely many vectors $\theta_u = A_F u/u \in V_F$ serve, with constants bounded by (i). For bounded T, f_T is continuous and positive on $[1,T_2] \times \mathcal{P} \times \mathcal{K}_0$. □

6 The boundary layer at a facet

The local law of Theorem 5.4(a) holds on compact subsets of the open face. Hence it says nothing near the boundary. In this section we follow the occupation density of the continuous-time chain from the inside of a face F up to a facet $G = F \setminus \{i\}$, where the time spent in i is small. We state the main results for the density in Section 6.2, after the setting, and prove them in Sections 6.3 to 6.8. Section 6.9 gives the same law for the counts, with a lattice kernel.

6.1 Setting

Let F be an irreducible face with $k = |F| \geq 2$, let $i \in F$, and put $G = F \setminus \{i\}$. We assume that G satisfies (H), so G is irreducible and $\mu(G) > 0$. The start mass μ_i may be zero. For $j \in G$ put

$$c_j^- = q_{ji}, \qquad c_j^+ = q_{ij},$$

the rates into i and out of i. Both vectors are nonnegative and nonzero, since F is irreducible. For $z \in \mathbb{C}$ let

$$Q_G(z) = Q_G + z\, c^-(c^+)^\top, \qquad \mu^{(z)} = \mu_G + z\mu_i c^+, \qquad e^{(z)} = \mathbf{1} + zc^-,$$

and write $Q_G(z) = A(z) - \operatorname{diag} d(z)$ with $A(z)$ the off-diagonal part and $d(z) = q_G - z\, c^- \circ c^+$. A jump $j \to j'$ of $Q_G(z)$ of rate $zc_j^- c_{j'}^+$ stands for a visit to i that is entered from j and left to j'.

Occupation coordinates. Let E_F be the event of Lemma 2.4. On E_F let $s \in (0,\infty)^G$ be the times spent in the states of G, $S = \sum_{j\in G} s_j$, and $x = T - S$ the time spent in i. We write

$$f_F(s,x) = e^{-\langle q_G, s\rangle - q_i x} H_F(s,x), \qquad f_G(s) = e^{-\langle q_G, s\rangle} H_G(s),$$

where H_F and H_G are the sums H of Lemma 2.4 for the faces F and G. By that lemma f_F is the density of the occupation times on E_F in any $k-1$ of the coordinates, and $f_G(s)$ is the same density for G at total time S. In this notation Theorem 5.4(a) for the face G reads

$$f_G(S\beta) = S^{-(k-2)/2} C_G(\beta) e^{-S J_G(\beta)} \big(1 + O_{\mathcal{K}}(1/S)\big) \tag{13}$$

uniformly for β in a compact $\mathcal{K} \subset \Delta_G^\circ$ and $S \geq 1$, with $C_G(\beta) = \mu_j$ and $J_G = q_j$ when $G = \{j\}$. The profile on the facet is $\beta = s/S \in \Delta_G^\circ$.

The weighted chain. For real $z \geq 0$ the matrix $Q_G(z)$ is Metzler and irreducible. We let $J_G(\beta, z)$ be its rate, the supremum in Proposition 3.3 with $Q_G(z)$ in place of Q_F. Its local-law constant $C_G(\beta, z)$ is Definition 5.1 for the off-diagonal part $A(z)$, with the start $\mu^{(z)}$ in place of μ and the end vector $e^{(z)}$ in place of $\mathbf{1}$. Namely

$$C_G(\beta, z) = \frac{(\mu^{(z)} \cdot r_\beta(z))(l_\beta(z) \cdot e^{(z)})}{(2\pi)^{(k-2)/2} \sqrt{\det \Sigma_\beta(z)}}.$$

So $J_G(\beta,0)=J_G(\beta)$ and $C_G(\beta,0)=C_G(\beta)$.

The facet constants. Let $r_\beta=r_\beta(0)$ and $l_\beta=\beta/r_\beta$ be the vectors of Definition 5.1 for the face G, so $l_\beta\cdot r_\beta=1$. Put

$$a_\beta=c^-\cdot l_\beta,\qquad b_\beta=c^+\cdot r_\beta,\qquad \omega_\beta=a_\beta b_\beta,\qquad \pi^{\mathrm{in}}_{\beta,j}=\frac{c^-_j l_{\beta,j}}{a_\beta},\qquad \pi^{\mathrm{out}}_{\beta,j}=\frac{c^+_j r_{\beta,j}}{b_\beta}.$$

Note that only ω_β is intrinsic, since a_β and b_β move in opposite ways when r_β is scaled. The law π^{in}_β is the law of the state from which the tilted chain enters i, and π^{out}_β the law of the state into which it leaves i. Let $\mathcal{L}_\beta$ be the Laplacian on G with the conductances $\kappa_{jj'}=l_{\beta,j}q_{jj'}r_{\beta,j'}+l_{\beta,j'}q_{j'j}r_{\beta,j}$ for $j\neq j'$, which are twice those of the additive reversibilization $\frac12(\widehat{Q}_\beta+\widehat{Q}^*_\beta)$ of the tilted generator $\widehat{Q}_\beta$, where $\widehat{Q}^*_\beta$ is its time reversal with respect to β. Its kernel is $\mathbb{R}\mathbf{1}$, since G is irreducible. Let $\mathcal{L}^+_\beta$ be its pseudo-inverse and

$$\gamma_\beta=q_i+\frac{\omega_\beta}{2}\,(\pi^{\mathrm{out}}_\beta-\pi^{\mathrm{in}}_\beta)^\top\mathcal{L}^+_\beta(\pi^{\mathrm{out}}_\beta-\pi^{\mathrm{in}}_\beta).\tag{14}$$

The quadratic form is the effective resistance of the network κ between the laws π^{in}_β and π^{out}_β. Hence $\gamma_\beta\geq q_i$, with equality exactly when the two laws agree. Finally put

$$\zeta=2\sqrt{\omega_\beta S x},\qquad z_*=\sqrt{\frac{x}{\omega_\beta S}},\qquad \mathbf{B}_\beta(x,S)=e^{-\gamma_\beta x}\sqrt{\frac{\omega_\beta S}{x}}\,I_1(\zeta),$$

with the modified Bessel function I_1. We call $\mathbf{B}_\beta$ the Bessel kernel.

We extend the rate on F to occupation times by homogeneity of degree one, and write $\bar J_F(s,x)=TJ_F((s,x)/T)$ with $T=S+x$. So $\bar J_F(S\beta,x)=S\bar J_F(\beta,x/S)$.

The generalized kernel. Fix a compact set $\mathcal{K}\subset\Delta^\circ_G$. Theorem 6.8 below extends $J_G(\beta,\cdot)$ and $C_G(\beta,\cdot)$ to holomorphic functions on a disc $|z|<2z_0$, for $\beta\in\mathcal{K}$, with $z_0>0$ given there. With these extensions we define, for $\beta\in\mathcal{K}$ and $x,S>0$, the generalized kernel

$$\mathbf{K}_\beta(x,S)=e^{-q_i x}\,\frac{1}{2\pi\mathrm{i}}\oint_{|z|=\rho}e^{x/z}\,\frac{C_G(\beta,z)}{C_G(\beta,0)}\,e^{S(J_G(\beta)-J_G(\beta,z))}\,\frac{dz}{z^2},$$

with $0<\rho<2z_0$. The integrand is holomorphic on the punctured disc, so the integral does not depend on ρ.

6.2 The main results

Theorem 6.1 (the facet layer) *Let F be irreducible with $k=|F|\geq 2$, let $i\in F$, let $G=F\setminus\{i\}$ satisfy* (H)*, and let $\mathcal{K}\subset\Delta^\circ_G$ be compact. Then there are $\delta>0$, T_0 and C such that the following hold uniformly for $\beta\in\mathcal{K}$, $T\geq T_0$ and $0<x\leq\delta T$, with $S=T-x$ and $s=S\beta$.*

(a) $f_F(s,x)=f_G(s)\,\mathbf{K}_\beta(x,S)\,(1+E)$ *with* $|E|\leq C(T^{-2}+x^{1/2}T^{-3/2})$*, so* $|E|\leq 2C/T$.

(b) $f_F(s,x)=f_G(s)\,\mathbf{B}_\beta(x,S)\,e^E$ *with* $|E|\leq C\big(1/T+\sqrt{x/T}\,(1+x)\big)$.

(c) *For the $\beta \in \mathcal{K}$ with $\pi^{\mathrm{in}}_\beta = \pi^{\mathrm{out}}_\beta$, and so for all $\beta \in \mathcal{K}$ if Q is reversible on F, we have $\gamma_\beta = q_i$, (b) holds with $|E| \le C(1/T + \sqrt{x/T})$, and*

$$f_F(s,x) = f_G(s)\,\mathbf{B}_\beta(x,S)\Big(1 + c_1(\beta)\,z_*\frac{I_0(\zeta)}{I_1(\zeta)} + E'\Big), \qquad |E'| \le C\,\frac{1+x}{T},$$

where $c_1(\beta) = \partial_z \log C_G(\beta, z)$ at $z = 0$.

For $k = 2$, (a) holds with $E = 0$, and Proposition 6.5 gives (b) for all $x > 0$ with an explicit bound.

For reversible Q, (18) below gives $c_1(\beta)$ explicitly. Part (b) gives the relative Bessel form $f_F = f_G\mathbf{B}_\beta(1 + o(1))$ for $x = o(T^{1/3})$, and beyond that a logarithmic form. To find the exact range of the relative form we put

$$\mathfrak{r}_\beta(\varepsilon) = \bar{J}_F(\beta, \varepsilon^2) - q_i\varepsilon^2 - J_G(\beta) + 2\sqrt{\omega_\beta}\,\varepsilon - (\gamma_\beta - q_i)\varepsilon^2.$$

Hence $S\mathfrak{r}_\beta(\sqrt{x/S})$ is the part of the rate $\bar{J}_F(s,x)$ that the exponent $SJ_G(\beta) + \gamma_\beta x - 2\sqrt{\omega_\beta S x}$ of $f_G\mathbf{B}_\beta$ does not capture.

Proposition 6.2 (the range of the Bessel form) *Let F, i, G and $\mathcal{K}$ be as in Theorem 6.1.*

(a) *The function $\mathfrak{r}_\beta$ is real-analytic near $\varepsilon = 0$, jointly with β, and $\mathfrak{r}_\beta(\varepsilon) = \varkappa_\beta\varepsilon^3 + O(\varepsilon^4)$ uniformly on $\mathcal{K}$, where*

$$\varkappa_\beta = \frac{\partial_z^3 J_G(\beta,0)}{6\,\omega_\beta^{3/2}} + \frac{(\gamma_\beta - q_i)^2}{\omega_\beta^{1/2}}.$$

(b) *There are $\delta > 0$, T_0 and C such that, uniformly for $\beta \in \mathcal{K}$, $T \ge T_0$ and $0 < x \le \delta T$, with $S = T - x$ and $s = S\beta$,*

$$\Big|\log\frac{f_F(s,x)}{f_G(s)\,\mathbf{B}_\beta(x,S)} + S\,\mathfrak{r}_\beta\Big(\sqrt{\frac{x}{S}}\Big)\Big| \le C\Big(\frac{1}{T} + \sqrt{\frac{x}{T}}\Big).$$

In particular the logarithm is $-\varkappa_\beta x^{3/2}S^{-1/2} + O(1/T + \sqrt{x/T} + x^2/T)$.

(c) *If $\pi^{\mathrm{in}}_\beta = \pi^{\mathrm{out}}_\beta$, then $\mathfrak{r}_\beta = 0$ near 0. This holds for every β if Q is reversible on F.*

(d) *Fix $\beta \in \mathcal{K}$ with $\mathfrak{r}_\beta \ne 0$, and let $m \ge 3$ be the order of the first nonzero Taylor coefficient of $\mathfrak{r}_\beta$ at 0. There is $\delta_\beta > 0$ such that for $0 < x = x(T) \le \delta_\beta T$, the ratio $f_F(s,x)/(f_G(s)\mathbf{B}_\beta(x,S))$ tends to 1 as $T \to \infty$ if and only if $x/T^{1-2/m} \to 0$. If $\mathfrak{r}_\beta = 0$ near 0, the ratio tends to 1 whenever $x/T \to 0$.*

(e) *For the cycle with $q_{12} = q_{23} = q_{31} = 2$ and $q_{21} = q_{32} = q_{13} = 1$, a start with $\mu_1 + \mu_2 > 0$, $i = 3$ and $\beta = (\frac12, \frac12)$,*

$$\omega_\beta = 2 + \tfrac94\sqrt{2}, \qquad \partial_z^3 J_G(\beta,0) = -\tfrac{1323}{64}\sqrt{2}, \qquad \varkappa_\beta = -\frac{49}{64\sqrt{\omega_\beta}} = -0.3363\ldots,$$

so $m = 3$ there, and the ratio tends to infinity when $x/T^{1/3} \to \infty$ with $x \le \delta_\beta T$.

The condition in (c) is weaker than reversibility. Let G be the 3-state cycle with the rates of (e), $q_{12} = q_{23} = q_{31} = 2$ and $q_{21} = q_{32} = q_{13} = 1$, and add a fourth state i with $q_{ji} = q_{ij} = c > 0$ for $j \in G$, and a start with $\mu(G) > 0$. The chain on $F = G \cup \{i\}$

is not reversible, since the products of the rates around the cycle are 8 and 1. At the uniform β, the criterion in the proof of Proposition 3.3 gives $r_\beta \propto \mathbf{1}$, since the uniform law is stationary for the cycle, and then $l_\beta \propto \beta$. Hence π_β^{in} and π_β^{out} are both uniform, and the relative Bessel form holds for $x = o(T)$ there. At other β the two laws differ.

Strategy of the proofs. We prove the results in three steps. First, pulling the visits to i out of the run expansion writes f_F as a contour integral of the density $f_G^{(z)}$ of a weighted chain on G, in which z counts the visits to i (Lemma 6.3). Second, a local law for the weighted chain, uniform for complex z near 0, replaces $f_G^{(z)}(S\beta)$ by $S^{-(k-2)/2} C_G(\beta, z) e^{-S J_G(\beta,z)}$ (Theorem 6.8). Third, a saddle point on the circle $|z| = z_*$ evaluates the integral, and the expansion of $J_G(\beta, z)$ at $z = 0$ (Proposition 6.6) compares the result with the Bessel kernel. Sections 6.3 to 6.6 give the exact formulas, the constants and the matching of the prefactor, Section 6.7 the second step, and Section 6.8 the third.

6.3 The exact representation

Lemma 6.3 (the runs in i pulled out) *For $z \in \mathbb{C}$ and $s \in (0,\infty)^G$ let*

$$f_G^{(z)}(s) = e^{-\langle d(z), s\rangle} \sum_{m \in \mathbb{Z}_{\geq 1}^G} W_z(m) \prod_{j \in G} \frac{s_j^{m_j - 1}}{(m_j - 1)!},$$

$$W_z(m) = \sum_w \mu_{w_1}^{(z)} \prod_{l < |w|} A(z)_{w_l w_{l+1}} \, e_{w_{|w|}}^{(z)},$$

the inner sum over the words w in G with no two equal neighbors and m_j letters j. Then $z \mapsto f_G^{(z)}(s)$ is entire, $f_G^{(0)} = f_G$, and for $x > 0$ and every $\rho > 0$

$$f_F(s,x) = e^{-q_i x} \sum_{K \geq 1} \frac{x^{K-1}}{(K-1)!} [z^K] f_G^{(z)}(s) \tag{15}$$

$$= e^{-q_i x} \frac{1}{2\pi \mathrm{i}} \oint_{|z| = \rho} e^{x/z} f_G^{(z)}(s) \, \frac{dz}{z^2}. \tag{16}$$

Hence $f_G^{(z)}$ is the killed density on the facet of the chain in which each visit to i is replaced by an instantaneous jump $j \to j'$ of rate $z c_j^- c_{j'}^+$, and z counts the visits. The K visits have total length x, with density $q_i^K x^{K-1} e^{-q_i x}/(K-1)!$, and each return carries a factor $c_{j'}^+/q_i$. Thus the powers of q_i cancel.

Proof Fix a word c in F with no equal neighbors, every letter of F, and $K = m_i(c) \geq 1$ letters i. Its term in $H_F(s,x)$ is $\mu_{c_1} \prod q_{c_l c_{l+1}}$ times $x^{K-1}/(K-1)!$ times $\prod_{j \in G} s_j^{m_j - 1}/(m_j - 1)!$. The letters i are isolated in c. We delete them and mark the places where they were. The result is a marked word c' in G, in which two equal neighbors occur exactly where c had $j\,i\,j$. A mark between letters j and j' carries $q_{ji} q_{ij'} = c_j^- c_{j'}^+$. A mark before the first letter j' carries $\mu_i c_{j'}^+$, and a mark after the last letter j carries c_j^-. Two neighbors without a mark are different and carry $q_{jj'}$. So the weight of c is the product of these weights. Between two different letters of c' there is at most one mark, and between two equal letters exactly one.

We group the marked words by their reduced word w, obtained by merging each block of equal letters of c' into one letter, and by the numbers $n_1, n_2, \ldots$ of marks inside the blocks.

If j forms ϱ blocks with n inner marks in all, it has $\varrho+n$ letters in c', and there are $\binom{n+\varrho-1}{\varrho-1}$ ways to spread the marks. Since $\binom{n+\varrho-1}{\varrho-1}/(n+\varrho-1)! = 1/(n!(\varrho-1)!)$,

$$\sum_{n\ge 0}\binom{n+\varrho-1}{\varrho-1}\frac{y^n s^{n+\varrho-1}}{(n+\varrho-1)!} = e^{ys}\frac{s^{\varrho-1}}{(\varrho-1)!}.$$

With $y = zc_j^- c_j^+$, one power of z for each mark, the inner marks give the factor $e^{zc_j^- c_j^+ s_j}$, which accounts for the term $-z\,c^-\circ c^+$ of $d(z)$. The marks between different letters give $A(z)$, and the first and last marks give $\mu^{(z)}$ and $e^{(z)}$. Collecting the powers of z gives (15). The entries of $A(z)$, $\mu^{(z)}$ and $e^{(z)}$ are at most $q_{\max}(1+|z|q_{\max})$, $1+|z|q_{\max}$ and $1+|z|q_{\max}$ in modulus. So the bound $W(m)\le k^M q_{\max}^{M-1}$ of Section 2 holds for W_z with $q_{\max}$ replaced by $q_{\max}(1+|z|q_{\max})$, up to a factor $(1+|z|q_{\max})^2$, and the series for $f_G^{(z)}(s)$ converges locally uniformly in z. Hence it is entire. For $z=0$ no word carries a mark, and $f_G^{(0)}$ is the sum of Lemma 2.4 for G. Finally we expand $e^{x/z}=\sum_{n\ge 0}x^n z^{-n}/n!$ and integrate term by term on the circle. This gives $\sum_{n\ge 0}(x^n/n!)[z^{n+1}]f_G^{(z)}(s)$, which is (15). □

The discrete law has the same structure. The only change is that the visits to i come in blocks of steps, and the lengths of the blocks form a composition of n_i.

Lemma 6.4 (the discrete form) *Let $hq_{\max}\le 1$, $\operatorname{supp} n = F$, $N' = N - n_i$, and let $\hat n$ be n restricted to G. For $z\in\mathbb{C}$ let*

$$P_{N'}^{(z)}(\hat n) = \sum_x \mu_{x_1}^{h,(z)} \prod_{t<N'} \left(I + hQ_G + zh^2 c^-(c^+)^\top\right)_{x_t x_{t+1}} e_{x_{N'}}^{h,(z)},$$

the sum over the words $x_1\cdots x_{N'}$ in G with composition $\hat n$, where $\mu^{h,(z)} = \mu_G + zh\mu_i c^+$ and $e^{h,(z)} = \mathbf{1} + zhc^-$. Then $P_{N'}^{(0)}(\hat n)$ is the probability of the composition $\hat n$ for the chain run for N' steps, and

$$P_N(n) = \sum_{K\ge 1}\binom{n_i-1}{K-1}(1-hq_i)^{n_i-K}\,[z^K]P_{N'}^{(z)}(\hat n).$$

Proof We delete the maximal blocks of i from a path with composition n. What is left is a path $x_1\cdots x_{N'}$ in G with composition $\hat n$. A block of length ℓ between $x_t = j$ and $x_{t+1} = j'$ replaces the factor $(I+hQ)_{jj'}$ by $hq_{ji}(1-hq_i)^{\ell-1}hq_{ij'}$. A block at the start replaces μ_{x_1} by $\mu_i(1-hq_i)^{\ell-1}hq_{ix_1}$, and a block at the end adds the factor $hq_{x_{N'}i}(1-hq_i)^{\ell-1}$. We count the blocks by z. If there are K blocks, the factors $(1-hq_i)^{\ell-1}$ give $(1-hq_i)^{n_i-K}$, and the block lengths form one of the $\binom{n_i-1}{K-1}$ compositions of n_i into K positive parts. □

6.4 Pair faces

For $k=2$ the facet is one state, so the weighted chain has no words to sum. Hence the contour formula gives the law in closed form.

Proposition 6.5 (pair faces) *Let $F=\{j,i\}$ with $q_{ji},q_{ij}>0$ and $\mu_j>0$, and put $\omega=q_{ji}q_{ij}$ and $\zeta=2\sqrt{\omega s x}$. For all $s,x>0$*

$$f_F(s,x)=\mu_j e^{-q_j s-q_i x}\Big[\sqrt{\frac{\omega s}{x}}\,I_1(\zeta)+\Big(q_{ji}+\frac{\mu_i}{\mu_j}q_{ij}\Big)I_0(\zeta)+\frac{\mu_i}{\mu_j}\sqrt{\frac{\omega x}{s}}\,I_1(\zeta)\Big].$$

Here $\omega_\beta=\omega$, $\gamma_\beta=q_i$ and $f_G(s)=\mu_j e^{-q_j s}$, and for all $s,x>0$

$$0\le\frac{f_F(s,x)}{f_G(s)\mathbf{B}_\beta(x,s)}-1\le\Big(q_{ji}+\frac{\mu_i}{\mu_j}q_{ij}\Big)\Big(\frac{1}{\omega s}+\sqrt{\frac{x}{\omega s}}\Big)+\frac{\mu_i}{\mu_j}\,\frac{x}{s}.$$

This is the facet form of Pedler's formula for two-state chains [37]. For $x\le T/2$ the bound is $O(1/T+\sqrt{x/T})$, and the term $\sqrt{x/T}$ is attained, since the middle term of the formula has exactly this size.

Proof For $G=\{j\}$ the only word is j, so $f_G^{(z)}(s)=(\mu_j+z\mu_i q_{ij})(1+zq_{ji})e^{-(q_j-z\omega)s}$. With $z=z_*t$ and $z_*=\sqrt{x/(\omega s)}$ we have $x/z+\omega sz=\frac{\zeta}{2}(t+1/t)$, and the generating function $e^{(\zeta/2)(t+1/t)}=\sum_{n\in\mathbb{Z}}I_n(\zeta)t^n$ [36, Eq. 10.35.1] gives

$$\frac{1}{2\pi\mathrm{i}}\oint e^{x/z+\omega sz}z^{n-2}\,dz=z_*^{n-1}I_{n-1}(\zeta)\qquad(n\in\mathbb{Z}).\tag{17}$$

We put this into (16) for $n=0,1,2$ and use $I_{-1}=I_1$. This gives the formula. Each term is nonnegative, which gives the lower bound. For the upper bound we divide the last two terms by the first. This gives $(q_{ji}+\frac{\mu_i}{\mu_j}q_{ij})z_*I_0(\zeta)/I_1(\zeta)+\frac{\mu_i}{\mu_j}x/s$. The lower bound $I_1(\zeta)/I_0(\zeta)\ge\zeta/(1+\sqrt{1+\zeta^2})$ of Amos [1] gives $I_0/I_1\le(1+\sqrt{1+\zeta^2})/\zeta\le 1+2/\zeta$, and $z_*(1+2/\zeta)=\sqrt{x/(\omega s)}+1/(\omega s)$. For a pair face, $\beta=(1)$ and the vectors of the facet are $r_\beta=l_\beta=1$, so $\omega_\beta=\omega$ and $\pi_\beta^{\mathrm{in}}=\pi_\beta^{\mathrm{out}}$, which gives $\gamma_\beta=q_i$. □

6.5 The constants for every irreducible chain

For $u\in(0,\infty)^G$ put

$$\Psi_\beta(u)=\sum_{j\in G}\beta_j\Big(q_j-\sum_{j'\in G\setminus\{j\}}q_{jj'}\frac{u_{j'}}{u_j}\Big),\qquad P_\beta(u)=(c^+\cdot u)\sum_{j\in G}\frac{c_j^-\beta_j}{u_j}.$$

Since $\sum_j\beta_j(-(Q_G(z)u)_j/u_j)=\Psi_\beta(u)-zP_\beta(u)$, the rate of the weighted chain is $J_G(\beta,z)=\sup_u[\Psi_\beta(u)-zP_\beta(u)]$ for real $z\ge 0$.

Proposition 6.6 (the facet expansion of the rate) *Let F be irreducible, $i\in F$, and $G=F\setminus\{i\}$ irreducible.*

(i) For $s=S\beta$ with $\beta\in\Delta_G^\circ$ and $x>0$,

$$\bar J_F(s,x)=q_i x+\sup_{z>0}\big[S\,J_G(\beta,z)-x/z\big].$$

(ii) The map $z\mapsto J_G(\beta,z)$ is real-analytic near $z=0$, jointly with β, and

$$J_G(\beta,0)=J_G(\beta),\qquad\partial_z J_G(\beta,0)=-\omega_\beta,\qquad\partial_z^2 J_G(\beta,0)=2\omega_\beta(\gamma_\beta-q_i).$$

(iii) *There are $\delta > 0$ and C such that, uniformly for β in a compact subset of Δ_G° and $0 < x \le \delta S$,*

$$\left| \bar{J}_F(s,x) - \left(S\, J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x} + (\gamma_\beta - q_i)x \right) \right| \le C\, x^{3/2} S^{-1/2}.$$

(iv) *If $\nu_j q_{jj'} = \nu_{j'} q_{j'j}$ on F for some $\nu \in (0,\infty)^F$, then $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$, $\gamma_\beta = q_i$, $J_G(\beta, z) = J_G(\beta) - z\omega_\beta$ for all real $z \ge 0$, $\omega_\beta = \left(\sum_{j\in G} \sqrt{q_{ij} q_{ji} \beta_j}\right)^2$, and for all $s = S\beta$ and $x > 0$*

$$\bar{J}_F(s,x) = S\, J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x}.$$

(v) *The map $z \mapsto J_G(\beta, z)$ is affine on $[0,\infty)$ if and only if $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$. Then $J_G(\beta, z) = J_G(\beta) - z\omega_\beta$ for all real $z \ge 0$, and $\bar{J}_F(s,x) = S\, J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x}$ for all $s = S\beta$ and $x > 0$.*

Hence in the occupation fraction α_i of i the rate has a square-root singularity $-2\sqrt{\omega_\beta \alpha_i}$ at the facet, and the time in i is charged at the rate γ_β, not q_i.

Proof (i) In the supremum that defines $\bar{J}_F(s,x)$ we write $u = (u_G, \epsilon)$ with $\epsilon = u_i$. The terms with ϵ are $-\epsilon \sum_j s_j c_j^- / u_j - (x/\epsilon)(c^+ \cdot u_G)$, and both coefficients are positive. Their supremum over $\epsilon > 0$ is $-2\sqrt{xS\, P_\beta(u_G)}$. So $\bar{J}_F(s,x) = q_i x + \sup_u [S\Psi_\beta(u) - 2\sqrt{xSP_\beta(u)}]$. Now $2\sqrt{xSP} = \inf_{z>0}[SzP + x/z]$, and we exchange the two suprema.

(ii) Write $u = e^\psi$. For real $z \ge 0$ the function $\psi \mapsto \Psi_\beta(e^\psi) - zP_\beta(e^\psi)$ is concave, since $P_\beta(e^\psi)$ is a positive combination of the convex functions $e^{\psi_m - \psi_j}$, and it does not change when a constant is added to ψ. At $z = 0$ its maximizer is $\psi_0 = \log r_\beta$, unique up to a constant (Proposition 3.3(b)). The Hessian of $\sum_{j\neq j'} \beta_j q_{jj'} e^{\psi_{j'} - \psi_j}$ at ψ_0 is the Laplacian with the conductances $\kappa_{jj'}$, because $\beta_j q_{jj'} r_{\beta,j'} / r_{\beta,j} = l_{\beta,j} q_{jj'} r_{\beta,j'}$. So the Hessian of Ψ_β at ψ_0 is $-\mathcal{L}_\beta$, which is negative definite on the complement of the constants. By the implicit function theorem on that complement, the critical point $\psi(z)$ is real-analytic in (β, z) near $(\beta, 0)$, and for real $z \ge 0$ it is the maximizer by concavity. So $J_G(\beta, z) = \Psi_\beta(\psi(z)) - zP_\beta(\psi(z))$ is real-analytic. The envelope theorem gives $\partial_z J_G = -P_\beta(\psi(z))$, so $\partial_z J_G(\beta, 0) = -(c^+ \cdot r_\beta)(c^- \cdot l_\beta) = -\omega_\beta$. Differentiating the critical equation gives $-\mathcal{L}_\beta \psi'(0) = \nabla P$, with ∇P the gradient of $P_\beta(e^\psi)$ at ψ_0. Its entries are

$$\partial_{\psi_m} P = c_m^+ r_{\beta,m}\, (c^- \cdot l_\beta) - (c^+ \cdot r_\beta)\, c_m^- l_{\beta,m} = \omega_\beta(\pi_{\beta,m}^{\mathrm{out}} - \pi_{\beta,m}^{\mathrm{in}}),$$

which sum to zero. So $\partial_z^2 J_G(\beta, 0) = -\nabla P \cdot \psi'(0) = \nabla P^\top \mathcal{L}_\beta^+ \nabla P$, and this is $2\omega_\beta(\gamma_\beta - q_i)$ by (14).

(iii) By the proof of (i), $\bar{J}_F(s,x) - q_i x = S \sup_\psi [\Psi_\beta(\psi) - 2\varepsilon P_\beta(\psi)^{1/2}]$ with $\varepsilon = \sqrt{x/S}$. The maximizer lies in a bounded set of ψ modulo constants, uniformly for β in a compact set, as in the proof of Proposition 3.3(b). Comparing its value with the value at ψ_0 gives $\Psi_\beta(\psi_0) - \Psi_\beta(\psi^*) \le 2\varepsilon \sup P_\beta^{1/2}$ on that set. We normalize ψ and ψ_0 to sum to zero. The Hessian $-\mathcal{L}_\beta$ at ψ_0 is negative definite on these vectors and continuous in β, so Taylor's theorem gives $c, r > 0$ with $\Psi_\beta(\psi_0) - \Psi_\beta(\psi) \ge c|\psi - \psi_0|^2$ for $|\psi - \psi_0| \le r$, uniformly on the compact set of β. For $|\psi - \psi_0| > r$, concavity along the segment from ψ_0 to ψ gives $\Psi_\beta(\psi_0) - \Psi_\beta(\psi) \ge cr|\psi - \psi_0| \ge cr^2$. Hence $\Psi_\beta(\psi_0) - \Psi_\beta(\psi) \ge c \min(|\psi - \psi_0|^2, r^2)$, which gives $\psi^* - \psi_0 = O(\varepsilon^{1/2})$. The critical equation $\nabla \Psi_\beta(\psi^*) = \varepsilon \nabla P_\beta(\psi^*) / \sqrt{P_\beta(\psi^*)}$ has a right side of size $O(\varepsilon)$, and $\nabla^2 \Psi_\beta$ is invertible near ψ_0 on the complement of the constants, so $\psi^* - \psi_0 = O(\varepsilon)$. We expand to second order around ψ_0. With $g = \nabla(P^{1/2}) = \nabla P / (2\sqrt{\omega_\beta})$,

the maximizer is $\psi^* = \psi_0 - 2\varepsilon\mathcal{L}_\beta^+ g + O(\varepsilon^2)$, and the value is $J_G(\beta) - 2\varepsilon\sqrt{\omega_\beta} + 2\varepsilon^2 g^\top \mathcal{L}_\beta^+ g + O(\varepsilon^3)$. Here $2g^\top \mathcal{L}_\beta^+ g = \nabla P^\top \mathcal{L}_\beta^+ \nabla P/(2\omega_\beta) = \gamma_\beta - q_i$ by (ii). Multiplying by S gives (iii), since $S\varepsilon^3 = x^{3/2}S^{-1/2}$.

(iv) By Proposition 5.2(b) for the face G we may take $r_{\beta,j} = \sqrt{\beta_j/\nu_j}$ and $l_{\beta,j} = \sqrt{\beta_j\nu_j}$. Then $c_j^- l_{\beta,j} = q_{ji}\sqrt{\beta_j\nu_j} = \nu_i q_{ij}\sqrt{\beta_j/\nu_j} = \nu_i c_j^+ r_{\beta,j}$, so $\pi_\beta^{\text{in}} = \pi_\beta^{\text{out}}$ and $\gamma_\beta = q_i$. More is true. For every real $z \geq 0$ the matrix $Q_G(z)$ is reversible with respect to ν_G, since $\nu_j c_j^- c_{j'}^+ = \nu_i q_{ij} q_{ij'}$ is symmetric in j, j'. So by Proposition 5.2(b) its optimal vector is r_β for every z, and $J_G(\beta, z) = \Psi_\beta(r_\beta) - zP_\beta(r_\beta) = J_G(\beta) - z\omega_\beta$. Next $\omega_\beta = \nu_i\big(\sum_j q_{ij}\sqrt{\beta_j/\nu_j}\big)^2 = \big(\sum_j \sqrt{q_{ij}q_{ji}\beta_j}\big)^2$, using $q_{ji} = q_{ij}\nu_i/\nu_j$. Finally (i) gives $\sup_{z>0}[-Sz\omega_\beta - x/z] = -2\sqrt{\omega_\beta S x}$.

(v) If the two laws agree, the gradient ∇P in the proof of (ii) vanishes. So ψ_0 is a critical point of the concave function $\psi \mapsto \Psi_\beta(e^\psi) - zP_\beta(e^\psi)$ for every real $z \geq 0$, hence a maximizer, and $J_G(\beta, z) = \Psi_\beta(r_\beta) - zP_\beta(r_\beta) = J_G(\beta) - z\omega_\beta$. Then (i) gives $\bar{J}_F$ as in (iv). Conversely, if $J_G(\beta, \cdot)$ is affine, then $\partial_z^2 J_G(\beta, 0) = 0$, so $\gamma_\beta = q_i$ by (ii), and the two laws agree, since the effective resistance in (14) vanishes only when they do. □

For $1/T \ll x \ll T$ the Bessel kernel is $(4\pi)^{-1/2}(\omega_\beta S)^{1/4}x^{-3/4}\exp(2\sqrt{\omega_\beta S x} - \gamma_\beta x)(1 + O(1/\zeta))$, by $I_1(\zeta) = e^\zeta(2\pi\zeta)^{-1/2}(1 + O(1/\zeta))$ [36, Eq. 10.40.1]. Hence Proposition 6.6(iii) says that $f_G(s)\mathbf{B}_\beta(x, S)$ and the local law formula of F, which in occupation coordinates is $T^{-(k-1)/2}C_F(\alpha)e^{-TJ_F(\alpha)}$ (proved only on compact subsets of the open face), have the same exponent up to $O(x^{3/2}S^{-1/2})$. For their powers of x and their constants to agree as well we need $C_F((1-\epsilon)\beta, \epsilon) \sim (4\pi)^{-1/2}\omega_\beta^{1/4}\epsilon^{-3/4}C_G(\beta)$ as $\epsilon \to 0$. We prove this for reversible Q in the next proposition.

6.6 Prefactor matching

Proposition 6.7 (prefactor matching) *Let Q be reversible on F with respect to ν, and let G satisfy* (H)*, with $|G| \geq 1$. Uniformly for β in a compact subset of Δ_G°, as $\epsilon \to 0$,*

$$C_F((1-\epsilon)\beta, \epsilon) = (4\pi)^{-1/2}\,\omega_\beta^{1/4}\,\epsilon^{-3/4}\,C_G(\beta)\big(1 + O(\sqrt{\epsilon})\big),$$

with $C_G(\beta) = \mu_j$ when $G = \{j\}$.

Proof We use Proposition 5.2(b) at $\alpha = ((1-\epsilon)\beta, \epsilon)$. On G the vectors $r_\alpha = \sqrt{\alpha/\nu}$ and $l_\alpha = \sqrt{\alpha\nu}$ are $\sqrt{1-\epsilon}$ times r_β and l_β, and at i they are $O(\sqrt{\epsilon})$. So $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = (\mu_G \cdot r_\beta)(l_\beta \cdot \mathbf{1})(1 + O(\sqrt{\epsilon}))$, where $\mu_G \cdot r_\beta > 0$ by (H). Next $\prod_{j\in F}\alpha_j = \epsilon(1-\epsilon)^{k-1}\prod_{j\in G}\beta_j$. The conductances are $c_{jj'}(\alpha) = (1-\epsilon)c_{jj'}(\beta)$ on G, and at i they are $c_{ij}(\alpha) = \sqrt{\epsilon(1-\epsilon)}\sqrt{q_{ij}q_{ji}\beta_j}$. A spanning tree of F with m edges at i has weight $O(\epsilon^{m/2})$. The trees with one edge at i are a spanning tree of G and one edge $\{i, j\}$. So $\text{tree}(c(\alpha)) = \sqrt{\epsilon}\sqrt{\omega_\beta}\,\text{tree}(c_G(\beta))(1 + O(\sqrt{\epsilon}))$, by $\sum_j \sqrt{q_{ij}q_{ji}\beta_j} = \sqrt{\omega_\beta}$ from Proposition 6.6(iv). We put these into the formula for C_F and divide by the same formula for C_G, which has one factor $(4\pi)^{1/2}$ fewer. The powers of ϵ are $\epsilon^{1/4}$ from the tree and ϵ^{-1} from the product, which gives $\epsilon^{-3/4}$. For $G = \{j\}$ we use $C_{\{j\}} = \mu_j$ and tree $= 1$ on one vertex. □

The first amplitude correction. Put $c_1(\beta) = \partial_z \log C_G(\beta, z)$ at $z = 0$. For reversible Q,

$$c_1(\beta) = \frac{\mu_i b_\beta}{\mu_G \cdot r_\beta} + \frac{a_\beta}{l_\beta \cdot \mathbf{1}} + \frac{\omega_\beta}{2} \sum_{\{j,j'\} \subseteq G} R_{jj'} \pi_{\beta,j} \pi_{\beta,j'}, \tag{18}$$

with $\pi_\beta = \pi_\beta^{\text{in}} = \pi_\beta^{\text{out}}$ and $R_{jj'}$ the effective resistance for the conductances $l_{\beta,j} q_{jj'} r_{\beta,j'}$. To see this, we use that $Q_G(z)$ is reversible with respect to ν_G for real $z \geq 0$, by the proof of Proposition 6.6(iv). So its vectors $r_\beta(z) = r_\beta$ and $l_\beta(z) = l_\beta$ do not move with z. By Proposition 5.2(b) and the formula for $C_G(\beta, z)$ in Section 6.1, for real $z \geq 0$

$$C_G(\beta, z) = \frac{(\mu^{(z)} \cdot r_\beta)(l_\beta \cdot e^{(z)})}{(4\pi)^{(k-2)/2}} \frac{\sqrt{\operatorname{tree}(c(z))}}{\prod_{j \in G} \beta_j}, \qquad c_{jj'}(z) = l_{\beta,j} A(z)_{jj'} r_{\beta,j'}.$$

Since $l_{\beta,j} c_j^- = a_\beta \pi_{\beta,j}$ and $c_{j'}^+ r_{\beta,j'} = b_\beta \pi_{\beta,j'}$, we have $c_{jj'}(z) = c_{jj'}(0) + z\omega_\beta \pi_{\beta,j} \pi_{\beta,j'}$. The derivatives at $z = 0$ of the logarithms of $\mu^{(z)} \cdot r_\beta$ and $l_\beta \cdot e^{(z)}$ are the first two terms of (18). The third term is the derivative of $\frac{1}{2} \log \operatorname{tree}(c(z))$, by the rule of Kirchhoff that $\partial \log \operatorname{tree}(c)/\partial c_e = R_e$, which says that $c_e R_e$ is the weighted fraction of the spanning trees that contain the edge e. The function $C_G(\beta, z)$ of Theorem 6.8 below is holomorphic and agrees with this one for real $z \geq 0$, so its derivative at 0 is also given by (18).

6.7 The weighted facet chain

The layer needs the local law of the weighted chain $Q_G(z)$, uniformly for complex z near 0. We prove it in this section.

Theorem 6.8 (the weighted facet chain) *Let G satisfy* (H) *with $|G| = k - 1 \geq 1$, let $i \notin G$ with $F = G \cup \{i\}$ irreducible, and let $\mathcal{K} \subset \Delta_G^\circ$ be compact. There are a number $z_0 > 0$, not given explicitly, and a constant C, both depending only on Q, μ, i and $\mathcal{K}$, and functions $J_G(\beta, z)$ and $C_G(\beta, z)$ on $\mathcal{K} \times \{|z| < 2z_0\}$, holomorphic in z, jointly continuous, real for real z, equal to the rate and the local-law constant of Section 6.1 for real $0 \leq z < 2z_0$, and with $C_G(\beta, 0) = C_G(\beta) > 0$, such that uniformly for $\beta \in \mathcal{K}$, $S \geq 1$ and complex $|z| \leq z_0$*

$$f_G^{(z)}(S\beta) = S^{-(k-2)/2} C_G(\beta, z) e^{-S J_G(\beta, z)} (1 + \varepsilon), \qquad |\varepsilon| \leq C/S. \tag{19}$$

For $|G| = 1$ this holds with $\varepsilon = 0$.

At $z = 0$ the display is the local law (13), that is Theorem 5.4(a) for the face G. The proof inherits every constant from $z = 0$ by continuity, which is why z_0 is not explicit. The rate $J_G(\beta, z)$ is defined for all real $z \geq 0$, but for complex z the proof needs a critical point of the torus phase near the real one and a Hessian with positive definite real part, and neither need persist far from $z = 0$. The number δ of Theorem 6.1 is chosen so that the saddle radius z_* stays below $z_0/2$.

Lemma 6.9 (one and two states) *Theorem 6.8 holds when $|G| \leq 2$, with the functions J_G and C_G given in the proof.*

Proof For $G = \{j\}$ the proof of Proposition 6.5 gives $f_G^{(z)}(s) = (\mu_j + z\mu_i q_{ij})(1 + zq_{ji})e^{-(q_j - z\omega)s}$, which is (19) with $\varepsilon = 0$, $J_G(\beta, z) = q_j - z\omega$ and $C_G(\beta, z) = (\mu_j + z\mu_i q_{ij})(1 + zq_{ji})$. Now let $G = \{1, 2\}$, and write $a_{12}(z) = q_{12} + zc_1^- c_2^+$ and $a_{21}(z) = q_{21} + zc_2^- c_1^+$ for the entries of $A(z)$, $\varpi(z) = a_{12}(z)a_{21}(z)$, $p = \mu^{(z)}$ and $e = e^{(z)}$. A word in G with no equal neighbors alternates. The words that start with 1 are $1(21)^n$, with weight $p_1\varpi^n e_1$ and $(n+1, n)$ letters, and $(12)^n$, with weight $p_1\varpi^{n-1}a_{12}e_2$ and (n, n) letters, for $n \geq 1$. With $\xi = 2\sqrt{\varpi s_1 s_2}$ the power series of I_1 and I_0 give

$$\sum_{n\geq 1} \frac{\varpi^n s_1^n s_2^{n-1}}{n!\,(n-1)!} = \sqrt{\frac{\varpi s_1}{s_2}}\, I_1(\xi), \qquad \sum_{n\geq 1} \frac{(\varpi s_1 s_2)^{n-1}}{(n-1)!^2} = I_0(\xi).$$

We do the same for the words that start with 2 and obtain

$$f_G^{(z)}(s) = e^{-\langle d(z), s\rangle}\Big[\Big(p_1 e_1\sqrt{\frac{\varpi s_1}{s_2}} + p_2 e_2\sqrt{\frac{\varpi s_2}{s_1}}\Big) I_1(\xi) + \big(p_1 a_{12} e_2 + p_2 a_{21} e_1\big) I_0(\xi)\Big].$$

Both sides are entire in z, so the branch of the square root does not matter. Since G is irreducible, $\varpi(0) = q_{12}q_{21} > 0$. We choose z_0 so that $|\varpi(z) - \varpi(0)| \leq \varpi(0)/2$ for $|z| \leq 2z_0$ and take the principal branch of $\sqrt{\varpi}$. Then $\xi = 2S\sqrt{\beta_1\beta_2\varpi}$ at $s = S\beta$ satisfies $|\arg \xi| \leq \pi/12$ and $|\xi| \geq cS$, uniformly on $\mathcal{K}$. The expansion $I_\nu(\xi) = e^\xi(2\pi\xi)^{-1/2}(1 + O(1/|\xi|))$ holds for $\nu = 0, 1$, uniformly in the sector $|\arg \xi| \leq \pi/4$ and $|\xi| \geq 1$ [36, Eq. 10.40.1]. Since $(2\pi\xi)^{-1/2} = S^{-1/2}(4\pi)^{-1/2}(\varpi\beta_1\beta_2)^{-1/4}$, this gives (19) with

$$J_G(\beta, z) = \langle d(z), \beta\rangle - 2\sqrt{\beta_1\beta_2\varpi}, \qquad C_G(\beta, z) = \frac{U + V}{\sqrt{4\pi}\,(\varpi\beta_1\beta_2)^{1/4}},$$

where $U = p_1 e_1\sqrt{\varpi\beta_1/\beta_2} + p_2 e_2\sqrt{\varpi\beta_2/\beta_1}$ and $V = p_1 a_{12} e_2 + p_2 a_{21} e_1$, and with $\varepsilon = (U\varepsilon_1 + V\varepsilon_0)/(U + V)$, where $|\varepsilon_\nu| \leq C/S$ are the errors of the two expansions. Both functions are holomorphic in z, jointly continuous and real for real z. At $z = 0$ we have $p = \mu_G$ and $e = \mathbf{1}$, so $U + V > 0$ because $\mu(G) > 0$. After shrinking z_0, $|U + V|$ is bounded below and $|U| + |V|$ above on $\mathcal{K} \times \{|z| \leq 2z_0\}$, which gives $|\varepsilon| \leq C/S$ once $cS \geq 1$. For $1 \leq S \leq 1/c$ the error ε is continuous on a compact set, so it is bounded, and we enlarge C. For real $z \geq 0$ the rate of the two-state matrix $Q_G(z)$ is $\sup_u[\beta_1(d_1 - a_{12}u_2/u_1) + \beta_2(d_2 - a_{21}u_1/u_2)] = \langle d, \beta\rangle - 2\sqrt{\beta_1\beta_2\varpi}$, by the inequality of the arithmetic and geometric means. The constant agrees with that of Section 6.1, because for real $z \geq 0$ both are the limit of $S^{1/2}e^{SJ_G(\beta,z)} f_G^{(z)}(S\beta)$. For the first this is the computation above. For the second it is Lemma A.4 on G for the weights $A(z) \geq 0$, $\mu^{(z)}$ and $e^{(z)}$, since $f_G^{(z)}(s) = e^{-\langle d(z), s\rangle} H_{A(z), \mu^{(z)}, e^{(z)}}(s)$ in the notation of Appendix A. □

Facets with two or more states. Let $|G| = k - 1 \geq 2$. We number the states of G as $1, \dots, k-1$ and put $m = k-2$. For $z \in \mathbb{C}$ let $A'(z)$ be $A(z)$ without the row and column of state 1, and let X_z, g_z and b_z be the functions of Lemma A.3 on G for the weights $A(z)$, $\mu^{(z)}$ and $e^{(z)}$. Since c^-, c^+ and μ_i are nonnegative, the weights $A(|z|)$, $\mu^{(|z|)}$ and $e^{(|z|)}$ dominate these in the sense of that lemma. For fixed $\beta \in \Delta_G^\circ$ and $z' \in \mathbb{C}^m$ put

$$\Phi_z(z') = \beta_1 g_z(z') + \sum_{j\geq 2} \beta_j z_j.$$

Let $\theta(\beta)$ be the tilt of the face G at β, let $\theta(\beta, z)$ be that of $Q_G(z)$ for real $z \geq 0$, and let $\mathcal{H}(\beta)$ be the Hessian $\mathcal{H}$ of the proof of Theorem 5.4(a) for the face G at $\alpha = \beta$. On $\mathbb{C}^m$ we use the norm $\|w\| = \max_j |w_j|$ with closed balls $\bar{B}(w, r)$, and $we^{\mathrm{i}\varphi}$ is the vector with entries $w_j e^{\mathrm{i}\varphi_j}$. The proof of Theorem 5.4(a) for G gives, for every $\beta \in \Delta_G^\circ$,

(P1) $\varrho(A'(0) - \operatorname{diag}\theta'(\beta)) < 0$ and $g_0(\theta'(\beta)) = \theta_1(\beta)$, by (A3),
(P2) $\nabla\Phi_0(\theta'(\beta)) = 0$, and $\nabla^2\Phi_0(\theta'(\beta)) = \mathcal{H}(\beta)$ is real, symmetric and positive definite,
(P3) $\operatorname{Re}\Phi_0(\theta'(\beta)e^{\mathrm{i}\varphi}) \le \Phi_0(\theta'(\beta)) - \sum_{j\ge 2}\beta_j\theta_j(\beta)(1-\cos\varphi_j)$ for $\varphi \in \mathbb{T}^m$,
(P4) $b_0(\theta'(\beta)) > 0$.

Also θ and $\mathcal{H}$ are continuous, by Proposition 5.2(a).

Lemma 6.10 (the complex critical point) *Let $|G| \ge 2$ and let $\mathcal{K} \subset \Delta_G^\circ$ be compact. There are $z_1 > 0$, $r_0 \ge r_1 > 0$, $\sigma_0 > 0$ and B_0, which depend only on Q, μ, i and $\mathcal{K}$, such that the following hold for every $\beta \in \mathcal{K}$.*

(a) $r_0 < \frac{1}{4}\min_j\theta_j(\beta)$, and $\varrho(A'(|z|) - \operatorname{diag}s') < 0$ for $|z| \le z_1$ and real s' with $\|s' - \theta'(\beta)\| \le 3r_0$.

(b) The functions $g_z(z')$ and $b_z(z')$ are holomorphic in (z, z') and bounded in modulus by B_0 on $\{|z| < z_1\} \times U_\beta$, where U_β is the set of z' with $|z_j| > \theta_j(\beta) - 3r_0$ for all j.

(c) For $|z| < z_1$ there is exactly one $\zeta'(\beta, z) \in \bar{B}(\theta'(\beta), r_1)$ with $\nabla\Phi_z(\zeta'(\beta, z)) = 0$. It is holomorphic in z and jointly continuous in (β, z), $\zeta'(\beta, 0) = \theta'(\beta)$, $\|\zeta'(\beta, z) - \theta'(\beta)\| \le B_0|z|$, and $\zeta'(\beta, \bar{z}) = \overline{\zeta'(\beta, z)}$.

(d) The matrix $\mathcal{J}(\beta, z) = Z_c\nabla^2\Phi_z(\zeta'(\beta, z))Z_c$, where $Z_c = \operatorname{diag}\zeta'(\beta, z)$, is complex symmetric and jointly continuous, and $\mathcal{J}(\beta, 0) = \Theta'\mathcal{H}(\beta)\Theta' \ge \sigma_0 I$, where $\Theta' = \operatorname{diag}\theta'(\beta)$.

(e) For real $0 \le z < z_1$ we have $\zeta'(\beta, z) = \theta'(\beta, z)$, $\Phi_z(\zeta'(\beta, z)) = \langle\beta, \theta(\beta, z)\rangle$, $\beta_1^2 \det\mathcal{J}(\beta, z) = \prod_{j\ge 2}\theta_j(\beta, z)^2 \det\Sigma_\beta(z)$ and $\beta_1 b_z(\zeta'(\beta, z)) = (\mu^{(z)}\cdot r_\beta(z))(l_\beta(z)\cdot e^{(z)})$.

Proof (a) The Perron root of a Metzler matrix is a continuous function of its entries. By (P1) the function $(\beta, y, s') \mapsto \varrho(A'(y) - \operatorname{diag}s')$ is negative on the compact set of the points $(\beta, 0, \theta'(\beta))$ with $\beta \in \mathcal{K}$. So it is negative on a neighborhood of this set, and the neighborhood contains the points with $\beta \in \mathcal{K}$, $0 \le y \le z_1$ and $\|s' - \theta'(\beta)\| \le 3r_0$ when z_1 and r_0 are small. We also take $r_0 < \frac{1}{4}\min_{\mathcal{K}}\min_j\theta_j$, which is positive.

(b) Put $s'_\beta = \theta'(\beta) - 3r_0\mathbf{1} > 0$. By (a) we may use Lemma A.3 with the dominating weights at $|z|$ and the radii s'_β. So $\mathcal{Z}' - A'(z)$ is invertible on $\{|z| \le z_1\} \times U_\beta$, and there $|g_z| \le \bar{g}(s'_\beta)$ and $|b_z| \le \bar{b}(s'_\beta)$ for the functions $\bar{g}$ and $\bar{b}$ of the weights at $|z|$. The terms of their series do not decrease with $|z|$, so we may take the weights at z_1. Then the bounds are continuous in β, since $(\operatorname{diag}s' - A'(z_1))^{-1}$ is continuous where $\varrho(A'(z_1) - \operatorname{diag}s') < 0$. So they are bounded on $\mathcal{K}$. The weights are affine in z, so the last claim of Lemma A.3 gives the holomorphy.

(c) For $|z| \le z_1/2$ and $z' \in \bar{B}(\theta'(\beta), r_0)$ the polydisc of radius $\min(z_1/2, r_0)$ about (z, z') lies in $\{|z| < z_1\} \times U_\beta$, since its points have $|z_j| \ge \theta_j - 2r_0$. So by (b) and Cauchy's estimates, the derivatives of $g_z(z')$ and $b_z(z')$ in (z, z') up to order 4 are bounded on $\{|z| \le z_1/2\} \times \bar{B}(\theta'(\beta), r_0)$ by a constant B_1 that does not depend on β. Let $\mathcal{G}(z, z') = \nabla\Phi_z(z') = \beta_1\nabla g_z(z') + \beta'$. By (P2), $\mathcal{G}(0, \theta') = 0$ and $D_{z'}\mathcal{G}(0, \theta') = \mathcal{H}(\beta)$, whose inverse has norm at most B_2 on $\mathcal{K}$ by continuity. Put $\mathcal{T}_z(z') = z' - \mathcal{H}(\beta)^{-1}\mathcal{G}(z, z')$. For $|z| \le z_1/2$, $r \le r_0$ and $z' \in \bar{B}(\theta', r)$ we get

$$\|D_{z'}\mathcal{T}_z(z')\| = \|\mathcal{H}^{-1}(D_{z'}\mathcal{G}(0, \theta') - D_{z'}\mathcal{G}(z, z'))\| \le B_3(|z| + r), \qquad \|\mathcal{T}_z(\theta') - \theta'\| \le B_3|z|,$$

with B_3 depending only on m, B_1 and B_2. Choose $r_1 \le r_0$ and $z_2 \le z_1/2$ with $B_3(z_2 + r_1) \le \frac{1}{2}$ and $B_3 z_2 \le \frac{1}{2}r_1$. For $|z| \le z_2$ the map $\mathcal{T}_z$ sends $\bar{B}(\theta', r_1)$ into itself and is a contraction with

constant $\frac{1}{2}$ there. So $\mathcal{G}(z,\cdot)$ has exactly one zero $\zeta'(\beta,z)$ in this ball, the fixed point of $\mathcal{T}_z$, and $\|\zeta'-\theta'\| \le 2\|\mathcal{T}_z(\theta')-\theta'\| \le 2B_3|z|$. The iterates $\mathcal{T}_z^n(\theta')$ are holomorphic in z, continuous in (β,z) and converge uniformly, which gives the regularity of ζ', and $\zeta'(\beta,0)=\theta'$. The data A_G, $c^{\pm}$, μ and β are real, so $\mathcal{G}(\bar z,\bar w)=\overline{\mathcal{G}(z,w)}$. Hence $\overline{\zeta'(\beta,z)}$ is a zero of $\mathcal{G}(\bar z,\cdot)$ in the ball, and it equals $\zeta'(\beta,\bar z)$. We rename z_2 as z_1 and enlarge B_0 to at least $2B_3$.

(d) The Hessian $\nabla^2\Phi_z$ is symmetric and Z_c is diagonal, so $\mathcal{J}$ is complex symmetric, and it is continuous by (b) and (c). At $z=0$, $Z_c=\Theta'$ and (P2) gives $\mathcal{J}(\beta,0)=\Theta'\mathcal{H}(\beta)\Theta'$, which is positive definite and continuous in β. So its least eigenvalue has a positive minimum σ_0 on $\mathcal{K}$.

(e) Let $0\le z<z_1$ be real and write $\zeta'=\zeta'(\beta,z)$. By (c), ζ' is real, and it is positive since $\|\zeta'-\theta'\|\le r_0<\frac14\min_j\theta_j$. The matrix $A(z)=A(|z|)$ is nonnegative, and its graph contains that of A_G, so it is strongly connected. By (a) and the first fact of Appendix A, $X_z(\zeta')\ge 0$, and as in the proof of Theorem 5.4(a) the vector $\tilde r=(1,X_z(\zeta')b_1(z))\ge 0$, with $b_1(z)=(A(z)_{j1})_{j\ge2}$, lies in the kernel of $A(z)-\operatorname{diag}\tilde\theta$ with $\tilde\theta=(g_z(\zeta'),\zeta')$. By the second fact of Appendix A, $\tilde r>0$ and 0 is the Perron root. Let $\tilde l>0$ be the left Perron vector with $\tilde l\cdot\tilde r=1$, and $\pi=\tilde l\circ\tilde r$. With $\tilde R=\operatorname{diag}\tilde r$, the matrix $\tilde R^{-1}(A(z)-\operatorname{diag}\tilde\theta)\tilde R$ is an irreducible generator with stationary law π, so Lemma A.1 gives $\nabla\lambda(\tilde\theta)=-\pi$ for $\lambda(y)=\varrho(A(z)-\operatorname{diag}y)$. For real z' near ζ' the same argument gives $\lambda(g_z(z'),z')=0$, and differentiating at ζ' gives $\pi_1\partial_j g_z(\zeta')+\pi_j=0$. The critical equation $\beta_1\partial_j g_z(\zeta')+\beta_j=0$ then gives $\pi_j/\pi_1=\beta_j/\beta_1$ for $j\ge2$, and since π and β both sum to 1, $\pi=\beta$. So $\tilde l=\beta/\tilde r$, and $\tilde l^{\top}(A(z)-\operatorname{diag}\tilde\theta)=0$ says that the derivative of $u\mapsto\sum_{j\ne j'}\beta_jA(z)_{jj'}u_{j'}/u_j$ vanishes at $\tilde r$, as before (7). This function is convex in $\log u$, so $\tilde r$ is a minimizer, and Proposition 3.3(b) for $Q_G(z)$ shows that $\tilde r$ is a multiple of $r_\beta(z)$. Hence $\tilde\theta=A(z)\tilde r/\tilde r=\theta(\beta,z)$, and $\Phi_z(\zeta')=\beta_1\theta_1+\sum_{j\ge2}\beta_j\theta_j=\langle\beta,\theta(\beta,z)\rangle$. The last two claims are the identities of Lemma A.4 on G for $A(z)$, $\mu^{(z)}$ and $e^{(z)}$ at $\alpha=\beta$, since $\mathcal{J}(\beta,z)$ is the Hessian $\mathcal{H}$ of that lemma multiplied on both sides by $\operatorname{diag}\theta'(\beta,z)$. □

Proof of Theorem 6.8 For $|G|=1$ this is Lemma 6.9. Let $|G|\ge2$, and let z_1, r_0, r_1, σ_0 and B_0 be as in Lemma 6.10. For $\beta\in\mathcal{K}$ and $|z|<z_1$ write $\zeta'=\zeta'(\beta,z)$ and $\rho_j=|\zeta_j|$, and put

$$\Psi(\beta,z,\varphi)=\Phi_z(\zeta' e^{\mathrm{i}\varphi}),\qquad \chi(\beta,z,\varphi)=b_z(\zeta' e^{\mathrm{i}\varphi})\prod_{j\ge2}\zeta_j e^{\mathrm{i}\varphi_j}.$$

They play the roles of the phase Ψ and the amplitude a of Lemma B.3.

Step 1. The constants. Since θ_j is real, $|\rho_j-\theta_j|\le|\zeta_j-\theta_j|\le r_1$. So the points $\zeta' e^{\mathrm{i}\varphi}$ lie in U_β, and by Lemma 6.10(b), (c) the φ-derivatives of Ψ and χ are jointly continuous on the compact set $P_1\times\mathbb{T}^m$, where $P_1=\mathcal{K}\times\{|z|\le z_1/2\}$. Let B_Ψ and B_a bound those of Ψ up to order 4 and those of χ up to order 2. Since $\partial/\partial\varphi_j=\mathrm{i}w_j\,\partial/\partial w_j$ at $w=\zeta' e^{\mathrm{i}\varphi}$ and $\nabla\Phi_z(\zeta')=0$, we have $\nabla_\varphi\Psi(\beta,z,0)=0$ and $-\nabla_\varphi^2\Psi(\beta,z,0)=\mathcal{J}(\beta,z)$, as in the proof of Theorem 5.4(a). Fix $\varepsilon_0\in(0,1)$ with $m^3B_\Psi\varepsilon_0\le\frac34\sigma_0$, and put $2\eta=\beta_{\min}\theta_{\min}(1-\cos\varepsilon_0)$, where $\beta_{\min}$ and $\theta_{\min}$ are the least entries of β and $\theta(\beta)$ over $\mathcal{K}$. By (P3), $\operatorname{Re}\Psi(\beta,0,\varphi)-\Psi(\beta,0,0)\le-2\eta$ for $\|\varphi\|\ge\varepsilon_0$. By uniform continuity on $P_1\times\mathbb{T}^m$ we can choose $z_0\in(0,z_1/4]$ such that for $\beta\in\mathcal{K}$ and $|z|\le2z_0$

(i) $\operatorname{Re}\mathcal{J}(\beta,z)\ge\frac12\sigma_0 I$, since $\|\mathcal{J}(\beta,z)-\mathcal{J}(\beta,0)\|\le\frac12\sigma_0$ in the operator norm and $\mathcal{J}(\beta,0)$ is real,

(ii) $|\chi(\beta,z,0)|\ge c_\chi$, where $2c_\chi>0$ is the minimum over $\mathcal{K}$ of $\chi(\beta,0,0)=b_0(\theta')\prod_{j\ge2}\theta_j$, which is positive by (P4),

(iii) $\operatorname{Re}\Psi(\beta,z,\varphi)-\operatorname{Re}\Psi(\beta,z,0)\le-\eta$ for $\|\varphi\|\ge\varepsilon_0$.

The numbers z_0, ε_0, η and c_χ depend only on Q, μ, i and $\mathcal{K}$.

Step 2. The torus through the critical point. Let $\beta \in \mathcal{K}$, $|z| \le z_0$ and $S \ge 1$. Since $\|\rho' - \theta'\| \le r_0$, Lemma 6.10(a) shows that the radii ρ' are admissible in Lemma A.3 with the dominating weights at $|z|$. With $t = S\beta$ it gives

$$e^{S\langle d(z),\beta\rangle} f_G^{(z)}(S\beta) = \frac{1}{(2\pi)^m} \int_{\mathbb{T}^m} \exp\bigl(S\Phi_z(w)\bigr)\, b_z(w) \prod_{j\ge 2} w_j \, d\varphi, \qquad w_j = \rho_j e^{\mathrm{i}\varphi_j}.$$

The translation of φ by the arguments of the ζ_j preserves the measure and turns w into $\zeta' e^{\mathrm{i}\varphi}$. So the integral is $\int_{\mathbb{T}^m} e^{S\Psi} \chi \, d\varphi$. No contour is moved. The torus representation holds at every admissible radius, and we use it at the radii ρ' of the complex critical point.

Step 3. The Laplace lemma. By Step 1, Lemma B.3 applies on $\mathcal{P} = \mathcal{K} \times \{|z| \le z_0\}$ with $\sigma_{\min} = \frac{1}{2}\sigma_0$, the cutoff ε_0 in place of ε, and η. Since $\Psi(\beta, z, 0) = \Phi_z(\zeta')$, it gives (19) with

$$J_G(\beta, z) = \langle d(z), \beta\rangle - \Phi_z(\zeta'), \qquad C_G(\beta, z) = (2\pi)^{-m/2} \det \mathcal{J}(\beta, z)^{-1/2}\, b_z(\zeta') \prod_{j\ge 2} \zeta_j,$$

and with $|\varepsilon| \le C/(c_\chi S)$, where C is the constant of Lemma B.3.

Step 4. The functions. We define J_G and C_G on $\mathcal{K} \times \{|z| < 2z_0\}$ by these formulas. Since ζ' is holomorphic in z and $g_z(z')$ and $b_z(z')$ are holomorphic in (z, z'), the numbers $\Phi_z(\zeta')$, $b_z(\zeta')$, $\prod_j \zeta_j$ and the entries of $\mathcal{J}(\beta, z)$ are holomorphic in z and jointly continuous. By (i) the matrix $\mathcal{J}(\beta, z)$ lies in the set $\mathcal{S}_m$ of Appendix B, so $\det \mathcal{J}(\beta, z)^{-1/2}$ is continuous, and as a continuous square root of a holomorphic function without zeros it is holomorphic in z. For real z, ζ' is real by Lemma 6.10(c), so J_G and C_G are real, and $\mathcal{J}(\beta, z)$ is real and positive definite, so the branch is the positive root. For real $0 \le z < 2z_0$, Lemma 6.10(e) gives $J_G(\beta, z) = \langle d(z) - \theta(\beta, z), \beta\rangle$, which is the rate of $Q_G(z)$ by the identity $J = \langle q - \theta, \alpha\rangle$ after Definition 5.1, since the diagonal of $Q_G(z)$ is $-d(z)$. It also gives $C_G(\beta, z) = (\mu^{(z)} \cdot r_\beta(z))(l_\beta(z) \cdot e^{(z)})/((2\pi)^{m/2}\sqrt{\det \Sigma_\beta(z)})$, the constant of Section 6.1. At $z = 0$ it is $C_G(\beta) > 0$, by (H). □

The proof does not use the irreducibility of F, which only makes c^- and c^+ nonzero.

6.8 Proofs of the main results

Proof of Theorem 6.1(a) Theorem 6.8 gives (19), with $\varepsilon = 0$ for $k = 2$. Take $\delta \le \frac{1}{2}$, so $S \ge T/2$. Write $\Lambda(z) = S^{-(k-2)/2} C_G(\beta, z) e^{-SJ_G(\beta,z)}$ and $\varepsilon(z) = f_G^{(z)}(s)/\Lambda(z) - 1$. By Step 1(i), (ii) of the proof of Theorem 6.8, $C_G(\beta, z) \ne 0$ for $|z| \le 2z_0$ if $|G| \ge 2$, so ε is holomorphic there (for $|G| = 1$, $\varepsilon = 0$), and $|\varepsilon| \le C/S$ for $|z| \le z_0$. The Schwarz lemma for $(\varepsilon(z) - \varepsilon(0))S/(2C)$ on $|z| < z_0$ gives $|\varepsilon(z) - \varepsilon(0)| \le 2C|z|/(z_0 S)$ for $|z| \le z_0$. Since $f_G(s) = \Lambda(0)(1 + \varepsilon(0))$, (16) gives for every $\rho \le z_0$

$$e^{q_i x} f_F(s, x) = (1 + \varepsilon(0))\mathcal{A} + \mathcal{D}, \qquad \mathcal{A} = \frac{1}{2\pi\mathrm{i}} \oint_{|z|=\rho} e^{x/z} \Lambda(z) \frac{dz}{z^2}, \qquad |\mathcal{D}| \le \frac{2C\rho}{z_0 S} \mathcal{A}^{\mathrm{abs}},$$

where $\mathcal{A}^{\mathrm{abs}}$ is the integral of $|e^{x/z}\Lambda(z)||z|^{-2}$ against $|dz|/(2\pi)$. By definition $\mathbf{K}_\beta(x, S) = e^{-q_i x}\mathcal{A}/\Lambda(0)$, and $|\varepsilon(0)| \le \frac{1}{2}$ for T_0 large, so $f_F = f_G \mathbf{K}_\beta(1 + E)$ with $|E| \le 2|\mathcal{D}|/\mathcal{A}$ once $\mathcal{A} > 0$. So it suffices to show that $\mathcal{A} > 0$, $\mathcal{A}^{\mathrm{abs}} \le C'\mathcal{A}$ and $\rho/S \le C'(T^{-2} + x^{1/2}T^{-3/2})$. Put

$$\Xi(z) = \frac{C_G(\beta, z)}{C_G(\beta, 0)}\, e^{S(J_G(\beta) - J_G(\beta,z) - \omega_\beta z)},$$

so that $\Lambda(z)/\Lambda(0) = e^{S\omega_\beta z}\Xi(z)$. By Proposition 6.6(ii), the holomorphy of C_G, and Cauchy estimates on the compact set $\mathcal{K} \times \{|z| \le \frac{3}{2}z_0\}$, the function $R = \log \Xi$ is holomorphic on $|z| \le z_0$ after shrinking z_0, and

$$R(z) = c_1 z - S\omega_\beta(\gamma_\beta - q_i)z^2 + O(|z|^2 + S|z|^3), \qquad |R''(z)| \le C(1+S).$$

We take δ so small that $z_* \le \sqrt{2\delta/\omega_{\min}} \le z_0/2$, where $\omega_{\min} = \min_{\mathcal{K}} \omega_\beta > 0$.

Few visits, $\zeta \le Z_0$ for a fixed large Z_0. Take $\rho = \max(z_*, 1/S)$, which is at most C/S. On the circle $|x/z| \le Z_0/2$, $|S\omega_\beta z| \le C$ and $|\Xi(z) - 1| \le C/S$. Let $\mathcal{A}_0$ be $\mathcal{A}$ with $\Xi = 1$. By (17) it equals $\Lambda(0)\sqrt{\omega_\beta S/x}\, I_1(\zeta)$, which is at least $\Lambda(0)\omega_\beta S$, because $\sqrt{\omega S/x}\, I_1(\zeta) = \omega S \sum_{n\ge 0}(\zeta/2)^{2n}/(n!(n+1)!)$. Also $|\mathcal{A} - \mathcal{A}_0| \le C\Lambda(0)\rho^{-1}/S \le C\Lambda(0)$ and $\mathcal{A}^{\mathrm{abs}} \le C\Lambda(0)\rho^{-1} \le C\Lambda(0)S$. So $\mathcal{A} \ge \Lambda(0)(\omega_\beta S - C) > 0$ and $\mathcal{A}^{\mathrm{abs}} \le C'\mathcal{A}$ for S large, and $\rho/S \le C/S^2 \le 4C/T^2$.

Many visits, $\zeta \ge Z_0$. Let $\lambda(z) = x/z + S\omega_\beta z + R(z)$, so that $e^{x/z}\Lambda(z) = \Lambda(0)e^{\lambda(z)}$. It is real on the real axis, and $\lambda'(z) = -x/z^2 + S\omega_\beta + O(1+S|z|)$. By the implicit function theorem it has a real critical point $\rho_s = z_*(1 + O(\sqrt{x/S} + 1/\zeta))$, and we take $\rho = \rho_s$. For $z = \rho e^{\mathrm{i}\phi}$,

$$\operatorname{Re}\lambda(z) - \lambda(\rho) = (x/\rho + S\omega_\beta\rho)(\cos\phi - 1) + \operatorname{Re}[R(z) - R(\rho)].$$

Since R is real on the real axis, $\operatorname{Re}[R'(\rho)(z-\rho)] = R'(\rho)\rho(\cos\phi - 1)$, and $|R''| \le C(1+S)$ bounds the rest by $2C(1+S)\rho^2(1-\cos\phi)$. Also $|R'(\rho)|\rho \le C(\rho + S\rho^2)$. Now $S\rho^2 \le Cx$, $x/\rho + S\omega_\beta\rho \ge \zeta$, and $x/\zeta \le C\sqrt{\delta}$. So for δ small $\operatorname{Re}\lambda(z) - \lambda(\rho) \le -\frac{\zeta}{2}(1-\cos\phi)$, and

$$\mathcal{A}^{\mathrm{abs}} \le \Lambda(0)\rho^{-1}e^{\lambda(\rho)}\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-\frac{\zeta}{2}(1-\cos\phi)}d\phi \le C\Lambda(0)\rho^{-1}e^{\lambda(\rho)}\zeta^{-1/2}.$$

For the lower bound we write $\mathcal{A} = \Lambda(0)\rho^{-1}\frac{1}{2\pi}\int e^{\zeta\ell(\phi)}e^{-\mathrm{i}\phi}d\phi$ with $\ell(\phi) = \lambda(\rho e^{\mathrm{i}\phi})/\zeta$. Here $\ell(0)$ is real, $\ell'(0) = 0$, $-\ell''(0) = \rho^2\lambda''(\rho)/\zeta$ is real and lies in $[\frac{1}{2}, 2]$, the derivatives of ℓ up to order four are bounded uniformly in all parameters, and the gap above holds. The parameters (β, x, S) do not range over a compact set, so we use the uniform form of Lemma B.1 in Remark B.2, with $\sigma_{\min} = \frac{1}{2}$, a fixed $\varepsilon_0 \in (0,1)$ with $B_\Psi\varepsilon_0 \le \frac{3}{4}$, where B_Ψ bounds the derivatives of ℓ up to order four, the cutoff ε_0 in place of ε, and the gap $\eta = \frac{1}{2}(1-\cos\varepsilon_0)$. With $m = 1$, $\Psi = \ell$, $a = e^{-\mathrm{i}\phi}$ and ζ in place of T it gives $\mathcal{A} = \Lambda(0)\rho^{-1}e^{\lambda(\rho)}(2\pi\zeta(-\ell''(0)))^{-1/2}(1 + O(1/\zeta))$, which is positive and at least $c\,\mathcal{A}^{\mathrm{abs}}$. Finally $\rho_s \le 2z_*$ for δ small and Z_0 large, so $\rho/S \le 2x^{1/2}(\omega_{\min}S^3)^{-1/2} \le Cx^{1/2}T^{-3/2}$. □

Proof of Theorem 6.1(b) *and* (c) We use two facts on Bessel functions besides (17). For integers $n \ge 1$, $I_n(\zeta) = \frac{1}{\pi}\int_0^\pi e^{\zeta\cos t}\cos(nt)\,dt$ [36, Eq. 10.32.3]. Also $I_n \le I_1$ for $n \ge 1$ and $\zeta > 0$, since $I_\nu(\zeta)$ decreases in $\nu > 0$ [36, §10.37]. The bound $|1 - \cos nt - (1-\cos t)| \le n^2t^2/2$ and a Laplace estimate give $|I_1(\zeta) - I_n(\zeta)| \le Cn^2 I_1(\zeta)/\zeta$ for $\zeta \ge 1$. Also $I_0/I_1 \le 1 + 2/\zeta$, as in the proof of Proposition 6.5.

The case $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$. By Proposition 6.6(v), $J_G(\beta, z) = J_G(\beta) - z\omega_\beta$ for real $0 \le z < 2z_0$, hence on the whole disc by holomorphy, so $\Xi(z) = C_G(\beta,z)/C_G(\beta,0) = 1 + c_1 z + \sum_{n\ge 2} c_n z^n$ with $|c_n| \le Cz_0^{-n}$. By (17) the term $c_n z^n$ contributes $c_n z_*^n I_{n-1}(\zeta)/I_1(\zeta)$ relative to the main term $z_*^{-1}I_1(\zeta)$. For $n \ge 2$ this is at most $|c_n|z_*^n$, and the sum over $n \ge 2$ is $O(z_*^2) = O(x/T)$, since $z_* \le z_0/2$. The term $n = 1$ is $c_1 z_* I_0(\zeta)/I_1(\zeta) = O(z_* + 1/(\omega_\beta S)) = O(\sqrt{x/T} + 1/T)$. With (a) this proves both claims of (c), for all $0 < x \le \delta T$.

The general case, $x \le 1$. Write $\Xi(z) = e^{-S\omega_\beta(\gamma_\beta - q_i)z^2}(1 + c_1 z + R_2(z))$ with $|R_2(z)| \le C(|z|^2 + S|z|^3)$. On $|z| = z_*$ the Gaussian factor has modulus at most $e^{(\gamma_\beta - q_i)x} \le C$. We

expand it in powers of z^2 and use (17). With $y = (\gamma_\beta - q_i)x$ the term 1 gives the relative factor $\sum_{m\ge0}(-y)^m I_{2m-1}(\zeta)/(m!\,I_1(\zeta))$. For $\zeta \ge 1$ it differs from e^{-y} by at most $C\sum_{m\ge1} y^m m^2/(m!\,\zeta) \le Cx/\zeta = O(\sqrt{x/S})$. For $\zeta < 1$ we have $x < 1/(4\omega_\beta S)$, and $|I_{2m-1}/I_1 - 1| \le 1$ bounds the difference by $e^y - 1 = O(1/S)$. The term $c_1 z$ times the Gaussian factor gives, by the same expansion, the relative factor $c_1 z_* \sum_{m\ge0}(-y)^m I_{2m}(\zeta)/(m!\,I_1(\zeta))$, since $S\omega_\beta z_*^2 = x$. As $I_{2m} \le I_0$, its modulus is at most $|c_1| z_* e^y I_0(\zeta)/I_1(\zeta) \le C z_*(1 + 2/\zeta) = O(\sqrt{x/S} + 1/S)$. The term R_2 is bounded through the modulus of the integrand, and $\frac{1}{2\pi}\int e^{\zeta\cos\phi}d\phi = I_0(\zeta)$ gives $O\big((z_*^2 + Sz_*^3)I_0(\zeta)/I_1(\zeta)\big) = O(1/S + \sqrt{x/S})$. So $\mathbf{K}_\beta = \mathbf{B}_\beta(1 + O(1/T + \sqrt{x/T}))$ for $x \le 1$.

The general case, $1 \le x \le \delta T$. Here $\zeta \ge 2\sqrt{\omega_{\min} S} \ge Z_0$. We use the evaluation of $\mathcal{A}$ in the proof of (a), and the same evaluation with $\Xi = 1$, whose integral is exactly $\Lambda(0) z_*^{-1} I_1(\zeta)$. For $\Xi = 1$ the critical point is z_*, the critical value is ζ, and $-\ell''(0) = 1$. The ratio of the two evaluations is $e^{\lambda(\rho_s)-\zeta}(1 + O(\sqrt{x/S} + 1/\zeta))$, since $\rho_s/z_* - 1$ and the relative change of $-\ell''(0)$ are $O(\sqrt{x/S} + 1/\zeta)$. Now $x/z + S\omega_\beta z$ takes the value ζ at its critical point z_*, and the shift from z_* to ρ_s changes λ only by $O(\lambda''(\rho_s)(\rho_s - z_*)^2) = O(\zeta x/S + 1/\zeta)$. So

$$\lambda(\rho_s) - \zeta = R(z_*) + O(\zeta x/S + 1/\zeta) = -(\gamma_\beta - q_i)x + O\big(\sqrt{x/S}\,(1 + x)\big),$$

because $c_1 z_* = O(\sqrt{x/S})$, $S\omega_\beta z_*^2 = x$, $Sz_*^3 = O(x^{3/2}S^{-1/2})$, $\zeta x/S = O(x^{3/2}S^{-1/2})$ and $1/\zeta \le C\sqrt{x/S}$ for $x \ge 1$. Hence $\mathbf{K}_\beta/\mathbf{B}_\beta = e^{(\gamma_\beta - q_i)x}\mathcal{A}/\mathcal{A}_0 = \exp(O(\sqrt{x/T}(1 + x)))$, where the factor $e^{(\gamma_\beta - q_i)x}$ comes from $e^{-q_i x}$ in $\mathbf{K}_\beta$ against $e^{-\gamma_\beta x}$ in $\mathbf{B}_\beta$, and $\mathcal{A}_0$ is $\mathcal{A}$ with $\Xi = 1$. With (a) this proves (b). □

Proof of Proposition 6.2 (a) We prove this first. By the proof of Proposition 6.6(iii) with $S = 1$, $\mathfrak{r}_\beta(\varepsilon) = j(\varepsilon) - J_G(\beta) + 2\sqrt{\omega_\beta}\,\varepsilon - (\gamma_\beta - q_i)\varepsilon^2$ with $j(\varepsilon) = \sup_\psi[\Psi_\beta(\psi) - 2\varepsilon P_\beta(\psi)^{1/2}]$, and the maximizer is within $O(\varepsilon)$ of ψ_0 modulo constants. The critical equation $\nabla\Psi_\beta(\psi) = \varepsilon\nabla P_\beta(\psi)/\sqrt{P_\beta(\psi)}$ is real-analytic in $(\psi, \varepsilon, \beta)$ near $(\psi_0, 0, \beta)$, and its derivative in ψ at $\varepsilon = 0$ is $-\mathcal{L}_\beta$, which is invertible on the vectors that sum to zero. So the implicit function theorem gives an analytic solution $\psi(\varepsilon)$, which is the maximizer for small $\varepsilon \ge 0$. Hence j and $\mathfrak{r}_\beta$ are real-analytic near 0, jointly in β, and by Proposition 6.6(iii)

$$j(\varepsilon) = J_G(\beta) - 2\sqrt{\omega_\beta}\,\varepsilon + (\gamma_\beta - q_i)\varepsilon^2 + j_3\varepsilon^3 + O(\varepsilon^4)$$

uniformly on $\mathcal{K}$. So $\mathfrak{r}_\beta(\varepsilon) = j_3\varepsilon^3 + O(\varepsilon^4)$, and it remains to show $j_3 = \varkappa_\beta$. For each ψ the supremum of $-zP_\beta(\psi) - \varepsilon^2/z$ over $z > 0$ is $-2\varepsilon P_\beta(\psi)^{1/2}$, attained at $z = \varepsilon/\sqrt{P_\beta(\psi)}$. So $z(\varepsilon) = \varepsilon/\sqrt{P_\beta(\psi(\varepsilon))} = z_*(1 + O(\varepsilon))$ maximizes $J_G(\beta, z) - \varepsilon^2/z$ over $z > 0$, with the value $j(\varepsilon)$. Write $z = \varepsilon w$, $a = \frac{1}{2}\partial_z^2 J_G(\beta, 0) = \omega_\beta(\gamma_\beta - q_i)$ and $c = \frac{1}{6}\partial_z^3 J_G(\beta, 0)$. Then $(j(\varepsilon) - J_G(\beta))/\varepsilon$ is the maximum over w near $w_0 = \omega_\beta^{-1/2}$ of $\phi(w) + p(w) + O(\varepsilon^3)$, where $\phi(w) = -\omega_\beta w - 1/w$ and $p(w) = a\varepsilon w^2 + c\varepsilon^2 w^3$. Here $\phi(w_0) = -2\sqrt{\omega_\beta}$, $\phi'(w_0) = 0$ and $\phi''(w_0) = -2\omega_\beta^{3/2}$, so the maximum is $\phi(w_0) + p(w_0) + p'(w_0)^2/(4\omega_\beta^{3/2}) + O(\varepsilon^3)$. This gives $j_3 = c\,\omega_\beta^{-3/2} + a^2\omega_\beta^{-5/2} = \varkappa_\beta$.

(b) Take δ at most the δ of Theorem 6.1, and so small that $\sqrt{2\delta}$ lies in the disc where (a) holds uniformly. Then $\varepsilon = \sqrt{x/S}$ lies there, since $S \ge T/2$. First let $x \le 1$. Theorem 6.1(b) bounds the logarithm by $C(1/T + \sqrt{x/T})$, and by (a) $S|\mathfrak{r}_\beta(\varepsilon)| \le CS\varepsilon^3 = Cx^{3/2}S^{-1/2} \le C\sqrt{x/S}$. Now let $1 \le x \le \delta T$. The proof of Theorem 6.1(b) gives $\log(\mathbf{K}_\beta/\mathbf{B}_\beta) = (\gamma_\beta - q_i)x + \lambda(\rho_s) - \zeta + O(\sqrt{x/S})$, since $1/\zeta \le C\sqrt{x/S}$ for $x \ge 1$. On the real interval $I = [z_*/2, 2z_*]$ we have $\lambda = \lambda_0 + g$ with $\lambda_0(z) = x/z + S(J_G(\beta) - J_G(\beta, z))$ and $g(z) = \log(C_G(\beta, z)/C_G(\beta, 0)) =$

$O(z_*)$. Also $\lambda''(z) = 2x/z^3 + R''(z) \geq \omega_\beta S/(4z_*) - C(1+S) > 0$ on I for δ small, so λ is convex on I, and $\rho_s \in I$. Hence $\lambda(\rho_s) = \min_I \lambda$, which differs from $\min_I \lambda_0$ by $O(z_*)$. By the proof of (a) the point $z(\varepsilon) \in I$ maximizes $SJ_G(\beta,z) - x/z$ over $z > 0$, so $\min_I \lambda_0 = SJ_G(\beta) + q_i x - \bar{J}_F(s,x)$ by Proposition 6.6(i). Since $\bar{J}_F(s,x) = q_i x + Sj(\varepsilon)$, $\zeta = 2\sqrt{\omega_\beta}\,S\varepsilon$ and $(\gamma_\beta - q_i)x = (\gamma_\beta - q_i)S\varepsilon^2$, this gives

$$\log \frac{\mathbf{K}_\beta}{\mathbf{B}_\beta} = -S\bigl(j(\varepsilon) - J_G(\beta) + 2\sqrt{\omega_\beta}\,\varepsilon - (\gamma_\beta - q_i)\varepsilon^2\bigr) + O\Bigl(\sqrt{\frac{x}{S}}\Bigr) = -S\,\mathfrak{r}_\beta(\varepsilon) + O\Bigl(\sqrt{\frac{x}{S}}\Bigr).$$

With Theorem 6.1(a) and $S \geq T/2$ this proves the bound. The last claim follows from (a), since $S\varepsilon^4 = x^2/S$.

(c) If the two laws agree, then $\gamma_\beta = q_i$, and Proposition 6.6(v) gives $\bar{J}_F(\beta,\varepsilon^2) = J_G(\beta) + q_i\varepsilon^2 - 2\sqrt{\omega_\beta}\,\varepsilon$. So $\mathfrak{r}_\beta = 0$. For reversible Q the laws agree by Proposition 6.6(iv).

(d) Let $\varrho_m \neq 0$ be the coefficient of ε^m in $\mathfrak{r}_\beta$. Choose $\delta_\beta \leq \delta$ so small that $|\mathfrak{r}_\beta(\varepsilon) - \varrho_m\varepsilon^m| \leq \frac12|\varrho_m|\varepsilon^m$ for $\varepsilon \leq \sqrt{2\delta_\beta}$. Put $X = x^{m/2}S^{1-m/2} = S\varepsilon^m$. By (b) the logarithm ℓ of the ratio satisfies $\bigl||\ell| - |\varrho_m|X\bigr| \leq \frac12|\varrho_m|X + C(1/T + \sqrt{x/T})$. Since $T/2 \leq S \leq T$, X tends to 0 if and only if $x/T^{1-2/m}$ does. If it does, then $x/T \to 0$ too, and $\ell \to 0$. If it does not, there are $\eta > 0$ and arbitrarily large T with $X \geq \eta$. For these T, with $u = x/S$, we have $X = Su^{m/2} \geq \eta$, so $u \geq (\eta/S)^{2/m}$ and $\sqrt{u}/X = u^{-(m-1)/2}/S \leq \eta^{-(m-1)/m}S^{-1/m}$, which tends to 0. So $|\ell| \geq \frac12|\varrho_m|X - C(1/T + \sqrt{2u}) \geq \frac14|\varrho_m|\eta$ for these T when T is large, and the ratio does not tend to 1. If $\mathfrak{r}_\beta = 0$, then $|\ell| \leq C(1/T + \sqrt{x/T})$ by (b).

(e) For the cycle, $c^- = (q_{13}, q_{23}) = (1,2)$, $c^+ = (q_{31}, q_{32}) = (2,1)$ and $q_1 = q_2 = q_3 = 3$. So $d(z) = (3-2z)\mathbf{1}$, $a_{12}(z) = 2+z$ and $a_{21}(z) = 1+4z$, and Lemma 6.9 gives $J_G(\beta,z) = 3 - 2z - \sqrt{(2+z)(1+4z)}$. Its derivatives at 0 are $-\omega_\beta = -2 - \frac94\sqrt{2}$, $\frac{49}{16}\sqrt{2}$ and $-\frac{1323}{64}\sqrt{2}$. Then $c + a^2/\omega_\beta = -\frac{441}{128}\sqrt{2} + \frac{2401}{128(8+9\sqrt{2})} = -\frac{49}{64}\,\omega_\beta$, and dividing by $\omega_\beta^{3/2}$ gives $\varkappa_\beta$. Since $\varkappa_\beta < 0$, (b) and the choice of δ_β as in the proof of (d) give $\ell \geq \frac12|\varkappa_\beta|X - C(1/T + \sqrt{x/T})$, which tends to infinity when $x/T^{1/3} \to \infty$. □

6.9 The counts near a facet

Now let n have support F. Put $N' = N - n_i$, let $\hat{n}$ be n restricted to G, and put $\beta = \hat{n}/N'$, $S = hN'$, $x = hn_i$ and $p_i = 1 - hq_i$. So $T = S + x$. Let $\widetilde{\Lambda}_\beta(z) = C_G(\beta,z)C_G(\beta,0)^{-1}e^{S(J_G(\beta)-J_G(\beta,z))}$ be the integrand of $\mathbf{K}_\beta$ without the factors $e^{-q_i x}$, $e^{x/z}$ and z^{-2}, and define the lattice kernel

$$\mathbf{K}^h_\beta(n_i,S) = \frac{h}{2\pi\mathrm{i}}\oint_{|z|=\rho}\Bigl(p_i + \frac{h}{z}\Bigr)^{n_i-1}\widetilde{\Lambda}_\beta(z)\,\frac{dz}{z^2} = \sum_{K=1}^{n_i}\binom{n_i-1}{K-1}p_i^{n_i-K}h^K\,[z^K]\widetilde{\Lambda}_\beta(z),$$

with $0 < \rho < 2z_0$. It is $h\mathbf{K}_\beta$ with $e^{x/z-q_ix}$ replaced by $(p_i + h/z)^{n_i-1}$, as Lemma 6.4 replaces Lemma 6.3. If $\widetilde{\Lambda}_\beta(z) = e^{S\omega_\beta z}\sum_{r\geq0}\xi_r z^r$, then $\mathbf{K}^h_\beta(n_i,S) = p_i^{n_i}\sum_r \xi_r(h/p_i)^r B^{(r)}_{n_i}(u)$ with $u = hS\omega_\beta/p_i$ and $B_n(w) = \sum_{m=1}^n \binom{n-1}{m-1}w^m/m! = \frac{w}{n}L^{(1)}_{n-1}(-w)$, where $L^{(1)}_{n-1}$ is the Laguerre polynomial [36, Eq. 18.5.12]. So Laguerre polynomials replace the Bessel kernel. The ratio of the two laws below carries one power of h for every k. Indeed $P_{N'}(\hat{n})$ is close to $h^{k-2}f_G(s)$, by Lemma 5.5 for the face G if $k \geq 3$ and by the path $j^{N'}$ if $G = \{j\}$, and Theorems 6.1(a) and 6.12(b) show that $P_N(n)$ is then within a factor $1 + O(1/T + T^2/N)$ of $h^{k-1}f_F(s,x)$, also when n_i is small.

Theorem 6.11 (the counts on a pair face) *Let $F = \{j, i\}$ with $q_{ji}, q_{ij} > 0$ and $\mu_j > 0$. Put $\omega = q_{ji}q_{ij}$, $c_0 = 1$, $c_1 = q_{ji} + \frac{\mu_i}{\mu_j} q_{ij}$, $c_2 = \frac{\mu_i}{\mu_j}\omega$ and $p_l = 1 - hq_l$, and let $hq_{\max} \le \frac{1}{2}$ and $N', n_i \ge 1$.*

(a) (The discrete Pedler formula.) *Exactly, $P_{N'}(\hat{n}) = \mu_j p_j^{N'-1}$ and*

$$\frac{P_N(n)}{P_{N'}(\hat{n})} = R := \sum_{K=1}^{n_i} \binom{n_i - 1}{K - 1} p_i^{n_i - K} \sum_{e=0}^{2} c_e\, h^e \binom{N' - 1}{K - e} \Big(\frac{h^2\omega}{p_j}\Big)^{K-e}.$$

(b) $\mathbf{K}^h(n_i, S) = p_i^{n_i}\big[B_{n_i}(u) + c_1 \frac{h}{p_i} B'_{n_i}(u) + c_2 \frac{h^2}{p_i^2} B''_{n_i}(u)\big]$ *with $u = hS\omega/p_i$.*

(c) With $U = n_i u$,

$$1 - \frac{U + 2\sqrt{U} + 2}{2N'} \le \frac{R}{\mathbf{K}^h(n_i, S)} \le e^{2hq_j(1+2\sqrt{U})}.$$

So if $n_i \le N'$ and $1 \le T \le N^{1/2}$, then $R = \mathbf{K}^h(n_i, S)(1 + O(h + h^2\sqrt{N'n_i}))$.

(d) If $n_i \le N'$ and $hS \le 1$, then $\mathbf{K}^h(n_i, S) = h\,\mathbf{K}(x, S)(1 + O(h + hS))$, where $\mathbf{K}(x, S) = f_F(S, x)/f_G(S)$ is given by Proposition 6.5.

Part (c) has no term $1/T$ and holds at every T, also when T^2/N is of order one, where (d) does not apply and Section 8 of the arXiv version of this paper [40] treats the complete graph.

Theorem 6.12 (the counts near a facet) *Let F, i, G and $\mathcal{K}$ be as in Theorem 6.1. There are $\delta > 0$, $\varepsilon_1 > 0$, T_0 and C such that the following hold uniformly for $\operatorname{supp} n = F$ with $\beta \in \mathcal{K}$, $1 \le n_i \le \delta N$ and $T_0 \le T \le \varepsilon_1 N^{1/2}$.*

(a) $P_N(n) = P_{N'}(\hat{n})\,\mathbf{K}^h_\beta(n_i, S)\,(1 + E)$ with $|E| \le C(T^{-2} + x^{1/2}T^{-3/2} + T^2/N) \le \frac{1}{2}$.

(b) $\mathbf{K}^h_\beta(n_i, S) = h\,\mathbf{K}_\beta(x', S)(1 + O(T^2/N))$ for $x' = x$ and for $x' = x - h$, where $\mathbf{K}_\beta(0, S) = S\omega_\beta + c_1(\beta)$ by the same formula. Hence $P_N(n) = hP_{N'}(\hat{n})\,\mathbf{K}_\beta(x, S)(1 + O(1/T + T^2/N))$.

(c) Parts (b) *and* (c) *of Theorem 6.1 hold with $P_N(n)/(hP_{N'}(\hat{n}))$ in place of $f_F(s, x)/f_G(s)$ and with T^2/N added to each error bound.*

For $k = 2$, Theorem 6.11(a) gives the ratio $P_N(n)/P_{N'}(\hat{n})$ exactly. By Theorem 6.12(b) the lattice kernel matters only when T^2/N is not small. For $k \ge 3$ that regime is open, since Theorem 6.12(a) then has an error of order one. We prove both theorems in Appendix E. The second one discretizes the states of G as in the proof of Lemma 5.5, keeps the state i exact, and evaluates the contour integral in z as in the proof of Theorem 6.1(a).

9 Verification and open questions

Every theorem, proposition, lemma and corollary of this paper is proved in the text or in the appendices, apart from the results of [38] and [39] that we quote where we say so. No proof rests on a computation. Hence the formal proofs and the scripts below are independent checks of parts of the paper. All of them are in the repository named under Code availability.

Formal proofs

The folders lean/PaperC and lean/PaperB prove many finite and algebraic steps in Lean 4 (v4.33.1, with Mathlib), with no `sorry` or `admit`. The file lean/PaperC/MAP.md gives for each statement the Lean theorems that prove it or its finite core, and lists what is not in Lean. Mathlib has no Perron and Frobenius theorem for irreducible Metzler matrices, so a positive Perron vector is a hypothesis where one is needed, and the asymptotic results are not in Lean.

Computations

The command `py -I verify_all.py` in the folder paperC runs the 9 scripts that check this paper, in about four minutes, and writes each output, headed by the sha256 of its script, to paperC/data. Each check is an assertion, and the checks in floating point use a tolerance without interval certification. The scripts `verify_first_order.py` and `verify_second_order.py` compute the law in two independent ways, by the run expansion and by a dynamic program or a discrete torus integral, and check Sections 2 to 5, including the gaps quoted after Theorem 5.8. The script `verify_full_range.py` checks Lemma A.2, Lemma A.6 and Theorems 5.4(b) and 5.7(a). For $q_{12} = 1$, $q_{21} = 2$ and $N = 4000$ the ratio R of the exact law to Theorem 5.4(b) has $T|R-1| \le 0.25$ at points of $[0.45, 0.85]$ for $T = 16$, 63, 200, 400 and $1000 = N/4$, while the continuum form has $T|R-1|$ of order T from $T = 200$ on. The script `verify_examples.py` finds the first-order winners on complete graphs, cycles and small graphs in two ways, and the finite-N modes from two exact codes. The script `verify_rate_flows.py` computes no law and checks the rate of Proposition 3.3 against a form through flows. The script `verify_complete.py` checks the exact results on the complete graph in Section 7 of the arXiv version of this paper [40], in exact arithmetic for small N. The script `verify_layer.py` checks the constants and kernels of Section 6 on three chains with three states. One of them is the cycle of Proposition 6.2(e), with $\omega_\beta = 5.1820$ and $\gamma_\beta = 3.4179$ against $q_3 = 3$, and the script recomputes the derivatives of $J_G(\beta, z)$ and $\varkappa_\beta$ for it exactly. The generalized kernel matches f_F/f_G to $O(1/T)$ for x from $0.01/T$ to $T/4$. The script also checks the example after Proposition 6.2, where $J_G(\beta, \cdot)$ is affine at the uniform β to 10^{-15}. The script `verify_layer_general.py` checks Theorem 6.8 on two facets with four states, namely a facet of a non-reversible face with five states and a path whose jump chain is periodic. It computes the word sum exactly and prints S times the relative error for S from 4 to 32. These products settle near 0.438 and 0.545 at $z = 0$, as (19) predicts. The script `verify_counts.py` checks Section 4.4 and Section 6.9. It checks Theorem 6.11(a) against an enumeration of all paths in exact arithmetic, for every $N \le 12$ and two values of h, and the bounds of Theorem 6.11(c) on a grid up to $N' = 10^4$. For the cycle of Proposition 6.2(e) with $i = 3$ and $\beta = (\frac{1}{2}, \frac{1}{2})$ it computes the exact law by a dynamic program and the lattice kernel by a contour integral, and the error E of Theorem 6.12(a) satisfies $0 < E \le 0.62\,T^2/N$ and falls by a factor of about 2.5 each time N' doubles, at $T = 8$, $n_3 = 10$ and $200 \le N' \le 1600$. On the face $\{1\} \to \{2, 3\}$ with $q_{12} = q_{23} = q_{32} = 1$ and the start in state 1, where $p_F = \frac{3}{2}$ and $K_F^{\mathrm{red}} = \sqrt{2/\pi}$, it computes M_F exactly at $N = 40000$, and the slope of $\log M_F$ in $\log T$ is within 10^{-3} of $\frac{3}{2}$ for $8 \le T \le 20$, far from the powers 1 and 2 of

Theorem 4.2(ii), while M_F divided by $N^{-2}T^{3/2}$ is within 0.05% of $\sqrt{2/\pi}$. The figures only illustrate the theorems. Hence no statement is deduced from them or from these numbers.

Open questions

A local law up to the lower-dimensional corners of a face is open. The results on reducible faces (Theorems 4.2(ii) and 4.5 and Corollary 4.6) are proved for $T \le N^{1/3}$, and we have not extended them to $T \le h_* N$. We do not know whether the zero in Corollary 4.6 is unique, or where the most likely counts of a reducible face lie. For a facet of three or more states, the law of the counts near the facet when T^2/N is of order one is open. Theorem 6.12 has an error of order one there, and a statement would need a discrete form of Theorem 6.8. We do not know whether $\mathfrak{r}_\beta = 0$ forces $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$ in Proposition 6.2.

Acknowledgments. The question grew out of Problem 131 of Peter Kagey's Open Problem Collection [28], and I thank him for posing it. I thank Chase Hameetman and Aidan Deshong for conversations about some of the proofs of the preprint [39].

A note on the use of AI

I wrote all of the proofs and text of this version. Apart from the addition described below, I developed its general theory myself and used Claude Opus 5.5 (Anthropic) only to check it and inform me of any mistakes.

The preprint [39] describes how AI was used for that version. I wrote its Sections 2 and 3 by hand. The model then checked the manuscript and drafted rewordings, which I reviewed and edited, and I also checked its proofs with GPT-6 Astra (OpenAI). Of the results that version credits to the model, one is reproved here: the tree form of the constants in its Proposition 3.6, which is Proposition 5.2(b) here and Lemma 7.1 of the arXiv version [40].

The model reviewed the manuscript in several referee-style passes and a proof-reading pass, ran numerical experiments that aided the revision, and searched the literature.

I wrote the verification scripts and the Lean files described in Section 9. The model checked the scripts and fixed errors in the Lean code. I take responsibility for all claims, proofs and computations.

Statements and Declarations

- **Funding.** No funding was received for this work.
- **Competing interests.** The author has no competing interests to declare.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** The outputs of the computations described in Section 9 are in the folder kagey-problems/problem131/paperC/data of the repository named under Code availability, each headed by the sha256 of the script that wrote it.

- **Materials availability.** Not applicable.
- **Code availability.** All code, data and Lean files supporting this paper are in the folder kagey-problems/problem131 of https://github.com/apemm/Kagey-Problems, with the scripts in the subfolder paperC and the Lean files in lean. The release `paperC1-arxiv-v2` of the repository is archived at https://doi.org/10.5281/zenodo.23661434. Section 9 says how to run them. The scripts need Python 3 with `numpy`, `scipy`, `sympy` and `mpmath`, and the option `--full` adds the largest cases of four of them.
- **Use of AI tools.** See the note on the use of AI just before these declarations.
- **Author contribution.** I am the sole author and am responsible for the whole paper.

Appendix A Proofs of the local law and of the first-order bounds

We prove Proposition 5.2(a), Lemma 4.1, Theorem 5.4, Lemma 5.5, Lemma A.5, Theorem 4.2(ii) and Lemma A.6. Except in Appendix A.7, F is a face with $k \geq 2$ states, numbered $1, \dots, k$. For $x \in \mathbb{C}^F$ we write $x' = (x_2, \dots, x_k)$. We write i for the imaginary unit, since i denotes a state, and $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. We use two standard facts [3]. First, let $B \geq 0$ be square and S diagonal with positive diagonal. Then $\varrho(B-S) < 0$ if and only if $\varrho(S^{-1}B) < 1$, and also if and only if $(S-B)v > 0$ for some $v > 0$. In that case $(S-B)^{-1} = \sum_{j\geq 0}(S^{-1}B)^j S^{-1} \geq 0$. Second, if M is an irreducible Metzler matrix, then every nonnegative eigenvector of M is positive and belongs to $\varrho(M)$, and $\varrho(M)$ is larger than the Perron root of each proper principal submatrix of M (Lemma 3.2).

A.1 The Perron root under a diagonal perturbation

An irreducible generator on F is a Metzler matrix $\widetilde{Q}$ with $\widetilde{Q}\mathbf{1} = 0$ whose graph is strongly connected. It has a unique stationary law $\pi > 0$. The argument after Definition 5.1, with $(\widetilde{Q}, \pi)$ in place of $(\widehat{Q}_\alpha, \alpha)$, shows that $Z = (\mathbf{1}\pi^\top - \widetilde{Q})^{-1} - \mathbf{1}\pi^\top$ exists and satisfies (8). Put $D = \operatorname{diag}\pi$ and $\Gamma = DZ + Z^\top D$.

Lemma A.1 (the Perron root) *Let $\widetilde{Q}$, π, Z and Γ be as above. Then $\kappa(\eta) = \varrho(\widetilde{Q} + \operatorname{diag}\eta)$ is real-analytic near 0, with $\kappa(0) = 0$, $\nabla\kappa(0) = \pi$ and $\nabla^2\kappa(0) = \Gamma$. For $\eta \in \mathbb{R}^F$ and $u = Z\eta$,*

$$\eta^\top \Gamma \eta = \sum_{i \neq j} \pi_i \widetilde{Q}_{ij} (u_i - u_j)^2. \tag{A1}$$

So Γ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$, and the matrix Σ obtained from Γ by deleting the row and column of state 1 is positive definite.

Proof We differentiate the eigenvalue equation twice at $\eta = 0$. Since $\widetilde{Q} + \operatorname{diag}\eta$ is irreducible, its Perron root is a simple eigenvalue. So by the implicit function theorem, κ and the right eigenvector $v(\eta)$ with $\pi^\top v = 1$ are real-analytic near 0, with $\kappa(0) = 0$ and $v(0) = \mathbf{1}$. We write subscripts for derivatives at $\eta = 0$. Differentiating $(\widetilde{Q} + \operatorname{diag}\eta)v = \kappa v$ in η_j gives $\widetilde{Q}v_j + e_j =$

$\kappa_j \mathbf{1}$, and multiplying by $\pi^\top$ gives $\kappa_j = \pi_j$, since $\pi^\top \widetilde{Q} = 0$. Then $v_j = Ze_j$, because by (8) both vectors solve $\widetilde{Q}x = \pi_j \mathbf{1} - e_j$ with $\pi^\top x = 0$, and the kernel of $\widetilde{Q}$ is $\mathbb{R}\mathbf{1}$. Differentiating again, in η_i, gives $\widetilde{Q}v_{ij} + \operatorname{diag}(e_i)v_j + \operatorname{diag}(e_j)v_i = \kappa_{ij}\mathbf{1} + \kappa_i v_j + \kappa_j v_i$. We multiply by $\pi^\top$ again. The terms with $\widetilde{Q}$, κ_i and κ_j vanish, and we get $\kappa_{ij} = \pi_i Z_{ij} + \pi_j Z_{ji} = \Gamma_{ij}$.

For (A1) let $\bar{\eta} = \eta - (\pi \cdot \eta)\mathbf{1}$. By (8), $-\widetilde{Q}u = \bar{\eta}$ and $\pi \cdot u = 0$, so $\eta^\top \Gamma \eta = 2\sum_i \pi_i \eta_i u_i = 2\sum_i \pi_i \bar{\eta}_i u_i = -2\sum_i \pi_i u_i (\widetilde{Q}u)_i$. Now expand the squares in (A1). By $\widetilde{Q}\mathbf{1} = 0$ and $\pi^\top \widetilde{Q} = 0$ the terms with u_i^2 and with u_j^2 each sum to $-\sum_i \pi_i \widetilde{Q}_{ii} u_i^2$, and the cross terms sum to $-2\sum_i \pi_i u_i (\widetilde{Q}u)_i + 2\sum_i \pi_i \widetilde{Q}_{ii} u_i^2$. So both sides of (A1) agree. Also $\Gamma\mathbf{1} = 0$ by (8). If $\eta^\top \Gamma \eta = 0$, then u is constant along every edge of the strongly connected graph of $\widetilde{Q}$, so u is constant. Then $u = 0$ because $\pi \cdot u = 0$, so $\bar{\eta} = 0$ and $\eta \in \mathbb{R}\mathbf{1}$. A nonzero vector that vanishes at state 1 is not in $\mathbb{R}\mathbf{1}$, which gives the claim about Σ. □

Proof of Proposition 5.2(a) Step 1. The derivatives of the Perron root. For $\vartheta \in \mathbb{R}^F$ let $\Lambda(\vartheta) = \varrho(Q_F + \operatorname{diag}\vartheta)$, so that $\Lambda(\vartheta + c\mathbf{1}) = \Lambda(\vartheta) + c$, which we call the shift property. The Perron root is a simple eigenvalue with positive right and left eigenvectors $\hat{r}, \hat{l}$, normalized by $\sum_i \hat{r}_i = 1$ and $\hat{l} \cdot \hat{r} = 1$, and by the implicit function theorem Λ, $\hat{r}$ and $\hat{l}$ are real-analytic in ϑ. Put $\pi = \hat{l} \circ \hat{r}$ and $\widetilde{Q} = \hat{R}^{-1}(Q_F + \operatorname{diag}\vartheta - \Lambda(\vartheta)I)\hat{R}$ with $\hat{R} = \operatorname{diag}\hat{r}$. Then $\widetilde{Q}\mathbf{1} = 0$ and $\pi^\top \widetilde{Q} = 0$ by the two eigenvector equations, so $\widetilde{Q}$ is an irreducible generator with stationary law π. Since $\hat{R}$ commutes with diagonal matrices, $\Lambda(\vartheta + \eta) = \Lambda(\vartheta) + \varrho(\widetilde{Q} + \operatorname{diag}\eta)$. So by Lemma A.1, $\nabla\Lambda(\vartheta) = \pi$ and $\nabla^2\Lambda(\vartheta) = \Gamma$, with Γ built from $(\widetilde{Q}, \pi)$.

Step 2. The gradient can be inverted. Let $\Lambda_0(\beta) = \Lambda(0, \beta)$ for $\beta \in \mathbb{R}^{k-1}$. Its Hessian is Γ without the row and column of state 1, which is positive definite by Lemma A.1. Hence Λ_0 is strictly convex, $\nabla\Lambda_0$ is injective, and by the inverse function theorem it has a real-analytic inverse on its open image.

Step 3. The tilt that belongs to α. Fix $\alpha \in \Delta_F^\circ$ and put $\vartheta(\alpha) = q - \theta(\alpha)$. Then $Q_F + \operatorname{diag}\vartheta(\alpha) = A_F - \operatorname{diag}\theta(\alpha)$ has the positive right and left eigenvectors r_α and l_α for the eigenvalue 0, by Definition 5.1 and (7). So its Perron root is 0, and at $\vartheta(\alpha)$ we have $\pi = \alpha$, $\widetilde{Q} = \widehat{Q}_\alpha$ and $\Gamma = \Gamma_\alpha$. Let $\beta(\alpha)$ be the last $k-1$ coordinates of $\vartheta(\alpha) - \vartheta_1(\alpha)\mathbf{1}$. A shift along $\mathbf{1}$ does not change the derivatives of Λ, so $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$ and $\nabla^2\Lambda_0(\beta(\alpha)) = \Sigma_\alpha$. Hence Σ_α is positive definite, and $\beta(\alpha) = (\nabla\Lambda_0)^{-1}(\alpha')$ is real-analytic. The shift property and $\Lambda(\vartheta(\alpha)) = 0$ give $\vartheta_1(\alpha) = -\Lambda_0(\beta(\alpha))$. So $\theta(\alpha)$ is real-analytic, and then so are its Perron vectors and Σ_α.

Step 4. The Hessian of J_F. Since $\vartheta(\alpha) = \vartheta_1(\alpha)\mathbf{1} + (0, \beta(\alpha))$ and $\sum_i \alpha_i = 1$, we have $J_F(\alpha) = \langle \vartheta(\alpha), \alpha \rangle = \langle \beta(\alpha), \alpha' \rangle - \Lambda_0(\beta(\alpha))$. We differentiate in α'. The terms with the derivative of β cancel because $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$, and $\nabla J_F = \beta$ remains. So $\nabla^2 J_F$ is the Jacobian of $(\nabla\Lambda_0)^{-1}$ at α', which is Σ_α^{-1}. The formula for C_F follows from Definition 5.1. □

A.2 The torus representation

The generating function of the words in H is Good's resolvent formula [23, (2)], and the torus integral below is his constant-term formula, here for the switching rates A_F of the killed chain and with the circle of one state integrated out. We split

$$A_F = \begin{pmatrix} 0 & a_1^\top \\ b_1 & A' \end{pmatrix}, \qquad \mu_F = (\mu_1, \mu'),$$

so $a_1 = (q_{1j})_{j\ge 2}$, $b_1 = (q_{i1})_{i\ge 2}$ and $A' = (q_{ij})_{i,j\ge 2}$. For $z' \in \mathbb{C}^{k-1}$ let $\mathcal{Z}' = \operatorname{diag}(z')$. Where $\mathcal{Z}' - A'$ is invertible we put $X(z') = (\mathcal{Z}' - A')^{-1}$, $g(z') = a_1^{\top} X(z') b_1$ and

$$b(z') = \bigl(\mu_1 + \mu'^{\top} X(z') b_1\bigr)\bigl(1 + a_1^{\top} X(z')\mathbf{1}\bigr).$$

Lemma A.2 (torus representation) *Let $s' \in (0,\infty)^{k-1}$ satisfy $\varrho(A' - \operatorname{diag} s') < 0$. On the set of $z' \in \mathbb{C}^{k-1}$ with $|z_i| \ge s_i$ for every $i \ge 2$, the matrix $\mathcal{Z}' - A'$ is invertible, and X, g and b are given by absolutely convergent power series in $1/z_2, \dots, 1/z_k$ with nonnegative coefficients. Hence $|g(z')| \le g(s')$ and $|b(z')| \le b(s')$ there. For every $t \in (0,\infty)^F$,*

$$H(t) = \frac{1}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \exp\Bigl(t_1 g(z') + \sum_{i\ge 2} t_i z_i\Bigr)\, b(z') \prod_{i\ge 2} z_i \, d\varphi, \qquad z_i = s_i e^{\mathrm{i}\varphi_i}.$$

Let moreover $h q_{\max} \le 1$, put $p_i = 1 - h q_i$, and let $n \in \mathbb{Z}_{\ge 1}^F$. Then the probability that $X_1, \dots, X_{|n|}$ have composition n is

$$P_{|n|}(n) = \frac{h^{k-1}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \bigl(p_1 + h g(z')\bigr)^{n_1 - 1} \prod_{i\ge 2} (p_i + h z_i)^{n_i - 1}\, b(z') \prod_{i\ge 2} z_i \, d\varphi.$$

Proof Step 1. The series. By the first fact, $\varrho(\operatorname{diag}(s')^{-1} A') < 1$. Hence the series $X(z') = \sum_{j\ge 0} (\mathcal{Z}'^{-1} A')^j \mathcal{Z}'^{-1}$ converges absolutely for $|z_i| \ge s_i$, since its terms are bounded entrywise in modulus by those at $z' = s'$. As a_1, b_1 and μ are nonnegative, this gives the claims about X, g and b.

Step 2. A radius for state 1. Choose $s_1 > g(s')$ and put $S = \operatorname{diag}(s_1, s')$. We show that $\varrho(S^{-1} A_F) < 1$. For $\varepsilon > 0$ let $v = (1, v')$ with $v' = X(s')(b_1 + \varepsilon \mathbf{1}) > 0$. Then $((S - A_F)v)_i = \varepsilon$ for $i \ge 2$, and $((S - A_F)v)_1 = s_1 - g(s') - \varepsilon a_1^{\top} X(s')\mathbf{1}$. For small ε both are positive, and the first fact gives the claim.

Step 3. The k circles. Let $\mathcal{Z} = \operatorname{diag}(z)$ with $|z_i| = s_i$ for every $i \in F$. The matrices $A_F S^{-1}$ and $S^{-1} A_F$ are similar, so $\varrho(A_F S^{-1}) = \varrho(S^{-1} A_F) < 1$, and expanding $(\mathcal{Z} - A_F)^{-1} = \mathcal{Z}^{-1}(I - A_F \mathcal{Z}^{-1})^{-1}$ gives

$$\mu_F^{\top} (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}} \prod_{i\in F} z_i^{-m_i(c)},$$

absolutely and uniformly on the torus. Here $c = c_1 \cdots c_M$ runs over the words in F with $M \ge 1$ and $c_j \ne c_{j+1}$, and $m_i(c)$ is the number of letters i in c. For an integer m and $t_i > 0$, Cauchy's formula gives $\frac{1}{2\pi}\int_0^{2\pi} e^{t_i z_i} z_i^{1-m}\, d\varphi_i = t_i^{m-1}/(m-1)!$ if $m \ge 1$ and 0 if $m \le 0$. We multiply the series by $e^{\langle t, z\rangle} \prod_i z_i$ and integrate term by term. The words that miss a state of F drop out, and grouping the others by $m(c)$ gives

$$H(t) = \frac{1}{(2\pi)^k} \int_{\mathbb{T}^k} e^{\langle t, z\rangle}\, \mu_F^{\top} (\mathcal{Z} - A_F)^{-1} \mathbf{1} \prod_{i\in F} z_i \, d\varphi. \tag{A2}$$

Step 4. The first circle. The Schur complement $z_1 - g(z')$ of $\mathcal{Z}' - A'$ in $\mathcal{Z} - A_F$ is not 0, since $|g(z')| \le g(s') < s_1$. Block inversion gives

$$\mu_F^{\top} (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \frac{b(z')}{z_1 - g(z')} + \mu'^{\top} X(z') \mathbf{1}.$$

We integrate over φ_1 in (A2) and divide by 2π. The second term does not depend on z_1, so it contributes $\frac{1}{2\pi}\int_0^{2\pi} e^{t_1 z_1} z_1 \, d\varphi_1 = 0$. In the first term $g(z')$ lies inside the circle $|z_1| = s_1$, so

Cauchy's formula gives $e^{t_1 g(z')} b(z')$. The factors without z_1 are unchanged, and this proves the formula for H.

The discrete formula. We repeat Steps 3 and 4 with $(p_i + hz_i)^{n_i - 1}$ in place of $e^{t_i z_i}$. The coefficient of z^{m-1} in the polynomial $(p+hz)^{n-1}$ is $h^{m-1}\binom{n-1}{m-1} p^{n-m}$, which is 0 for $m > n$. So $\frac{1}{2\pi}\int_0^{2\pi} (p_i + hz_i)^{n_i-1} z_i^{1-m}\, d\varphi_i$ is this coefficient if $m \geq 1$ and 0 if $m \leq 0$. The polynomial is bounded on the circle, so the term-by-term integration of Step 3 applies, and it gives

$$\frac{1}{(2\pi)^k} \int_{\mathbb{T}^k} \mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} \prod_{i \in F} (p_i + hz_i)^{n_i - 1} z_i \, d\varphi = \sum_m W(m) \prod_{i \in F} h^{m_i - 1} \binom{n_i - 1}{m_i - 1} p_i^{n_i - m_i}.$$

Since $h^{M-1} = h^{k-1} \prod_i h^{m_i - 1}$, this is $h^{-(k-1)} P_{|n|}(n)$ by Lemma 2.3 with $|n|$ in place of N. In Step 4 the second term gives $\frac{1}{2\pi}\int_0^{2\pi} (p_1 + hz_1)^{n_1 - 1} z_1 \, d\varphi_1 = 0$, and Cauchy's formula evaluates the polynomial $(p_1 + hz_1)^{n_1 - 1}$ at $z_1 = g(z')$. □

Section 6 needs the same formula for complex weights and an end vector other than $\mathbf{1}$. Let A be a complex matrix on F with zero diagonal and let $\nu, e \in \mathbb{C}^F$. Let $W_{A,\nu,e}(m)$ be the sum over the words c of $W(m)$ of $\nu_{c_1} \prod_{j<M} A_{c_j c_{j+1}} e_{c_M}$, and $H_{A,\nu,e}(t)$ the sum $H(t)$ with $W_{A,\nu,e}$ in place of W. So $H = H_{A_F,\mu,\mathbf{1}}$. We split A and ν as above and define X and g by the same formulas, and

$$b(z') = \big(\nu_1 + \nu'^\top X(z') b_1\big)\big(e_1 + a_1^\top X(z') e'\big).$$

Lemma A.3 (complex weights) *Let $\bar{A} \geq 0$ be a matrix on F with zero diagonal and $\bar{\nu}, \bar{e} \in [0,\infty)^F$ with $|A_{ij}| \leq \bar{A}_{ij}$, $|\nu_i| \leq \bar{\nu}_i$ and $|e_i| \leq \bar{e}_i$ for all i, j, and let $\bar{X}$, $\bar{g}$ and $\bar{b}$ be the functions above for $(\bar{A}, \bar{\nu}, \bar{e})$. Let $s' \in (0,\infty)^{k-1}$ satisfy $\varrho(\bar{A}' - \operatorname{diag} s') < 0$. Then the formula for H in Lemma A.2 holds for $H_{A,\nu,e}$ with these X, g and b, and their series no longer have nonnegative coefficients, but they are dominated term by term in modulus by those of $\bar{X}$, $\bar{g}$ and $\bar{b}$ at s'. In particular $|g(z')| \leq \bar{g}(s')$ and $|b(z')| \leq \bar{b}(s')$ when $|z_i| \geq s_i$ for every $i \geq 2$. Where $\mathcal{Z}' - A'$ is invertible, X, g and b are rational functions of z' and of the entries of A, ν and e.*

Proof We follow the four steps of the proof of Lemma A.2. In Step 1 the first fact gives $\varrho(\operatorname{diag}(s')^{-1}\bar{A}') < 1$, and for $|z_i| \geq s_i$ the entries of $\mathcal{Z}'^{-1}A'$ are bounded in modulus by those of $\operatorname{diag}(s')^{-1}\bar{A}'$. So the series of X converges absolutely, and its terms are dominated by those of $\bar{X}(s') \geq 0$. Since a_1, b_1, ν and e are dominated by their barred versions, the same holds for g and for the two factors of b. Step 2 applies to $\bar{A}$ as it stands and gives $s_1 > \bar{g}(s')$ with $\varrho(S^{-1}\bar{A}) < 1$. In Step 3 the entries of $A\mathcal{Z}^{-1}$ are bounded in modulus by those of $\bar{A}S^{-1}$, so the word expansion of $\nu^\top(\mathcal{Z} - A)^{-1} e$ converges absolutely and uniformly on the torus, and the same integration gives (A2) with A, ν and e in place of A_F, μ_F and $\mathbf{1}$. In Step 4 the Schur complement $z_1 - g(z')$ is not 0, since $|g(z')| \leq \bar{g}(s') < s_1$, and block inversion gives $\nu^\top(\mathcal{Z} - A)^{-1} e = b(z')/(z_1 - g(z')) + \nu'^\top X(z') e'$. The rest of Step 4 is unchanged. The last claim is Cramer's rule. □

A.3 Proof of Lemma 4.1

We bound the integrand of the discrete formula of Lemma A.2 by its modulus at the real point. Step 2 bounds the continuum sums in the same way, for later use.

Step 1. The radii. Let F be irreducible and $\theta \in V_F$. Then $A_F - \operatorname{diag}\theta$ is an irreducible Metzler matrix with Perron root 0. Hence it has a right eigenvector $u > 0$ for 0, and $\theta_i = (A_F u)_i / u_i$. Each row of A_F has a positive entry, since G_F is strongly connected and $k \geq 2$, so $\theta > 0$. By the second fact, $\varrho(A' - \operatorname{diag}\theta') < 0$. The rows $i \geq 2$ of $(A_F - \operatorname{diag}\theta)u = 0$ give $u' = u_1 X(\theta') b_1$, and then row 1, which reads $a_1^\top u' = \theta_1 u_1$, gives $\theta_1 = g(\theta')$. Hence

$$\theta > 0, \qquad \varrho(A' - \operatorname{diag}\theta') < 0, \qquad g(\theta') = \theta_1 \qquad \text{for every } \theta \in V_F. \tag{A3}$$

Step 2. Continuum sums. For $\sigma > 0$ let H_σ be H for σA_F in place of A_F. We use Lemma A.2 for σA_F with the radii $\sigma\theta'$, which is allowed since $\varrho(\sigma A' - \operatorname{diag}\sigma\theta') = \sigma\varrho(A' - \operatorname{diag}\theta') < 0$. The functions of Lemma A.2 for σA_F satisfy $X_\sigma(\sigma z') = \sigma^{-1} X(z')$, so $g_\sigma(\sigma z') = \sigma g(z')$ and $b_\sigma(\sigma z') = b(z')$. So on the torus with radii $\sigma\theta'$ we have $|g_\sigma| \leq \sigma g(\theta') = \sigma\theta_1$ by (A3), $|b_\sigma| \leq b(\theta')$ and $|z_i| = \sigma\theta_i$. Hence the integrand has modulus at most $e^{\sigma\langle t,\theta\rangle} b(\theta')\sigma^{k-1} \prod_{i\geq 2}\theta_i$, and for every $\sigma > 0$ and $t \in (0,\infty)^F$

$$H_\sigma(t) \leq \sigma^{k-1} b(\theta') \prod_{i\geq 2} \theta_i \, e^{\sigma\langle t,\theta\rangle}. \tag{A4}$$

Step 3. The bound. Let n have support F, let $hq_{\max} \leq 1$ and $p_i = 1 - hq_i$. We use the discrete formula of Lemma A.2 with the radii θ'. On that torus $|p_i + hz_i| \leq p_i + h\theta_i$ for $i \geq 2$, $|p_1 + hg(z')| \leq p_1 + h\theta_1$ since $|g(z')| \leq g(\theta') = \theta_1$, and $|b(z')| \leq b(\theta')$. So

$$P_{|n|}(n) \leq h^{k-1} b(\theta') \prod_{i\geq 2} \theta_i \prod_{i\in F} (p_i + h\theta_i)^{n_i - 1} \qquad (\theta \in V_F,\ hq_{\max} \leq 1). \tag{A5}$$

Now let $hq_{\max} \leq \frac{1}{2}$. Since $\log(1+x) \leq x$, we have $(p_i + h\theta_i)^{n_i} = (1 - h(q_i - \theta_i))^{n_i} \leq e^{-hn_i(q_i - \theta_i)}$, and $p_i + h\theta_i \geq p_i \geq \frac{1}{2}$. So the lemma holds with $C_\theta = 2^k b(\theta') \prod_{i\geq 2}\theta_i$. By Lemma A.2, b is continuous on the open set of s' with $\varrho(A' - \operatorname{diag} s') < 0$, which contains θ' for every $\theta \in V_F$ by (A3). So C_θ is bounded on compact subsets of V_F. □

A.4 Proof of Theorem 5.4(a)

The field cancels from the two sides of Theorem 5.4, so we take $v = 0$. For $k = 1$ the formula is part of Lemma 2.4. Let $k \geq 2$ and $\alpha \in \mathcal{K}$, and write $\theta = \theta(\alpha)$, $r = r_\alpha$, $l = l_\alpha$ and $\Theta' = \operatorname{diag}(\theta')$. By (A3) we may use Lemma A.2 with the radii θ' and $t = T\alpha$. With Lemma 2.4 this gives

$$f_T(\alpha) = \frac{T^{k-1} e^{-T\langle q,\alpha\rangle}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} e^{T\Psi(\alpha,\varphi)} \chi(\alpha,\varphi)\, d\varphi, \tag{A6}$$

where $z_i = \theta_i e^{\mathrm{i}\varphi_i}$, $\Psi(\alpha,\varphi) = \alpha_1 g(z') + \sum_{i\geq 2}\alpha_i z_i$ and $\chi(\alpha,\varphi) = b(z')\prod_{i\geq 2} z_i$. We check the hypotheses of Lemma B.1 with $m = k - 1$.

Hypothesis (iii). By (A3), $\Psi(\alpha,0)=\langle\alpha,\theta\rangle$ is real. By Lemma A.2, $|g(z')|\le g(\theta')=\theta_1$ on the torus, so

$$\operatorname{Re}\Psi(\alpha,\varphi)\le\Psi(\alpha,0)-\sum_{i\ge2}\alpha_i\theta_i(1-\cos\varphi_i).$$

All α_i and θ_i are positive, so the sum is positive unless $\varphi=0$.

Hypotheses (i) *and* (ii). We need the first two derivatives of g at θ'. We get them by differentiating a relation $\lambda(g(z'),z')=0$ for a Perron root λ, whose derivatives are known from Lemma A.1. For $z\in\mathbb{R}^F$ near θ let $\lambda(z)=\varrho(A_F-\operatorname{diag}z)$. With $R=\operatorname{diag}r$ we have $A_F-\operatorname{diag}(\theta-\eta)=R(\widehat{Q}_\alpha+\operatorname{diag}\eta)R^{-1}$, and $\widehat{Q}_\alpha$ is an irreducible generator with stationary law α. So $\lambda(\theta-\eta)=\varrho(\widehat{Q}_\alpha+\operatorname{diag}\eta)$, and Lemma A.1 gives $\nabla\lambda(\theta)=-\alpha$ and $\nabla^2\lambda(\theta)=\Gamma_\alpha$. For real z' near θ' we have $\varrho(A'-\operatorname{diag}z')<0$, so $(1,X(z')b_1)\ge0$ by the first fact. This vector lies in the kernel of $A_F-\operatorname{diag}(g(z'),z')$, in the rows $i\ge2$ by the definition of X and in row 1 by the definition of g. By the second fact a nonnegative eigenvector belongs to the Perron root, so $\lambda(g(z'),z')=0$.

Write g_j, g_{ij} for the derivatives of g at θ' and λ_j, λ_{ij} for those of λ at θ. Differentiating $\lambda(g(z'),z')=0$ in z_j gives $\lambda_1g_j+\lambda_j=0$, and differentiating again in z_i gives $\lambda_1g_{ij}+\lambda_{11}g_ig_j+\lambda_{1j}g_i+\lambda_{i1}g_j+\lambda_{ij}=0$. Put $w_j=e_j-(\alpha_j/\alpha_1)e_1\in\mathbb{R}^F$ for $j\ge2$. Then the first equation gives $g_j=-\alpha_j/\alpha_1$, so $w_j=e_j+g_je_1$, and the last four terms of the second equation are $w_i^\top\Gamma_\alpha w_j$. So

$$g_j=-\frac{\alpha_j}{\alpha_1},\qquad\alpha_1g_{ij}=w_i^\top\Gamma_\alpha w_j.$$

Hence $\partial\Psi/\partial z_j=\alpha_1g_j+\alpha_j=0$ at θ', and the Hessian $\mathcal{H}$ of Ψ in z' at θ' is $V^\top\Gamma_\alpha V$, where V has columns $w_2,\dots,w_k$. Since $\partial/\partial\varphi_j=\mathrm{i}z_j\,\partial/\partial z_j$ and the first derivatives vanish, $\nabla_\varphi\Psi(\alpha,0)=0$. We saw above that $\Psi(\alpha,0)$ is real, so this is hypothesis (i). Also $-\nabla^2_\varphi\Psi(\alpha,0)=\Theta'\mathcal{H}\Theta'$, which is real and symmetric, as (ii) requires.

For the rest of (ii), namely that this matrix is positive definite, let $B=I+\mathbf{1}\alpha'^\top/\alpha_1$. For $y\in\mathbb{R}^{k-1}$ the vector Vy has the entry $-\alpha'^\top y/\alpha_1$ at state 1 and the entries y elsewhere, so $Vy+(\alpha'^\top y/\alpha_1)\mathbf{1}=(0,By)$. Since $\Gamma_\alpha\mathbf{1}=0$, we may replace Vy by $(0,By)$ in $y^\top V^\top\Gamma_\alpha Vy$, which gives $\mathcal{H}=B^\top\Sigma_\alpha B$. The matrix determinant lemma gives $\det B=1+\alpha'^\top\mathbf{1}/\alpha_1=1/\alpha_1$. By Lemma A.1, Σ_α is positive definite, and hence so are $\mathcal{H}$ and $\Theta'\mathcal{H}\Theta'$. Also $\det\mathcal{H}=\det\Sigma_\alpha/\alpha_1^2$.

Regularity. By Proposition 5.2(a), $\theta(\alpha)$ is real-analytic. Fix $\alpha_0\in\mathcal{K}$. By (A3) and continuity there is $\delta\in(0,\min_i\theta_i(\alpha_0))$ with $\varrho(A'-\operatorname{diag}(\theta'(\alpha_0)-\delta\mathbf{1}))<0$. By Lemma A.2, X, g and b are holomorphic where $|z_i|>\theta_i(\alpha_0)-\delta$ for every $i\ge2$, and this holds on the torus of radii $\theta'(\alpha)$ for α near α_0. So Ψ and χ are real-analytic near $\{\alpha_0\}\times\mathbb{T}^{k-1}$. So their φ-derivatives are jointly continuous on $\mathcal{K}\times\mathbb{T}^{k-1}$, and covering $\mathcal{K}$ by finitely many such sets bounds them there.

The constant. The matrix $\mathcal{J} = \Theta'\mathcal{H}\Theta'$ of Lemma B.1 has $\det \mathcal{J} = \prod_{i\ge 2}\theta_i^2 \det \Sigma_\alpha/\alpha_1^2$. So the lemma gives, uniformly for $\alpha \in \mathcal{K}$ and $T \ge 1$,

$$\int_{\mathbb{T}^{k-1}} e^{T\Psi}\chi\, d\varphi = e^{T\langle \alpha,\theta\rangle}\Big(\frac{2\pi}{T}\Big)^{\frac{k-1}{2}} \frac{\alpha_1}{\prod_{i\ge 2}\theta_i\sqrt{\det \Sigma_\alpha}}\Big(\chi(\alpha,0) + O\Big(\frac{1}{T}\Big)\Big).$$

Here $\chi(\alpha,0) = b(\theta')\prod_{i\ge 2}\theta_i$. As in the derivation of (A3), $r' = r_1 X(\theta')b_1$. In the same way the columns $j \ge 2$ of $l^\top(A_F - \operatorname{diag}\theta) = 0$, which is (7), give $l' = l_1 X(\theta')^\top a_1$. Hence $\mu\cdot r = r_1(\mu_1 + \mu'^\top X(\theta')b_1)$ and $l\cdot \mathbf{1} = l_1(1 + a_1^\top X(\theta')\mathbf{1})$. Since $l_1 r_1 = \alpha_1$,

$$b(\theta') = \frac{(\mu\cdot r_\alpha)(l_\alpha\cdot \mathbf{1})}{\alpha_1}.$$

As $r_\alpha, l_\alpha > 0$ and $\mu(F) > 0$ by (H), $\chi(\alpha,0)$ is positive and continuous, hence bounded below on $\mathcal{K}$. So the last bracket is $\chi(\alpha,0)(1+O(1/T))$. Putting this into (A6) and using $J_F(\alpha) = \langle q-\theta,\alpha\rangle$ gives $f_T(\alpha) = T^{(k-1)/2}C_F(\alpha)e^{-TJ_F(\alpha)}(1+O(1/T))$, uniformly on $\mathcal{K}$. □

The proof uses A_F, μ and the end vector $\mathbf{1}$ only through their signs and the graph of A_F.

Lemma A.4 (general weights) *Let $A \ge 0$ be a matrix on F with zero diagonal and a strongly connected graph, let $\nu, e \in [0,\infty)^F$ be nonzero, and let $\theta(\alpha)$, r_α, l_α and Σ_α be the objects of Definition 5.1 for A in place of A_F. Let g and b be the functions of Lemma A.3 for (A,ν,e). Then the identities (A3), $g_j = -\alpha_j/\alpha_1$, $\mathcal{H} = B^\top\Sigma_\alpha B$, $\det\mathcal{H} = \det\Sigma_\alpha/\alpha_1^2$ and $\alpha_1 b(\theta') = (\nu\cdot r_\alpha)(l_\alpha\cdot e)$ of the proof above hold, and uniformly for α in a compact $\mathcal{K} \subset \Delta_F^\circ$ and $T \ge 1$*

$$H_{A,\nu,e}(T\alpha) = (2\pi T)^{-(k-1)/2} e^{T\langle\alpha,\theta(\alpha)\rangle}\,\frac{(\nu\cdot r_\alpha)(l_\alpha\cdot e)}{\sqrt{\det\Sigma_\alpha}}\,\big(1+O(1/T)\big).$$

Proof Definition 5.1, Proposition 5.2(a) and Lemma A.1 involve A_F only as a nonnegative matrix with zero diagonal and a strongly connected graph, so they hold for A. We repeat the proof above with Lemma A.3, for $(\bar A,\bar\nu,\bar e) = (A,\nu,e)$, in place of Lemma A.2. The paragraphs on the hypotheses and on the regularity do not change. In the paragraph on the constant the identities $r' = r_1X(\theta')b_1$ and $l' = l_1X(\theta')^\top a_1$ give $\nu\cdot r_\alpha = r_1(\nu_1 + \nu'^\top X(\theta')b_1)$ and $l_\alpha\cdot e = l_1(e_1 + a_1^\top X(\theta')e')$, and $l_1r_1 = \alpha_1$ gives the formula for $b(\theta')$. It is positive since $r_\alpha, l_\alpha > 0$ and $\nu, e \ge 0$ are nonzero. Then (A6) without the factor $T^{k-1}e^{-T\langle q,\alpha\rangle}$ and the Laplace lemma give the display. □

A.5 Proof of Lemma 5.5

The discrete law of Lemma 2.3 and the continuum density of Lemma 2.4 are sums over the same vectors m of run counts, and we compare them term by term.

Step 1. The ratio of the terms. Let $t = hn$ and $\alpha = n/N \in \mathcal{K}$. Lemma 2.4 gives the identity $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t) = \sum_m v_m$ with

$$v_m = W(m)\, h^{k-1} \prod_{i\in F} e^{-q_i t_i} \frac{t_i^{m_i-1}}{(m_i-1)!}, \qquad m \in \mathbb{Z}^F_{\geq 1}.$$

Since $h^{m_i-1}\binom{n_i-1}{m_i-1} = \frac{t_i^{m_i-1}}{(m_i-1)!}\prod_{1\leq j<m_i}(1-j/n_i)$, the m-th term of Lemma 2.3 is $\kappa_m v_m$ with

$$\kappa_m = \prod_{i\in F}\Big[\prod_{1\leq j<m_i}\Big(1-\frac{j}{n_i}\Big)\Big](1-hq_i)^{n_i-m_i}e^{hq_in_i}.$$

Let $w_m = v_m/\sum_{m'} v_{m'}$, a probability on the vectors m since $H(t) > 0$ under (H). We write $\mathbb{E}_w$ and $\mathbb{P}_w$ for it and regard $M = \sum_i m_i$ as a random variable. By Lemma 2.3 the ratio in the lemma is $\sum_m \kappa_m v_m/\sum_m v_m = \mathbb{E}_w\kappa_m$.

Step 2. Bounds for κ_m. Let $n_{\min} = \min_i n_i$, and let N be so large that $hq_{\max} \leq \frac{1}{2}$, which is possible since $h \leq N^{-1/2}$. We claim that for every m

$$\Big(1-\frac{M^2}{2n_{\min}}\Big)e^{-q_{\max}^2T^2/N} \leq \kappa_m \leq e^{hq_{\max}M}. \tag{A7}$$

First let $m_i \leq n_i$ for every i. The factors $1-j/n_i$ are at most 1, and $(1-hq_i)^{n_i-m_i} \leq e^{-hq_i(n_i-m_i)}$, so $\kappa_m \leq e^{hq_{\max}M}$. For the lower bound, $\log(1-x) \geq -x-x^2$ for $0 \leq x \leq \frac{1}{2}$ gives $(1-hq_i)^{n_i-m_i}e^{hq_in_i} \geq e^{-n_ih^2q_i^2}$, and the product of these over i is at least $e^{-q_{\max}^2T^2/N}$. The factors $1-j/n_i$ lie in $[0,1]$, so their product is at least 1 minus the sum of the j/n_i, and hence at least $1-\sum_i m_i^2/(2n_i) \geq 1-M^2/(2n_{\min})$. Now let $m_i > n_i$ for some i. Then the m-th term of Lemma 2.3 is 0, so $\kappa_m = 0$, while $M \geq n_{\min}+1$ gives $M^2/(2n_{\min}) \geq 2$, so the left side of (A7) is negative.

By (A7) it remains to bound $\mathbb{E}_w e^{hq_{\max}M}$ and $\mathbb{E}_w M^2$ from above.

Step 3. The number of runs under w. Since W for σA_F is $\sigma^{M-1}W(m)$, we have $\mathbb{E}_w\sigma^{M-1} = H_\sigma(t)/H(t)$. So the function $\Upsilon(c) = \log \mathbb{E}_w e^{c(M-1)} = \log H_{e^c}(t) - \log H(t)$ is finite. It is convex with $\Upsilon(0) = 0$, as the logarithm of a moment generating function, and $\Upsilon(1) \geq 0$ since $M \geq 1$. So $\Upsilon(c) \leq c\,\Upsilon(1)$ for $0 \leq c \leq 1$. We show that $\Upsilon(1) \leq C_{\mathcal{K}}T$, with an upper bound for H_e and a lower bound for H at $t = T\alpha$. By (A4) with $\sigma = e$ and $\theta = \theta(\alpha)$, $H_e(T\alpha) \leq e^{k-1}b(\theta')\prod_{i\geq 2}\theta_i\, e^{eT\langle\alpha,\theta\rangle}$. For the lower bound, $H(T\alpha)e^{-T\langle\alpha,\theta\rangle} = T^{-(k-1)}e^{TJ_F(\alpha)}f_T(\alpha)$, by the definition of f_T and $J_F(\alpha) = \langle q-\theta,\alpha\rangle$. By Theorem 5.4(a) this is at least $\frac{1}{2}T^{-(k-1)/2}C_F(\alpha)$ for $T \geq T_1$, and for $1 \leq T \leq T_1$ it is continuous and positive on $[1,T_1]\times\mathcal{K}$. So in both cases $H(T\alpha) \geq c_{\mathcal{K}}T^{-(k-1)/2}e^{T\langle\alpha,\theta\rangle}$. Since $\theta(\alpha)$ and $b(\theta'(\alpha))$ are bounded on $\mathcal{K}$, dividing the two bounds gives $\Upsilon(1) \leq C_{\mathcal{K}}T$.

Step 4. The two averages. Let $x = hq_{\max} \leq \frac{1}{2}$. Then $\mathbb{E}_w e^{xM} = e^{x+\Upsilon(x)} \leq e^{x(1+C_{\mathcal{K}}T)} = 1+O(T^2/N)$, because $x \leq q_{\max}T/N$ and $T^2/N \leq 1$. For the second moment we use the Chernoff bound $\mathbb{P}_w(M-1 \geq y) \leq e^{\Upsilon(1)-y}$ for $y \geq \Upsilon(1)$ and the bound 1 for $y < \Upsilon(1)$. This gives

$$\mathbb{E}_w(M-1)^2 = \int_0^\infty 2y\,\mathbb{P}_w(M-1 \geq y)\,dy$$

$$\le \Upsilon(1)^2 + \int_{\Upsilon(1)}^{\infty} 2ye^{\Upsilon(1)-y}\,dy = \Upsilon(1)^2 + 2\Upsilon(1) + 2,$$

so $\mathbb{E}_w M^2 = O(T^2)$. Finally $n_{\min} \ge \varepsilon_{\mathcal{K}} N$ with $\varepsilon_{\mathcal{K}} = \min_{\alpha\in\mathcal{K}} \min_i \alpha_i > 0$. So by (A7), $\mathbb{E}_w \kappa_m$ lies between $e^{-q_{\max}^2 T^2/N}(1 - \mathbb{E}_w M^2/(2\varepsilon_{\mathcal{K}} N))$ and $\mathbb{E}_w e^{hq_{\max} M}$, and both are $1 + O(T^2/N)$, uniformly for $\alpha \in \mathcal{K}$ and $1 \le T \le N^{1/2}$. □

A.6 The derivative in T

Theorem 5.8(iv) needs the derivative of the local law in T. For fixed N and n, $P_N(n)$ is a polynomial in T, and we differentiate it at fixed N and n.

Lemma A.5 (the derivative in T) *Let F satisfy* (H) *with $k \ge 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There are T_1, N_0 and C such that for $N \ge N_0$, $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ and $T_1 \le T \le N^{1/3}$,*

$$\Big|\partial_T \log P_N(n) + J_F(n/N)\Big| \le \frac{C}{T}.$$

Proof Put $\alpha = n/N$ and $t = hn$. Let $\tilde{w}$ be the probability on the vectors m proportional to the terms of Lemma 2.3. Since $T\partial_T = h\partial_h$ and the m-th term has the logarithmic derivative $(M-1) - \sum_i (n_i - m_i) hq_i/(1-hq_i)$ in $\log h$,

$$T\partial_T \log P_N(n) = \mathbb{E}_{\tilde{w}}(M-1) - \sum_{i\in F} \frac{hq_i}{1-hq_i}\,\mathbb{E}_{\tilde{w}}(n_i - m_i).$$

For $c \in \mathbb{R}$ let $P^{(c)}$ be the sum of Lemma 2.3 with $e^c A_F$ in place of A_F, that is with $e^{c(M-1)}W(m)$ in place of $W(m)$ and the same factors $(1-hq_i)^{n_i-m_i}$. Then $\Upsilon(c) = \log P^{(c)} - \log P^{(0)} = \log \mathbb{E}_{\tilde{w}} e^{c(M-1)}$ is convex, and $\Upsilon'(0) = \mathbb{E}_{\tilde{w}}(M-1)$.

Step 1. The asymptotics of Υ. The proof of Lemma 5.5 uses A_F only through the non-negative weights $W(m)$ and through bounds on its continuum sum H. For $e^c A_F$ this sum is $H_{e^c}(t) = e^{c(k-1)} H(e^c t)$, since a word with M letters has $M-1$ switches and $M-k$ is the total degree of $\prod_i t_i^{m_i - 1}$. Let $|c| \le 1$ and $T \ge e$. Step 3 of that proof needs the ratio $H_{e^{c+1}}(t)/H_{e^c}(t) = e^{k-1} H(e^{c+1}t)/H(e^c t)$ to be at most $e^{C_{\mathcal{K}} T}$. This follows from (A4) and from the lower bound for H at time $e^c T \ge 1$ in that step. So Lemma 5.5 holds for the weights $e^c A_F$, uniformly for $|c| \le 1$, and gives $P^{(c)} = h^{k-1} e^{-\langle q,t\rangle} e^{c(k-1)} H(e^c t)(1 + O(T^2/N))$. By Lemma 2.4 and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$, Theorem 5.4(a) at time $e^c T$ reads $H(e^c T\alpha) = (e^c T)^{-(k-1)/2} C_F(\alpha) e^{e^c T\langle\theta(\alpha),\alpha\rangle}(1 + O(1/T))$. Hence, uniformly for $|c| \le 1$, $\alpha \in \mathcal{K}$ and $e \le T \le N^{1/3}$,

$$\Upsilon(c) = a(c) + \varepsilon(c), \qquad a(c) = \frac{k-1}{2}\,c + (e^c - 1)\,T\langle\theta(\alpha),\alpha\rangle, \qquad |\varepsilon(c)| \le \frac{C_1}{T},$$

where we used $T^2/N \le 1/T$.

Step 2. Convexity. We have $a'(0) = \frac{k-1}{2} + T\langle\theta(\alpha),\alpha\rangle$ and $0 \le a'' \le C_2 T$ on $[-1,1]$, since θ is bounded on $\mathcal{K}$. Let $\eta = 1/T$. By convexity, $(\Upsilon(0) - \Upsilon(-\eta))/\eta \le \Upsilon'(0) \le (\Upsilon(\eta) - \Upsilon(0))/\eta$. By Taylor's theorem both quotients differ from $a'(0)$ by at most $C_2 T\eta/2 + 2C_1/(T\eta)$, which is $C_2/2 + 2C_1$. So $\mathbb{E}_{\tilde{w}}(M-1) = T\langle\theta(\alpha),\alpha\rangle + O(1)$, and in particular $\mathbb{E}_{\tilde{w}} M = O(T)$.

Step 3. The other term. Since $hn_i = T\alpha_i$ and $hq_{\max} \le \frac{1}{2}$, $\sum_i hq_i n_i/(1-hq_i) = T\langle q,\alpha\rangle + O(Th)$ and $0 \le \sum_i hq_i \mathbb{E}_{\tilde{w}} m_i/(1-hq_i) \le 2hq_{\max}\mathbb{E}_{\tilde{w}} M = O(T^2/N)$. Altogether $T\partial_T \log P_N(n) = T\langle\theta(\alpha),\alpha\rangle - T\langle q,\alpha\rangle + O(1) = -TJ_F(\alpha) + O(1)$, and we divide by T. □

A.7 Proof of Theorem 4.2(ii)

Let F be admissible, with components $C_1, \dots, C_r$. If F is irreducible, then $r = 1$, and the argument below uses only Lemma 4.1 and an explicit family of paths, not the local law.

For the upper bound we apply Lemma 4.1 to every block of a path. For the lower bound a component of least exit rate gets almost all the time, and every other component C gets only about N/T steps, the time scale of a single switch. Such a short block has probability bounded below and only about $(N/T)^{|C|-1}$ compositions instead of $N^{|C|-1}$.

If $Q = 0$, only singletons with positive start mass are admissible, and the bounds follow from $M_{\{i\}} = \mu_i$. Otherwise let a be the least positive rate q_{ij}. Since $T \le N^{1/3}$, we have $h \le N^{-2/3}$ and $T^2/N \le 1$, and we take N_0 so large that h is as small as we need. As in the proof of Lemma 2.2, a path of positive probability with support F visits each component in one block of consecutive times, and it passes from one block to the next by one switch. We call the order of the blocks, the first state and these switches the pattern of the path. There are finitely many patterns.

Upper bound. Fix a pattern whose j-th block lies in the component D_j and starts at y_j, and whose switch out of block j is $x_j \to y_{j+1}$ for $j < r$. So block j ends at x_j for $j < r$, and the pattern does not fix the end of the last block. Let $\operatorname{supp} n = F$. Block j has length $N_j = \sum_{i \in D_j} n_i$. By the Markov property the paths with this pattern and composition n have total probability $\mu_{y_1} \prod_{j<r} h q_{x_j y_{j+1}} \prod_{j \le r} \varpi_j$. Here ϖ_j is the probability that the chain started at y_j stays in D_j for N_j visits, with the composition that n prescribes on D_j, and for $j < r$ also ends at x_j. We drop the condition on the end. Then ϖ_j is at most the probability that $X_1, \dots, X_{N_j}$ have that composition for the chain started at y_j. If $|D_j| \ge 2$, Lemma 4.1 with start y_j and $\theta = q - \lambda_{D_j}\mathbf{1}$ gives $\varpi_j \le \kappa_1 h^{|D_j|-1} e^{-h\lambda_{D_j} N_j}$, where κ_1 is the largest of these finitely many constants. If $D_j = \{i\}$, then $\lambda_{D_j} = q_i$ and $\varpi_j = (1 - hq_i)^{N_j - 1} \le e^{1/2 - h\lambda_{D_j} N_j}$. We multiply over j. The powers of h add up to $\sum_j (|D_j| - 1) + r - 1 = k_F - 1$. In the exponent, $\lambda_{D_j} \ge \lambda_F$ and $\sum_j N_j = N$ give $h \sum_j \lambda_{D_j} N_j \ge T\lambda_F$. Summing over the patterns gives $P_N(n) \le \kappa_2 (T/N)^{k_F - 1} e^{-T\lambda_F}$.

Lower bound. We write down a family of paths with support F, bound its probability from below and count its compositions. By Lemma 2.2(a) we can number the components so that there are $y_1 \in C_1$ with $\mu_{y_1} > 0$ and $x_j \in C_j$, $y_{j+1} \in C_{j+1}$ with $q_{x_j y_{j+1}} > 0$ for $j < r$. Take any $x_r \in C_r$, choose j^* with $\lambda_{C_{j^*}} = \lambda_F$ and $|C_{j^*}| = k^*$, and put $m = \lceil N/T \rceil$, so that $1 \le hm \le 2$.

Claim A. There is $c_A > 0$ such that for every component C and all $y, x \in C$, the chain started at y stays in C for its first m visits, visits every state of C and has $X_m = x$ with probability at least c_A. Take a walk $y = v_0, \dots, v_J = x$ through every state of C with $q_{v_s v_{s+1}} > 0$ and $J \le |C|^2$. Consider the paths $X_1, \dots, X_m$ that make exactly these J switches, in this order, at any J of the $m - 1$ transition times. There are $\binom{m-1}{J}$ of them, and each has probability at least $(ha)^J (1 - hq_{\max})^m$. For large N we have $m > |C|^2 \ge J$. Then $\binom{m-1}{J} h^J \ge (hm - Jh)^J / J! \ge (1 - Jh)^J / J!$ and $(1 - hq_{\max})^m \ge e^{-2hq_{\max} m} \ge e^{-4q_{\max}}$, since $1 \le hm \le 2$. So their total probability is at least $(1 - Jh)^J a^J e^{-4q_{\max}} / J!$, and $(1 - Jh)^J \ge 2^{-J}$ for small h.

Claim B. There is $c_B > 0$ such that for all $y, x \in C_{j^*}$ and $\ell \geq 2m$, the event of Claim A, with C_{j^*} in place of C and ℓ visits in place of m, has probability at least $c_B e^{-h\lambda_F \ell - h^2 \lambda_F^2 \ell}$. Let $w > 0$ satisfy $Q_{C_{j^*}} w = -\lambda_F w$ and let $B = I + hQ_{C_{j^*}} \geq 0$. Then $(B^s)_{zz'}$ is the probability of going from z to z' in s steps inside C_{j^*}. Since $B^s w = (1-h\lambda_F)^s w$, the probability of staying in C_{j^*} for s steps from z is $\sum_{z'} (B^s)_{zz'} \geq (B^s w)_z / \max w \geq (1 - h\lambda_F)^s \min w / \max w$. We use Claim A for the first m visits, from y back to y, then $s = \ell - 2m + 1$ steps inside C_{j^*} to any state z', then Claim A for the last m visits, from z' to x. By the Markov property this gives at least $c_A^2 (\min w / \max w)(1 - h\lambda_F)^\ell$. Also $\lambda_F \leq q_{\max}$, because the Perron root of $Q_{C_{j^*}}$ is at least each diagonal entry. So $h\lambda_F \leq \frac{1}{2}$ and $\log(1 - h\lambda_F) \geq -h\lambda_F - h^2\lambda_F^2$.

The family of paths. Give block $j \neq j^*$ the length m and block j^* the length $\ell = N - (r-1)m$, which is at least $2m$ for $T \geq T_0 = r+2$ and $N \geq N_0$. Consider the paths that go from y_j to x_j in block j as in the claims and pass from block j to block $j+1$ by the switch $x_j \to y_{j+1}$. They all have support F. By the Markov property, and since $h\ell \leq T$ and $h^2 \ell \leq T^2/N \leq 1$, their total probability is at least $\mu_{y_1}(ha)^{r-1} c_A^{r-1} c_B e^{-T\lambda_F - q_{\max}^2}$. The composition of such a path is fixed by its restrictions to the blocks. On the block in C_j the restriction is a vector of $|C_j|$ positive integers whose sum is m for $j \neq j^*$ and ℓ for $j = j^*$. A vector of positive integers with a given sum is fixed by all but one of its entries, and each entry is at most the sum. Also $\ell \leq N$. So these paths have at most $(N+1)^{k^*-1} \prod_{j \neq j^*} (m+1)^{|C_j|-1}$ compositions, and $m+1 \leq 3N/T$. Since $\sum_{j \neq j^*} (|C_j| - 1) = k_F - k^* - (r-1)$ and $h = T/N$, dividing the total probability by the number of compositions gives $M_F \geq c\, N^{-(k_F-1)} T^{k_F - k^*} e^{-T\lambda_F}$. □

A.8 The local law over the whole switching range

Here F is irreducible with $k \geq 2$ states, h_* is fixed with $h_* q_{\max} < 1$, $p_i = 1 - hq_i$ and $p_{\min} = 1 - h_* q_{\max} > 0$. Let Ω be the open set of $s' \in (0, \infty)^{k-1}$ with $\varrho(A' - \operatorname{diag} s') < 0$, and for $s' \in \Omega$ put $s_1 = g(s')$ and

$$\Psi_h(\alpha; s') = -\sum_{i \in F} \alpha_i L_h(q_i - s_i),$$

with L_h from Section 3.3. For $h > 0$ this is $\frac{1}{h} \sum_i \alpha_i \log(p_i + hs_i)$.

Lemma A.6 (the discrete Hessian) *Let $(h, \alpha) \in [0, h_*] \times \Delta_F^\circ$, and let $\theta = \theta^{(h)}(\alpha)$, r, l and $c_h = c_h(\alpha)$ be as in the definition of $C_F^{(h)}$ in Section 5.2. Put $\pi = l \circ r$.*

(a) The point θ' is the only minimizer of $\Psi_h(\alpha; \cdot)$ on Ω, $\Psi_h(\alpha; \theta') = -J_F^{(h)}(\alpha)$, and the Hessian H_h of $\Psi_h(\alpha; \cdot)$ at θ' is positive definite.

(b) There is a $(k-1) \times (k-1)$ matrix K with $\det K = 1/(c_h \pi_1 \prod_{i \in F}(p_i + h\theta_i))$ and $\nabla^2 J_F^{(h)}(\alpha) = K^\top H_h^{-1} K$. So $\nabla^2 J_F^{(h)}(\alpha)$ is positive definite.

(c) $C_F^{(h)}(\alpha) = b(\theta') / \big((2\pi)^{(k-1)/2} \prod_{i \in F}(p_i + h\theta_i) \sqrt{\det H_h}\big)$, and $C_F^{(h)}$ is positive and real-analytic on $[0, h_*] \times \Delta_F^\circ$ up to $h = 0$.

Proof (a) The map $\psi \mapsto \theta(\psi)'$ is a real-analytic diffeomorphism from $\{\psi \in \mathbb{R}^F : \psi_1 = 0\}$ onto Ω. Indeed $\theta(\psi) = A_F e^{\psi}/e^{\psi}$ lies in V_F, so $\theta(\psi)' \in \Omega$ and $\theta_1(\psi) = g(\theta(\psi)')$ by (A3). Conversely, for $s' \in \Omega$ the vector $u = (1, X(s')b_1)$ is nonnegative by the first fact, and it is a null vector of $A_F - \operatorname{diag}(g(s'), s')$, as in the proof of Theorem 5.4(a). By the second fact it is positive, and $\psi = \log u$ has $\psi_1 = 0$ and $\theta(\psi) = (g(s'), s')$. Both maps are real-analytic. Since $\Psi_h(\alpha; \theta(\psi)') = -G_h(\alpha, \psi)$, part (a) follows from Proposition 3.5(b). The first derivatives vanish at the critical point, so there the two Hessians are congruent through the Jacobian of the diffeomorphism, which is invertible.

(b) Write $y_i(s') = L_h(q_i - s_i)$ with $s_1 = g(s')$, so that, in the coordinates α', $-\Psi_h = y_1 + \sum_{j \geq 2} \alpha_j (y_j - y_1)$. By (a), $J_F^{(h)}(\alpha) = -\Psi_h(\alpha; \theta^{(h)}(\alpha)')$, and $\theta^{(h)}(\alpha)'$ is real-analytic in α. So the envelope theorem gives $\partial J_F^{(h)}/\partial \alpha_j = y_j(\theta') - y_1(\theta')$ for $j \geq 2$. Let $K_{ij} = -\partial(y_j - y_1)/\partial s_i$ at θ', for $i, j \geq 2$. Differentiating $\nabla_{s'} \Psi_h(\alpha; \theta^{(h)}(\alpha)') = 0$ in α' gives $H_h \, \partial\theta'/\partial\alpha' = -K$, and hence $\nabla^2 J_F^{(h)} = -K^\top \partial\theta'/\partial\alpha' = K^\top H_h^{-1} K$.

To compute K put $e_i = L_h'(q_i - \theta_i) = 1/(p_i + h\theta_i)$, and let g_i be the derivatives of g at θ'. The vector $\psi = \log r$ attains the supremum defining $J_F(\pi)$ by the criterion in the proof of Proposition 3.3, since π is the stationary law of $R^{-1}(A_F - \operatorname{diag}\theta)R$ with $R = \operatorname{diag} r$. So $\theta = \theta(\pi)$, and the proof of Theorem 5.4(a) at the point π gives $g_i = -\pi_i/\pi_1$. Hence $K_{ij} = \delta_{ij} e_j + e_1 \pi_i/\pi_1$, that is $K = \operatorname{diag}(e') + (e_1/\pi_1)\pi' \mathbf{1}^\top$, and the matrix determinant lemma gives $\det K = \prod_{i \in F} e_i \cdot \sum_{i \in F} \pi_i (p_i + h\theta_i)/\pi_1$. By Proposition 3.5(c) with $\psi = \log r$, $\alpha_i = c\, \pi_i (p_i + h\theta_i)$ for a constant c. Summing $\alpha_i/(p_i + h\theta_i)$ over i gives $c = c_h$, and summing α_i gives $\sum_i \pi_i (p_i + h\theta_i) = 1/c_h$. This is the value of $\det K$, which is not 0, so $\nabla^2 J_F^{(h)}(\alpha)$ is positive definite by (a).

(c) The identity $\alpha_1 b(\theta') = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})$ of the proof of Theorem 5.4(a), at the point π, gives $b(\theta') = (\mu \cdot r)(l \cdot \mathbf{1})/\pi_1$, since $l \cdot r = 1$. By (b), $\sqrt{\det \nabla^2 J_F^{(h)}} = |\det K|/\sqrt{\det H_h}$. Putting both into the definition of $C_F^{(h)}$ gives the formula, which is positive. By Proposition 3.5(b), $\theta^{(h)}(\alpha)$ and $J_F^{(h)}$ are real-analytic up to $h = 0$, and so are c_h and the normalized null vectors r and l, since 0 is a simple eigenvalue of $A_F - \operatorname{diag}\theta$. As $\det \nabla^2 J_F^{(h)} > 0$, $C_F^{(h)}$ is real-analytic up to $h = 0$. □

Proof of Theorem 5.4(b) The field cancels from the two sides, so we take $v = 0$. Let $\mathcal{K}$ be as in the theorem and put $\mathcal{P} = [0, h_*] \times \mathcal{K}$. For $(h, \alpha) \in \mathcal{P}$ let $\theta = \theta^{(h)}(\alpha)$ and $\Theta' = \operatorname{diag}(\theta')$.

The integral. Let $1 \leq T \leq h_* N$, $\operatorname{supp} n = F$ and $\alpha = n/N \in \mathcal{K}$, so that $h = T/N \in (0, h_*]$ and $n_i = T\alpha_i/h$. By (A3) we may use the discrete formula of Lemma A.2 with the radii θ'. It gives $P_N(n) = h^{k-1}(2\pi)^{-(k-1)} \int_{\mathbb{T}^{k-1}} F_n \, d\varphi$, where F_n is the integrand. With $z_i = \theta_i e^{\mathrm{i}\varphi_i}$ and $z_1 = g(z')$ put

$$\Psi_h(\alpha, \varphi) = -\sum_{i \in F} \alpha_i L_h(q_i - z_i), \qquad a_h(\alpha, \varphi) = \frac{b(z') \prod_{i \geq 2} z_i}{\prod_{i \in F} (p_i + h z_i)},$$

where $\operatorname{Re}(p_i + h z_i) > 0$ for every i. There $F_n = e^{T\Psi_h} a_h$, since $(p_i + h z_i)^{n_i} = \exp(n_i \log(p_i + h z_i))$ with the principal logarithm and $n_i \log(p_i + h z_i) = -T\alpha_i L_h(q_i - z_i)$. The logarithm has no continuous branch on the whole torus when $h\theta_i > p_i$, so we use these formulas only near $\varphi = 0$.

Regularity on a cube. The map $(h, \alpha) \mapsto \theta^{(h)}(\alpha)$ is continuous on the compact set $\mathcal{P}$, so $\theta_i \in [\theta_{\min}, \theta_{\max}]$ with $\theta_{\min} > 0$, and $\varrho(A' - \operatorname{diag}\theta')$ is bounded above by a negative constant. As in the regularity paragraph of the proof of (a), X, g and b are holomorphic near the tori of radii θ', uniformly on $\mathcal{P}$, and $g(z')$ is uniformly continuous in (h, α, φ). So there is

$\varepsilon_1 \in (0,1)$ such that $|g(z') - \theta_1| \le \theta_{\min}$ when $\|\varphi\| \le \varepsilon_1$. Then $\operatorname{Re} g(z') \ge 0$ and $\operatorname{Re} z_i > 0$ for $i \ge 2$, so $\operatorname{Re}(p_i + hz_i) \ge p_{\min}$ for every i. Since $1 - shy = (1-s) + s(1-hy)$, the function $L_h(y) = \int_0^1 y\,ds/(1-shy)$ is holomorphic in y and real-analytic in h where $\operatorname{Re}(1-hy) > 0$, including $h = 0$. So Ψ_h and a_h are defined and real-analytic on $\mathcal{P} \times \{\|\varphi\| \le \varepsilon_1\}$, and their φ-derivatives up to orders 4 and 2 are bounded there by constants B_Ψ and B_a. At $h = 0$ they are $\Psi - \langle q, \alpha\rangle$ and χ of (A6).

Hypotheses (i) *and* (ii). At $\varphi = 0$, $\Psi_h(\alpha, 0) = \Psi_h(\alpha; \theta') = -J_F^{(h)}(\alpha)$ by Lemma A.6(a), which is real. Since $\partial/\partial\varphi_j = \mathrm{i} z_j\,\partial/\partial z_j$ and Ψ_h is holomorphic in z' with the derivatives of the real function $\Psi_h(\alpha;\cdot)$ at θ', Lemma A.6(a) gives $\nabla_\varphi \Psi_h(\alpha, 0) = 0$ and $-\nabla_\varphi^2 \Psi_h(\alpha, 0) = \Theta' H_h \Theta'$. This matrix is real, symmetric and positive definite, and it is continuous on $\mathcal{P}$, so its least eigenvalue is at least some $\sigma_{\min} > 0$ on $\mathcal{P}$.

The gap. Choose $\varepsilon \in (0, \varepsilon_1]$ with $(k-1)^3 B_\Psi \varepsilon \le \frac{3}{2}\sigma_{\min}$. For $i \ge 2$,

$$|p_i + h\theta_i e^{\mathrm{i}\varphi_i}|^2 = (p_i + h\theta_i)^2 - 2p_i h\theta_i(1 - \cos\varphi_i).$$

With $\log(1-x) \le -x$, and with $n_i - 1 \ge n_i/2 = T\alpha_i/(2h)$, which holds for $N \ge N_0$ since then $n_i \ge 2$, this gives $|p_i + hz_i|^{n_i - 1} \le (p_i + h\theta_i)^{n_i-1} e^{-c_* T(1-\cos\varphi_i)}$ with $c_* = \min_{\mathcal{K}} \min_i \alpha_i\, p_{\min}\theta_{\min}/(2(1 + h_*\theta_{\max})^2) > 0$. The h in n_i cancels the h in $p_i h\theta_i$, so c_* does not depend on h. Also $|p_1 + hg(z')| \le p_1 + h\theta_1$ and $|b(z')| \le b(\theta')$ by Lemma A.2. If $\|\varphi\| \ge \varepsilon$, then $1 - \cos\varphi_i \ge 1 - \cos\varepsilon$ for some $i \ge 2$, so

$$|F_n(\varphi)| \le e^{-\eta T} F_n(0) = e^{-\eta T} e^{T\Psi_h(\alpha,0)} a_h(\alpha, 0), \qquad \eta = c_*(1 - \cos\varepsilon),$$

and $a_h(\alpha, 0)$ is bounded on $\mathcal{P}$.

The expansion. By Remark B.2 in its localized form, with $m = k - 1$, uniformly in (h, α) and $T \ge 1$,

$$\int_{\mathbb{T}^{k-1}} F_n\, d\varphi = e^{-TJ_F^{(h)}(\alpha)} \Big(\frac{2\pi}{T}\Big)^{\frac{k-1}{2}} \det(\Theta' H_h \Theta')^{-1/2} \Big(a_h(\alpha, 0) + O\Big(\frac{1}{T}\Big)\Big).$$

Here $a_h(\alpha, 0) = b(\theta') \prod_{i \ge 2} \theta_i / \prod_i (p_i + h\theta_i)$ is positive and continuous on $\mathcal{P}$, so the bracket is $a_h(\alpha, 0)(1 + O(1/T))$. Since $\det(\Theta' H_h \Theta') = \prod_{i \ge 2} \theta_i^2 \det H_h$ and $h^{k-1} T^{-(k-1)/2} = N^{-(k-1)} T^{(k-1)/2}$, Lemma A.6(c) gives the display of (b) with a factor $1 + E$, $|E| \le C/T$.

Bounded T. It remains to bound $1 + E$ below for $1 \le T \le T_1 = 2C$. Let $N \ge T_1^2$. Then $T \le N^{1/2}$, and Lemma 5.5 and part (a) give $P_N(n) = N^{-(k-1)} f_T(\alpha)(1 + O(T_1^2/N))$. The ratio of $f_T(\alpha)$ to $T^{(k-1)/2} C_F(\alpha) e^{-TJ_F(\alpha)}$ is continuous and positive on $[1, T_1] \times \mathcal{K}$, and the ratio of $C_F^{(h)}(\alpha) e^{-TJ_F^{(h)}(\alpha)}$ to $C_F(\alpha) e^{-TJ_F(\alpha)}$ is $1 + O(T_1 h)$ by Proposition 3.5(b) and Lemma A.6(c). So for large N, $1 + E$ lies between two positive constants for $1 \le T \le T_1$, and for $T \ge T_1$ it lies in $[\frac{1}{2}, \frac{3}{2}]$. □

Appendix B A Laplace lemma with a parameter

We prove the Laplace lemma that Appendix A uses for Theorem 5.4, and its form with a complex Hessian that Section 6 uses for Theorem 6.8. The compact parameter space $\mathcal{K}$ can include the rates and the start when constants are needed uniformly over a class of chains. On $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$ we use $\|\varphi\| = \max_i \operatorname{dist}(\varphi_i, 2\pi\mathbb{Z})$. For $x \in \mathbb{R}^m$ we write $\|x\| = \max_i |x_i|$ and $|x|$ for the Euclidean norm. Repeated indices are summed from 1 to m.

Lemma B.1 (Laplace integrals with a parameter) *Let $\mathcal{K}$ be a compact metric space and $m \geq 1$. Let $\Psi, a : \mathcal{K} \times \mathbb{T}^m \to \mathbb{C}$ be such that the φ-derivatives of Ψ up to order 4 and of a up to order 2 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Assume that for every $\alpha \in \mathcal{K}$*

(i) the value $\Psi(\alpha, 0)$ is real and $\nabla_\varphi \Psi(\alpha, 0) = 0$,

(ii) the matrix $\mathcal{J}(\alpha) = -\nabla_\varphi^2 \Psi(\alpha, 0)$ is real, symmetric and positive definite, and

(iii) the inequality $\operatorname{Re} \Psi(\alpha, \varphi) < \Psi(\alpha, 0)$ holds for every $\varphi \neq 0$ in $\mathbb{T}^m$.

Then, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha,\varphi)} a(\alpha, \varphi)\, d\varphi = e^{T\Psi(\alpha,0)} \Big(\frac{2\pi}{T}\Big)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \Big(a(\alpha, 0) + O\Big(\frac{1}{T}\Big)\Big).$$

Proof By compactness $B_\Psi = \max_{|\beta|\leq 4} \sup |\partial_\varphi^\beta \Psi|$ and $B_a = \max_{|\beta|\leq 2} \sup |\partial_\varphi^\beta a|$ are finite, the least eigenvalue of $\mathcal{J}(\alpha)$ has a positive minimum $\sigma_{\min}$ over $\mathcal{K}$, and $\det \mathcal{J}(\alpha) \leq (mB_\Psi)^m$. Below, C depends only on m, $\sigma_{\min}$, B_Ψ, B_a and the numbers ε and η chosen below, and it may change from line to line. Hence every bound below is uniform on $\mathcal{K}$.

Localization. Choose $\varepsilon \in (0, 1)$ with $m^3 B_\Psi \varepsilon \leq \frac{3}{2}\sigma_{\min}$. The set of (α, φ) with $\|\varphi\| \geq \varepsilon$ is compact, and by (iii) the continuous function $\operatorname{Re} \Psi(\alpha, \varphi) - \Psi(\alpha, 0)$ is negative on it. Hence it is at most $-\eta$ there for some $\eta > 0$, and the integral over this set has modulus at most $(2\pi)^m B_a e^{T(\Psi(\alpha,0)-\eta)} \leq CT^{-1-m/2} e^{T\Psi(\alpha,0)}$.

Scaling. Reduction modulo 2π identifies the cube $\|x\| \leq \varepsilon$ in $\mathbb{R}^m$ with the set $\|\varphi\| \leq \varepsilon$, and it preserves measure. Let $R(\alpha, \varphi) = \Psi(\alpha, \varphi) - \Psi(\alpha, 0) + \frac{1}{2}\varphi^\top \mathcal{J}(\alpha)\varphi$. By (i) and (ii), R and its derivatives of order 1 and 2 vanish at $\varphi = 0$. Taylor's theorem with integral remainder, which holds for complex values, gives on the cube $R = \frac{1}{6}\Psi_{ijk}\varphi^i\varphi^j\varphi^k + \omega_0$ with $|\omega_0| \leq C\|\varphi\|^4$, where Ψ_{ijk} are the third derivatives at $(\alpha, 0)$. It also gives $|R| \leq \frac{m^3}{6} B_\Psi \|\varphi\|^3 \leq \frac{1}{4}\sigma_{\min}|\varphi|^2$, since $\sum_{|\beta|=3} 1/\beta! = m^3/6$ and $\|\varphi\|^3 \leq \varepsilon|\varphi|^2$. We substitute $\varphi = x/\sqrt{T}$ and put $w(x) = TR(\alpha, x/\sqrt{T})$. The integral over the cube is

$$T^{-m/2} e^{T\Psi(\alpha,0)} \int_{\|x\| \leq \varepsilon\sqrt{T}} e^{-\frac{1}{2}x^\top \mathcal{J} x}\, e^{w(x)} a(\alpha, x/\sqrt{T})\, dx,$$

and $|w(x)| \leq \frac{1}{4}\sigma_{\min}|x|^2$.

Taylor expansion. Let $P(x) = \frac{1}{6}\Psi_{ijk}x^i x^j x^k$, $a_0 = a(\alpha, 0)$ and $a_i = \partial_{\varphi_i} a(\alpha, 0)$. Then $w = T^{-1/2}P + \omega_1$ with $|\omega_1| \leq CT^{-1}|x|^4$, and $|w| \leq CT^{-1/2}|x|^3$. Since $|e^w - 1 - w| \leq \frac{1}{2}|w|^2 e^{|w|}$, this gives $|e^w - 1 - T^{-1/2}P| \leq CT^{-1}(|x|^4 + |x|^6)e^{\sigma_{\min}|x|^2/4}$. Taylor's theorem for a to orders 1 and 2 gives $|a(\alpha, x/\sqrt{T}) - a_0| \leq CT^{-1/2}|x|$ and $|a(\alpha, x/\sqrt{T}) - a_0 - T^{-1/2}a_i x^i| \leq CT^{-1}|x|^2$. Multiplying out,

$$e^w a(\alpha, x/\sqrt{T}) = a_0 + T^{-1/2}\big(a_0 P(x) + a_i x^i\big) + \omega(x), \qquad |\omega(x)| \leq CT^{-1}(1 + |x|)^6 e^{\sigma_{\min}|x|^2/4}.$$

Integration. Since $x^\top \mathcal{J} x \geq \sigma_{\min}|x|^2$, the term ω contributes at most CT^{-1} to the integral, because $(1 + |x|)^6 e^{-\sigma_{\min}|x|^2/4}$ is integrable. Outside the cube $|x| > \varepsilon\sqrt{T}$, so extending the integrals of the other two terms to $\mathbb{R}^m$ changes them by at most $Ce^{-\sigma_{\min}\varepsilon^2 T/4} \leq CT^{-1}$. On $\mathbb{R}^m$ we have $\int e^{-\frac{1}{2}x^\top \mathcal{J} x} dx = (2\pi)^{m/2} \det \mathcal{J}^{-1/2}$, and the odd polynomial $a_0 P(x) + a_i x^i$ integrates to 0 against this weight. This is why the error is $O(1/T)$ and not $O(T^{-1/2})$. So the integral over the cube is $e^{T\Psi(\alpha,0)} T^{-m/2} \big[(2\pi)^{m/2} \det \mathcal{J}^{-1/2} a_0 + O(T^{-1})\big]$. Since $\det \mathcal{J} \leq (mB_\Psi)^m$, this error and the one from the localization are $O(1/T)$ relative to $e^{T\Psi(\alpha,0)}(2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. □

Remark B.2 (the uniform form) The proof of Lemma B.1 uses compactness and joint continuity only to obtain the bounds B_Ψ and B_a on the derivatives, the lower bound $\sigma_{\min}$ on the least eigenvalue of $\mathcal{J}$, and a gap $\operatorname{Re}\Psi(\alpha,\varphi) - \Psi(\alpha,0) \le -\eta$ for $\|\varphi\| \ge \varepsilon$, where $\varepsilon \in (0,1)$ satisfies $m^3 B_\Psi \varepsilon \le \frac{3}{2}\sigma_{\min}$. So the following holds with the same proof. Let $\mathcal{P}$ be any set, and let $\Psi, a : \mathcal{P} \times \mathbb{T}^m \to \mathbb{C}$ satisfy (i) and (ii) for every $\alpha \in \mathcal{P}$, with φ-derivatives of Ψ up to order 4 and of a up to order 2 bounded by B_Ψ and B_a on $\mathcal{P} \times \mathbb{T}^m$, with $\mathcal{J}(\alpha) \ge \sigma_{\min} I$ for all α, and with the gap above for all α. Then the expansion of Lemma B.1 holds uniformly for $\alpha \in \mathcal{P}$ and $T \ge 1$, with a constant that depends only on m, B_Ψ, B_a, $\sigma_{\min}$, ε and η.

The proof uses the integrand off the cube $\|\varphi\| \le \varepsilon$ only in the localization, through a bound on its modulus, and it uses the bounds on the derivatives only on the cube. So the expansion also holds in the following localized form, with a constant that depends in addition on a number B. Let Ψ and a be defined only on $\mathcal{P} \times \{\|\varphi\| \le \varepsilon\}$, satisfy (i) and (ii), have φ-derivatives bounded by B_Ψ and B_a there, with $\mathcal{J}(\alpha) \ge \sigma_{\min} I$ and $m^3 B_\Psi \varepsilon \le \frac{3}{2}\sigma_{\min}$, and let the integrand be $e^{T\Psi} a$ on the cube and any measurable function of modulus at most $B e^{T(\Psi(\alpha,0)-\eta)}$ off it. Appendix A.8 needs this form, since there the logarithm in the exponent has no branch on the whole torus.

Section 6 needs the same expansion when $\Psi(\alpha,0)$ and the Hessian are complex. Let $\mathcal{S}_m$ be the set of complex symmetric $m \times m$ matrices M whose real part is positive definite. For $M \in \mathcal{S}_m$ and $x \in \mathbb{C}^m$ with $x \ne 0$ we have $\operatorname{Re}(\bar{x}^\top M x) = \bar{x}^\top (\operatorname{Re} M) x > 0$, because $\bar{x}^\top P x$ is real for a real symmetric P. So $\det M \ne 0$. The set $\mathcal{S}_m$ is convex, so $1/\det M$ has exactly two continuous square roots on it. We write $\det M^{-1/2}$ for the one that is positive on the real matrices of $\mathcal{S}_m$. Then

$$\int_{\mathbb{R}^m} e^{-\frac{1}{2} x^\top M x}\, dx = (2\pi)^{m/2} \det M^{-1/2} \qquad (M \in \mathcal{S}_m).$$

Indeed both sides are holomorphic functions of the entries M_{jk}, $j \le k$, on the open set $\mathcal{S}_m$. For the left side we may differentiate under the integral, since $|e^{-x^\top M x/2}| = e^{-x^\top (\operatorname{Re} M) x/2}$ is bounded by a fixed Gaussian for M near a point of $\mathcal{S}_m$. The right side is a continuous square root of a holomorphic function without zeros. The two sides agree on the real positive definite matrices, which form a nonempty open subset of the real points of $\mathcal{S}_m$. A holomorphic function on a connected open set that vanishes on such a subset vanishes everywhere, since its Taylor coefficients at a real point are real derivatives. So the two sides agree on $\mathcal{S}_m$.

Lemma B.3 (a complex Hessian) *Let $\mathcal{P}$ be a set and $m \ge 1$. Let $\Psi, a : \mathcal{P} \times \mathbb{T}^m \to \mathbb{C}$ have φ-derivatives up to order 4 and 2 that are continuous in φ and bounded in modulus by B_Ψ and B_a on $\mathcal{P} \times \mathbb{T}^m$. Let $\sigma_{\min} > 0$, $\eta > 0$ and $\varepsilon \in (0,1)$ with $m^3 B_\Psi \varepsilon \le \frac{3}{2}\sigma_{\min}$. Assume that for every $p \in \mathcal{P}$*

(i) $\nabla_\varphi \Psi(p,0) = 0$,

(ii) the complex symmetric matrix $\mathcal{J}(p) = -\nabla_\varphi^2 \Psi(p,0)$ *satisfies* $\operatorname{Re}\mathcal{J}(p) \ge \sigma_{\min} I$, *and*

(iii) $\operatorname{Re}\Psi(p,\varphi) - \operatorname{Re}\Psi(p,0) \le -\eta$ *for* $\|\varphi\| \ge \varepsilon$.

Then the expansion of Lemma B.1 holds uniformly for $p \in \mathcal{P}$ and $T \ge 1$, with the branch of $\det \mathcal{J}^{-1/2}$ defined above and a constant that depends only on m, B_Ψ, B_a, $\sigma_{\min}$, ε and η.

Proof The proof of Lemma B.1 in the form of Remark B.2 goes through with three changes. First, the factor $e^{T\Psi(p,0)}$ is now complex, and the localization uses only its modulus $e^{T\operatorname{Re}\Psi(p,0)}$. By (iii) the integral over $\|\varphi\| \geq \varepsilon$ has modulus at most $(2\pi)^m B_a e^{-\eta T} |e^{T\Psi(p,0)}|$. Second, for real x the number $x^\top(\operatorname{Im}\mathcal{J})x$ is real, so $|e^{-x^\top \mathcal{J}x/2}| = e^{-x^\top(\operatorname{Re}\mathcal{J})x/2} \leq e^{-\sigma_{\min}|x|^2/2}$. This is the only bound on the Gaussian weight that the integration step uses, and the weight is still even and integrable, so the odd polynomial still integrates to 0. Third, the Gaussian integral over $\mathbb{R}^m$ is $(2\pi)^{m/2} \det \mathcal{J}^{-1/2}$ by the identity above, and Hadamard's inequality gives $|\det \mathcal{J}| \leq (mB_\Psi)^m$, so the errors are $O(1/T)$ relative to the main term. The scaling and the Taylor expansion use only the bounds on the derivatives, Taylor's theorem for complex values and $|w(x)| \leq \frac{1}{4}\sigma_{\min}|x|^2$, and they do not change. □

Appendix E Proofs for reducible faces and for the counts near a facet

E.1 Blocks of a reducible face

In this subsection and the next, F is admissible, the notation is that of Section 4.4, and $hq_{\max} \leq \frac{1}{2}$. For an irreducible C, states $y, x \in C$ and $m \in \mathbb{Z}_{\geq 1}^C$ with $|m| = l$, let $G_C(m)_{yx}$ be the probability that the chain with $X_1 = y$ has composition m in $X_1, \dots, X_l$ and $X_l = x$. So $G_{\{i\}}(m) = (1 - hq_i)^{m-1}$. For n with support F let $n^{(j)}$ be n restricted to C_j, $N_j = |n^{(j)}|$, $s_j = hN_j$ and $G_j = G_{C_j}(n^{(j)})$. The blocks of a path with composition n occupy fixed times, a path that starts outside C_1 never visits C_1, and the Markov property at the $r-1$ switches gives

$$P_N(n) = \mu_{C_1}^\top G_1 \, (hR_1) \, G_2 \, (hR_2) \cdots (hR_{r-1}) \, G_r \, \mathbf{1}.$$

The bridge run expansion. Let $|C| = k \geq 2$ and write $W_{yx} = W_{A_C, e_y, e_x}$. The proof of Lemma 2.3, with the first and the last letter of the words fixed, gives $G_C(m)_{yx} = \sum_{m'} W_{yx}(m') h^{M'-1} \prod_i \binom{m_i - 1}{m_i' - 1} (1 - hq_i)^{m_i - m_i'}$. As in Step 1 of the proof of Lemma 5.5, with $t = hm$ the m'-th term is $\kappa_{m'} v_{m'}$, where $v_{m'} = W_{yx}(m') h^{k-1} \prod_i e^{-q_i t_i} t_i^{m_i' - 1}/(m_i' - 1)!$, so that $\sum_{m'} v_{m'} = h^{k-1} \Phi_C(t)_{yx}$, and $\kappa_{m'}$ is the factor of that proof. Its argument for (A7) gives

$$1 - \sum_i \frac{m_i'(m_i' - 1)}{2m_i} - q_{\max}^2 h|t| \leq \kappa_{m'} \leq e^{hq_{\max} M'}.$$

Since $W_{yx}(m') \leq k^{M'} q_{\max}^{M'-1}$, the series of Φ_C converges for every $t \in \mathbb{R}^C$, and so do the series with the extra factors $M' e^{M'/2}$ or $m_i'(m_i' - 1) t_i^{-1}$ in their terms. So Φ_C is real-analytic on $\mathbb{R}^C$, and these series are bounded on bounded sets. Also $\Phi_C(t)_{yx} > 0$ for $t \in (0, \infty)^C$, since some word from y to x uses every state.

Lemma E.1 (one bound for every block) *There are N_0 and C_0 such that for $N \geq N_0$, $T \leq N^{1/3}$, every j, all $y, x \in C_j$ and every m with $\operatorname{supp} m = C_j$ and $s = h|m| \leq T$, we have $G_{C_j}(m)_{yx} \leq C_0 h^{k_j - 1} w_j(s)$, where $w_j(s) = (1 \vee s)^{-(k_j - 1)/2} e^{-\lambda_F s}$ if $\lambda_j = \lambda_F$ and $k_j \geq 2$, and $w_j(s) = e^{-\lambda_j s}$ otherwise.*

Proof We prove this in two cases. For $k_j = 1$, $(1-hq_i)^{|m|-1} \le e^{hq_i} e^{-q_i s}$. Let $k_j \ge 2$. The bridge is at most the probability that the chain started at y has composition m in its first $|m|$ visits. Lemma 4.1 with the start e_y and $\theta = q - \lambda_j \mathbf{1} \in V_{C_j}$ bounds it by $Ch^{k_j-1}e^{-\lambda_j s}$. If $\lambda_j = \lambda_F$ and $s \ge 1$, we apply Theorem 5.7(b) with $v = 0$ to the chain started at y on the face C_j, which satisfies (H), with $l = |m|$ steps and time s in place of N and T, and the same h. Its range holds, since $l = sN/T \ge N^{2/3}$ and $s^3 \le l$ is $s^2 T \le N$. It gives $G_{C_j}(m)_{yx} \le C' l^{-(k_j-1)} s^{(k_j-1)/2} e^{-\lambda_F s} = C' h^{k_j-1} s^{-(k_j-1)/2} e^{-\lambda_F s}$ with a constant C' for the start e_y. There are finitely many starts y. □

Lemma E.2 (blocks of bounded time) *For every R there is C_R such that $|G_C(m)_{yx}/h^{|C|-1} - \Phi_C(hm)_{yx}| \le C_R h$ for all $y, x \in C$ and $\operatorname{supp} m = C$ with $h|m| \le R$. Hence, if $D = \nu^\top \Psi_1 R_1' \Psi_2 \cdots R_{p-1}' \Psi_p w$ with fixed nonnegative ν, R_c' and w and $\Psi_c = G_{C_c}(m^{(c)})/h^{|C_c|-1}$, and D^{c} is the same product with $\Phi_{C_c}(hm^{(c)})$, then $|D - D^{\mathrm{c}}| \le C_R' h$ when $\sum_c h|m^{(c)}| \le R$.*

Proof For a singleton, $e^{hq}e^{-qt} \ge (1-hq)^{m-1} \ge e^{-qt-q^2ht}$ with $t = hm$. Let $|C| = k \ge 2$. By the bound on $\kappa_{m'}$, $\kappa_{m'} - 1 \le hq_{\max} M' e^{M'/2}$, which gives $G/h^{k-1} - \Phi_C \le C_R h$. Also $1 - \kappa_{m'} \le \sum_i m_i'(m_i'-1)/(2m_i) + q_{\max}^2 hR$, and since $1/m_i = h/t_i$, the sum of $v_{m'} m_i'(m_i'-1)/(2m_i)$ over m' is h^k times a series with the extra factor $m_i'(m_i'-1)/(2t_i)$, which is bounded on $|t| \le R$. This gives $\Phi_C - G/h^{k-1} \le C_R h$. For the product we telescope, since every factor is bounded on $|t| \le R$. □

Lemma E.3 (the bridge law) *Let $|C| = k \ge 2$ and let $\mathcal{K} \subset \Delta_C^\circ$ be compact. Uniformly for $y, x \in C$, m with $m/l \in \mathcal{K}$ and $1 \le s = hl \le l^{1/2}$, $G_C(m)_{yx} = h^{k-1}\Phi_C(hm)_{yx}(1 + O(s^2/l))$.*

Proof We prove this as we prove Lemma 5.5 in Appendix A, with the bridge run expansion in place of Lemma 2.3, and l and s in place of N and T. Its Step 3 uses the weights only through an upper bound on H for eA_C and a lower bound on H at $t = s\alpha$. Step 2 of the proof of Lemma 4.1, with Lemma A.3 for (A_C, e_y, e_x) in place of Lemma A.2, gives (A4) for H_{A_C,e_y,e_x} with b_{yx} in place of b, and $b_{yx}(\theta'(\alpha))$ is bounded on $\mathcal{K}$. Lemma A.4 with $(\nu, e) = (e_y, e_x)$ gives $H_{A_C,e_y,e_x}(s\alpha)e^{-s\langle\alpha,\theta(\alpha)\rangle} \ge cs^{-(k-1)/2}$ for $s \ge s_1$, and for $1 \le s \le s_1$ the left side is continuous and positive on $[1, s_1] \times \mathcal{K}$. □

Lemma E.4 (bridge maxima) *Let $|C| = k \ge 2$, $\lambda = \lambda_C$ and $\Sigma = \Sigma_C$, and put $Y_{yx} = h^{k-1} s^{-(k-1)/2} r_{C,y} l_{C,x} e^{-\lambda s}/((2\pi)^{(k-1)/2}\sqrt{\det \Sigma})$. There are s_2, l_2 and C_3 such that the following hold for $y, x \in C$, $l \ge l_2$ and $s = hl \in [s_2, l^{1/3}]$. (i) $G_C(m)_{yx} \le Y_{yx}(1 + C_3(1/s + s^2/l))$ for every m with $|m| = l$ and $\operatorname{supp} m = C$. (ii) $G_C(m^0)_{yx} \ge Y_{yx}(1 - C_3(1/s + s^2/l))$ if $|m^0| = l$ and $|m_i^0 - l\alpha_{C,i}^*| < 1$ for all i.*

Proof Put $c_{yx}(\alpha) = r_{\alpha,y} l_{\alpha,x}/((2\pi)^{(k-1)/2}\sqrt{\det \Sigma_\alpha})$, which is real-analytic and positive on Δ_C° by Proposition 5.2(a), with $c_{yx}(\alpha_C^*) = r_{C,y} l_{C,x}/((2\pi)^{(k-1)/2}\sqrt{\det \Sigma})$. Let $J^0 = J_C - \lambda$, which is nonnegative and vanishes only at α_C^* (Proposition 3.3(a)), and let ξ be the

displacement $\alpha' - \alpha_C^{*\prime}$. Since $\nabla^2 J_C = \Sigma_\alpha^{-1}$, Taylor's theorem gives $\rho_0, c_U, C_U, C_1 > 0$ such that $U = \{|\xi| \le \rho_0\} \subset \Delta_C^\circ$ and on U, for all y, x, $c_U|\xi|^2 \le J^0 \le C_U|\xi|^2$, $|\log(c_{yx}(\alpha)/c_{yx}(\alpha_C^*)) - g_{yx}\cdot\xi| \le C_1|\xi|^2$ with $|g_{yx}| \le C_1$, and $\log c_{yx}$ is C_1-Lipschitz.

Outside U. The function J^0 is lower semicontinuous and positive on the compact set $\mathcal{E} = \Delta_C \setminus \{|\xi| < \rho_0\}$, so $J^0 \ge 3\eta$ there for some $\eta > 0$. Since $J_C = \sup_{\theta\in V_C}\langle q-\theta,\cdot\rangle$ (Section 4), each point of $\mathcal{E}$ has a neighborhood on which one $\theta \in V_C$ gives $\langle q-\theta,\cdot\rangle > \lambda + 2\eta$, and finitely many $\theta_1, \dots, \theta_p$ cover $\mathcal{E}$. For $m/l \in \mathcal{E}$, the bridge is at most the probability that the chain started at y has composition m in its first l visits, and Lemma 4.1 bounds it by $Ch^{k-1}e^{-s(\lambda+2\eta)} = Y_{yx}\,O(s^{(k-1)/2}e^{-2\eta s}) = Y_{yx}\,O(1/s)$.

Inside U. Lemma E.3 and Lemma A.4 with $(\nu, e) = (e_y, e_x)$ at $t = s\alpha$ give $G_C(m)_{yx} = Y_{yx}(c_{yx}(\alpha)/c_{yx}(\alpha_C^*))e^{-sJ^0(\alpha)}(1+E)$ with $|E| \le C_2(1/s + s^2/l)$, where $\alpha = m/l$. For $s \ge 2C_1/c_U$ the logarithm of the two middle factors is at most $g_{yx}\cdot\xi - \frac{1}{2}c_U s|\xi|^2 \le |g_{yx}|^2/(2c_U s)$. With the bound outside U this proves (i). For (ii), $|\xi| \le k/l$ at m^0, so $c_{yx}(m^0/l) \ge c_{yx}(\alpha_C^*)e^{-kC_1/l}$ and $sJ^0(m^0/l) \le C_U k^2 s/l^2$. □

E.2 Proofs of Theorem 4.5 and Corollary 4.6

Proof of Theorem 4.5 Write $a = a_F$, $K = K_F^{\mathrm{red}}$ and $X = N^{-(k_F-1)}T^{p_F}e^{-T\lambda_F} = h^{k_F-1}T^{-a}e^{-T\lambda_F}$. For $j \in S^*$ let $F_j^<(t^<)$ and $F_j^>(t^>)$ be the functions inside the suprema that define A_j and B_j. Call a block flat if it is a singleton $\{i\}$ with $q_i = \lambda_F$. Flat blocks exist only if $k^* = 1$, and then they are the blocks in S^*. Put $\tilde{w}_m(u) = e^{\lambda_F u}w_m(u) \le 1$, with w_m from Lemma E.1, let $\delta_0 > 0$ be the least positive $\lambda_m - \lambda_F$ ($\delta_0 = 1$ if there is none), and $\eta(u) = \max(e^{-\delta_0 u}, u^{-1/2})$ for $u \ge 1$. A block that is not flat and has time $u \ge 1$ has $\tilde{w}_m(u) \le \eta(u)$.

The constant. Since Φ_C is real-analytic on $\mathbb{R}^C$, $F_j^<$ and $F_j^>$ are Lipschitz on bounded parts of the closed orthants, and the suprema over the open and the closed orthants agree. Fix $t \in (0,\infty)^C$, let $h \to 0$ and $m = \lceil t/h \rceil$. By Lemma E.2, $G_C(m)/h^{|C|-1} \to \Phi_C(t)$, and Lemma E.1 with $T = |t| + 1$ and $N = T/h$ gives $e^{\lambda_F|t|}\Phi_{C_m}(t) \le C_0\tilde{w}_m(|t|)$ entrywise. So every factor in $F_j^<$ and $F_j^>$ is bounded, and $A_j, B_j < \infty$. They are positive, since $\mu_{C_1} \ge 0$ and every $R_m \ge 0$ are nonzero, $r, l > 0$ and every $\Phi_{C_m} > 0$ on the open orthant. If $k^* \ge 2$, no factor is flat, so $F_j^<$ and $F_j^>$ tend to 0 at infinity, and the suprema are attained at points $t_*^<$ and $t_*^>$ of the closed orthants.

Upper bound. Let $\operatorname{supp} n = F$, so $\sum_m s_m = T$. By the factorization and Lemma E.1, summed over the finitely many switch states, and since $\sum_m(k_m - 1) + r - 1 = k_F - 1$,

$$P_N(n) \le C_1 h^{k_F-1}e^{-\lambda_F T}\prod_m \tilde{w}_m(s_m).$$

Let $k^* \ge 2$ and take m_0 with $s_{m_0} \ge T/r$. (A) If $m_0 \notin S^*$, then $\lambda_{m_0} > \lambda_F$, or $\lambda_{m_0} = \lambda_F$ and $k_{m_0} \ge k^* + 1$, so $P_N(n) \le C_1 X T^a \max(e^{-\delta_0 T/r}, (T/r)^{-a-1/2}) \le KX$ for $T \ge T_1$. (B1) If $m_0 = j \in S^*$ and $T - s_j \ge R_0$, another block has time at least R_0/r, and $\tilde{w}_j(s_j) \le (T/r)^{-a}$, so $P_N(n) \le C_1 r^a \eta(R_0/r)X \le KX$ once R_0 is fixed large. (B2) If $m_0 = j \in S^*$ and $T - s_j < R_0$, write $P_N(n) = \mathbf{a}^\top G_j \mathbf{b}$ with $\mathbf{a}^\top = \mu_{C_1}^\top G_1 h R_1 \cdots G_{j-1} h R_{j-1}$ and $\mathbf{b} = hR_j G_{j+1}\cdots G_r \mathbf{1}$, both nonnegative. Lemma E.4(i) applies with $l = N_j \ge N/2$ and $s = s_j \ge T - R_0$, since $s_j^3 \le N_j$ means $s_j^2 T \le N$, and $s_j^2/N_j = s_j T/N$. It gives

$$P_N(n) \le h^{k^*-1}s_j^{-a}e^{-\lambda_F s_j}\,\frac{(\mathbf{a}\cdot\mathbf{r})(l'\cdot\mathbf{b})}{(2\pi)^a\sqrt{\det\Sigma_{C_j}}}\Big(1 + C\Big(\frac{1}{T} + \frac{T^2}{N}\Big)\Big),$$

where $\mathbf{r} = r_{C_j}$ and $l' = l_{C_j}$. Let $\varkappa_< = \sum_{m<j}(k_m - 1) + j - 1$ be the number of factors h in $\mathbf{a}$. Then $\mathbf{a}\cdot\mathbf{r}/h^{\varkappa_<}$ is a product as in Lemma E.2 with total time $\rho_< < R_0$, whose

continuum version is at most $e^{-\lambda_F \rho_<} A_j$. So $e^{\lambda_F \rho_<} \mathbf{a} \cdot \mathbf{r} \le h^{\varkappa_<}(A_j + Ch)$, and likewise $e^{\lambda_F \rho_>} l' \cdot \mathbf{b} \le h^{\varkappa_>}(B_j + Ch)$. Since $\varkappa_< + \varkappa_> + k^* - 1 = k_F - 1$, $\rho_< + \rho_> + s_j = T$, $s_j^{-a} \le T^{-a}(1 + CR_0/T)$ and $h \le T^2/N$, we get $P_N(n) \le X(K + C(1/T + T^2/N))$.

Let $k^* = 1$, and let ρ be the total time of the blocks that are not flat. If $\rho \ge R_0$, one of them has time at least R_0/r, and $P_N(n) \le C_1 \eta (R_0/r) X \le KX$. If $\rho < R_0$, each flat block has $(1 - h\lambda_F)^{N_m - 1} \le e^{h\lambda_F} e^{-\lambda_F s_m}$, and the other blocks form products as in Lemma E.2 with total time below R_0. So $P_N(n) \le h^{k_F - 1} e^{-\lambda_F T} e^{rhq_{\max}} (\Psi^* + Ch)$, where Ψ^* is the supremum of $e^{\lambda_F |t|} \mu_{C_1}^\top \Phi_{C_1} R_1 \cdots \Phi_{C_r} \mathbf{1}$, in which every flat factor equals 1. At a flat $j \in S^*$ this product splits into the products of A_j and B_j, in disjoint variables, so $\Psi^* = A_j B_j = K$.

Lower bound. Let $k^* \ge 2$, and take $j \in S^*$ that attains the maximum in K and the points $t_*^<$ and $t_*^>$, with total time τ_*. For $m \ne j$ put $n^{(m)} = \max(1, \lceil t_*^{(m)}/h \rceil)$ entrywise. Every entry of $hn^{(m)}$ is within h of that of $t_*^{(m)}$, so $F_j^<(hn^<) \ge A_j - Ch$ and $F_j^>(hn^>) \ge B_j - Ch$. Give block j the length $l = N - \sum_{m \ne j} N_m$ and a composition m^0 with $|m_y^0 - l\alpha^*_{C_j,y}| < 1$, which has full support for large N, so that $s = hl \in [T - \tau_* - Ch, T]$. By Lemma E.2, $\mathbf{a} \ge h^{\varkappa_<}(\mathbf{a}^{\mathrm{c}} - Ch)$ entrywise, with $e^{\lambda_F |hn^<|} \mathbf{a}^{\mathrm{c}} \cdot \mathbf{r} = F_j^<(hn^<)$, and likewise for $\mathbf{b}$. Lemma E.4(ii) and nonnegativity give $P_N(n) \ge h^{k_F - 1} s^{-a} e^{-\lambda_F T} (A_j - Ch)(B_j - Ch)((2\pi)^a \sqrt{\det \Sigma_{C_j}})^{-1}(1 - C(1/T + T^2/N))$, and $s^{-a} \ge T^{-a}$. So $M_F \ge X(K - C(1/T + T^2/N))$. Let $k^* = 1$. The supremum Ψ^* is attained at a point of the closed orthants. Indeed, with the flat factors equal to 1, every other factor is at most $C_0 \tilde{w}_m \le C_0$, and a factor whose time grows tends to 0, since it is not flat. So the function tends to 0 at infinity. We take a flat $j \in S^*$, treat the blocks that are not flat in the same way at such a point, give every other flat block the length 1, and use $(1 - h\lambda_F)^{l-1} \ge e^{-\lambda_F s}(1 - CT^2/N)$ for block j. □

Proof of Corollary 4.6 Theorem 4.5 for F and G gives, uniformly on $[T_1, N^{1/3}]$,

$$D_N(T) = -\Delta L + \Xi(T) + \kappa + O\Big(\frac{1}{T} + \frac{T^2}{N}\Big), \qquad \Xi(T) = \delta T + \pi \log T.$$

(i) For fixed N, M_F and M_G are positive maxima of finitely many polynomials in h, so D_N is continuous. At $T = \frac{\Delta}{2\delta} L$ the display gives $D_N = -\frac{\Delta}{2} L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta} L$ it gives $D_N = \Delta L + O(\log L) > 0$.

(ii) A zero has $\delta T_N + \pi \log T_N = \Delta L + O(1)$. If $\pi \ge 0$, then $\delta T_N \le \Delta L + O(1)$. If $\pi < 0$, then $\pi \log T_N \ge \pi L/3$, since $T_N \le N^{1/3}$, and $\delta T_N \le (\Delta + |\pi|/3)L + O(1)$. So $T_N = O(L)$, then $T_N = \frac{\Delta}{\delta} L + O(\log L)$, and the error term is $O(1/L)$.

(iii) Let $\Xi(T^\circ) = \Delta L - \kappa$. As in (ii), $T^\circ = \frac{\Delta}{\delta} L + O(\log L)$, and $\Xi' = \delta + \pi/T \ge \delta/2$ on $[\frac{\Delta}{2\delta} L, \frac{2\Delta}{\delta} L]$ for large N. So every zero is within $O(1/L)$ of T°. Putting $\log T^\circ = \log L + \log \frac{\Delta}{\delta} + O(\log L/L)$ into $\delta T^\circ = \Delta L - \pi \log T^\circ - \kappa$ gives the formula. □

E.3 Proof of Theorem 6.11

(a) For $G = \{j\}$ the only word in Lemma 6.4 is $j^{N'}$, so $P_{N'}^{(z)}(\hat{n}) = (\mu_j + zh\mu_i q_{ij})(1 + zhq_{ji})(p_j + zh^2\omega)^{N'-1}$ and $P_{N'}^{(0)}(\hat{n}) = \mu_j p_j^{N'-1}$. We divide by it, expand $(1 + zh\frac{\mu_i}{\mu_j} q_{ij})(1 + zhq_{ji}) = \sum_e c_e h^e z^e$ and $(1 + zh^2\omega/p_j)^{N'-1}$ by the binomial theorem, and use the lemma.

(b) By Lemma 6.9, $\widetilde{\Lambda}(z) = (1 + c_1 z + c_2 z^2) e^{S\omega z}$. So $\xi_e = c_e$ in the Laguerre form of Section 6.9, which follows from $\sum_K \binom{n-1}{K-1} p^{n-K} h^K (S\omega)^{K-e}/(K - e)! = p^n (h/p)^e B_n^{(e)}(u)$.

(c) Write $n = n_i$, $p = p_i$ and $m = K - e$. The (K, e) term of R is ρ_m times the (K, e) term $t_{K,e} = \binom{n-1}{K-1} p^{n-K} h^K c_e (S\omega)^m / m!$ of $\mathbf{K}^h$, where $\rho_m = \prod_{l=1}^{m} (1 - l/N')\, p_j^{-m}$, which is 0 for $m \ge N'$. All $t_{K,e} \ge 0$, so $R/\mathbf{K}^h = \mathbb{E}_\nu \rho_m$ for the probability ν proportional to $t_{K,e}$. Since $-\log(1-y) \le 2y$ for $y \le \frac{1}{2}$, $1 - m(m+1)/(2N') \le \rho_m \le e^{2hq_j m}$. Fix e. Then $t_{K,e}$ is proportional to $\beta_K = \binom{n-1}{K-1} u^m / m!$, and $\beta_{K+1}/\beta_K = \frac{n-K}{K} \cdot \frac{u}{K+1-e} \le \frac{U}{K(K+1-e)}$. Multiplying out and summing over K gives $\mathbb{E}[(K-1)(K-e)] \le U$ under the law proportional to β_K, that is $\mathbb{E}[(m-1)m] \le U$ for $e = 0$, $\mathbb{E}[m^2] \le U$ for $e = 1$ and $\mathbb{E}[m(m+1)] \le U$ for $e = 2$. So $\mathbb{E}m \le 1 + \sqrt{U}$ and $\mathbb{E}m(m+1) \le U + 2\sqrt{U} + 2$, which gives the lower bound. Since $\beta_K \propto u^m$, the logarithmic derivative of $\mathcal{S}_e(u) = \sum_K \beta_K$ is $\mathbb{E}^{(u)}[m]/u$. So with $c = 2hq_j \le 1$ and $\sigma = e^c$, $\log \mathbb{E} e^{cm} = \int_u^{\sigma u} \mathbb{E}^{(v)}[m]\, v^{-1} dv \le c(1 + \sqrt{\sigma U}) \le c(1 + 2\sqrt{U})$, which gives the upper bound. Mixing over e keeps both bounds. For the last claim, $U/N' \le 2\omega h^2 n$, $\sqrt{U}/N' \le h\sqrt{2\omega}$, $1/N' \le 2h$ since $hN' \ge T/2$, and $hq_j\sqrt{U} = O(h^2\sqrt{N'n})$. For $T \le N^{1/2}$ we have $h^2\sqrt{N'n} \le h^2 N = T^2/N \le 1$, so the exponent $2hq_j(1 + 2\sqrt{U})$ is bounded and the upper bound is $1 + O(h + h^2\sqrt{N'n})$.

(d) Let $a_K = [z^K]\widetilde{\Lambda} = \sum_e c_e (S\omega)^{K-e}/(K-e)!$. By (15), $\mathbf{K}(x, S) = e^{-q_i x} \sum_{K \ge 1} x^{K-1} a_K/(K-1)!$, and the (K, e) terms of $\mathbf{K}^h/h$ and $\mathbf{K}$ have the ratio $\prod_{l<K}(1 - l/n)\, p^{n-K} e^{hq_i n}$, which lies between $(1 - K(K-1)/(2n)) e^{-hq_i^2 x}$ and $e^{hq_i K}$, and is 0 for $K > n$. So $\mathbf{K}^h/(h\mathbf{K})$ is the mean of this ratio under the probability ν' proportional to the terms $t'_{K,e} = c_e x^{K-1} (S\omega)^{K-e}/((K-1)!(K-e)!)$ of $\mathbf{K}$. For fixed e, $t'_{K+1,e}/t'_{K,e} = U'/(K(K+1-e))$ with $U' = \omega S x$, and as in (c) this gives $\mathbb{E}_e[(K-1)(K-e)] = U'$. Since $m = K - e \ge 0$ is an integer, this gives $\mathbb{E}_e[K(K-1)] \le 3U'$ for $e \le 1$ and $\mathbb{E}_2[K(K-1)] \le 3U' + 2$. The total weights of $e = 0, 1, 2$ are $S\omega\mathcal{I}(U')$, at least $c_1\mathcal{I}(U')$ and $c_2 x\, \mathcal{I}(U')$, with $\mathcal{I}(y) = \sum_{j \ge 0} y^j/(j!\,(j+1)!)$, so $\nu'(e = 2) \le c_2 x/(c_1 + S\omega)$. Hence $\mathbb{E}_{\nu'}[K(K-1)]/(2n) \le \frac{3}{2}\omega h S + c_2 h/c_1$. As in (c), $\mathbb{E}_{\nu'} e^{hq_i K} \le e^{hq_i(3 + 2\sqrt{U'})} = 1 + O(h + hS)$, since $\sqrt{U'} \le \sqrt{\omega}\, S$ for $n \le N'$. Also $e^{-hq_i^2 x} \ge 1 - q_i^2 hS$. This proves (d). □

E.4 Proof of Theorem 6.12

Constants depend only on Q, μ, i and $\mathcal{K}$. Write $n = n_i$, $p = p_i$, $s = h\hat{n} = S\beta$, $\tilde{x} = (n-1)h$ and

$$w_d(z) = \left(p + \frac{h}{z}\right)^{n-1} = \sum_{K=1}^{n} b_K z^{-(K-1)}, \qquad b_K = \binom{n-1}{K-1} p^{n-K} h^{K-1},$$

so that $\mathbf{K}^h_\beta(n, S) = h \sum_K b_K [z^K]\widetilde{\Lambda}_\beta$. Let Λ, ε, Ξ and $R = \log \Xi$ be as in the proof of Theorem 6.1(a). So $|\varepsilon(z) - \varepsilon(0)| \le C|z|/S$ for $|z| \le z_0$, and R is real on the real axis with $|R(z)| \le C(|z| + S|z|^2)$, $|R'(z)| \le C(1 + S|z|)$ and $|R''| \le C(1 + S)$ on $|z| \le z_0$. Throughout, $hS \le T^2/N \le 1$, $\tilde{x} \le x \le \delta T$ and $S \ge (1 - \delta)T$.

The coefficient identity. The proof of Lemma 6.3 uses the rates only as weights of words. So for every matrix q' of nonnegative rates on F in place of Q, with the same

q_G in the exponent, the coefficient of $x^{K-1}/(K-1)!$ in (15) gives

$$e^{-\langle q_G,s\rangle}\sum_{m:\,m_i=K} W'(m)\prod_{j\in G}\frac{s_j^{m_j-1}}{(m_j-1)!} = [z^K]f_G'^{(z)}(s),$$

where W' and $f_G'^{(z)}$ are W and $f_G^{(z)}$ built from q', with $d'(z) = q_G - zc'^{-}\circ c'^{+}$.

The radius. Put $\tilde z_* = \sqrt{\tilde x/(\omega_\beta S)}$ and $\tilde\zeta = 2\sqrt{\omega_\beta S\tilde x}$. If $\tilde\zeta \le Z_0$, let $\rho = \max(\tilde z_*, 1/S)$. If $\tilde\zeta > Z_0$, let ρ be the critical point on $I = [\tilde z_*/2, 2\tilde z_*]$ of $\lambda_d(z) = (n-1)\log(p+h/z) + S\omega_\beta z + R(z)$. Let $\mathcal{A}_d = \frac{1}{2\pi\mathrm{i}}\oint_{|z|=\rho} w_d\Lambda\,dz/z^2$, let $\mathcal{A}_d^{\mathrm{abs}}$ be the integral of $|w_d\Lambda||z|^{-2}$ against $|dz|/(2\pi)$, and $\zeta_d = \max(1, S\omega_\beta\rho)$.

Lemma E.5 (the discrete contour) *For Z_0 large, then δ small and T_0 large, ρ is well defined, $\rho \le z_0/2$, $\mathcal{A}_d > 0$, $\mathcal{A}_d^{\mathrm{abs}} \le C\mathcal{A}_d$ and $\mathcal{A}_d \ge c\rho^{-1}w_d(\rho)\Lambda(\rho)\zeta_d^{-1/2}$. With $\mathbf{D} = \sum_K b_K[z^K]f_G^{(z)}(s) = \frac{1}{2\pi\mathrm{i}}\oint_{|z|=\rho} w_d f_G^{(z)}(s)\,dz/z^2$,*

$$\mathbf{D} = (1+\varepsilon(0))\mathcal{A}_d(1+O(\rho/S)), \qquad \rho/S \le C(T^{-2} + x^{1/2}T^{-3/2}).$$

Proof Few visits. Here $\rho \le C/S$. On the circle $|w_d| \le e^{\tilde x/\rho} \le e^{\tilde\zeta/2} \le e^{Z_0/2}$, $|e^{S\omega_\beta z}| \le C$ and $|\Xi - 1| \le C/S$. With $\Xi = 1$ the integral is $\Lambda(0)\sum_K b_K(S\omega_\beta)^K/K! \ge \Lambda(0)p^{n-1}S\omega_\beta \ge cS\Lambda(0)$, since $p^{n-1} \ge e^{-2q_i\tilde x}$ and $\tilde x \le Z_0^2/(4\omega_{\min}S)$. The change from $\Xi = 1$ is at most $C\Lambda(0)\rho^{-1}/S \le C\Lambda(0)$, and $\mathcal{A}_d^{\mathrm{abs}} \le C\Lambda(0)S$ and $\rho^{-1}w_d(\rho)\Lambda(\rho) \le CS\Lambda(0)$. This gives the claims.

Many visits. Put $a = h/\tilde z_* = 2h\omega_\beta S/\tilde\zeta \le 2\omega_{\max}/Z_0 \le \frac18$, where $\omega_{\max} = \max_{\mathcal{K}}\omega_\beta$. Here the lattice enters, and $hS \le 1$ makes it harmless. On I, $\lambda_d'(z) = -\tilde x/(z^2(p+h/z)) + S\omega_\beta + R'(z)$. The first term is at most $-4\omega_\beta S/(1+2a)$ at $\tilde z_*/2$ and at least $-\omega_\beta S/2$ at $2\tilde z_*$, since $p \ge \frac12$, and $|R'| \le C(1+S\tilde z_*)$ with $\tilde z_* \le \sqrt{2\delta/\omega_{\min}}$. So λ_d' changes sign on I. Also $\lambda_d''(z) = \tilde x(2p+h/z)z^{-3}(p+h/z)^{-2} + R''(z) \ge \omega_\beta S/(18\tilde z_*) - C(1+S) > 0$ on I, since $(2p+y)/(p+y)^2 \ge \frac49$ for $p \in [\frac12, 1]$ and $0 \le y \le \frac14$. So ρ is unique, and $\rho \le 2\tilde z_* \le z_0/2$ for δ small. On the circle $h/\rho \le \frac14$, so $\mathrm{Re}(p+h/z) \ge \frac14$, λ_d is holomorphic near the circle, real on $(0, z_0]$, and $e^{\lambda_d} = w_d\Lambda/\Lambda(0)$. Since $|p + (h/\rho)e^{-\mathrm{i}\phi}|$ is largest at $\phi = 0$, and $\mathrm{Re}[R(z) - R(\rho)] \le C(\rho + S\rho^2)(1-\cos\phi)$ as in the proof of Theorem 6.1(a), we get $\mathrm{Re}\,\lambda_d(\rho e^{\mathrm{i}\phi}) - \lambda_d(\rho) \le -\frac{\zeta_d}{2}(1-\cos\phi)$, with $\zeta_d \ge \tilde\zeta/4 \ge Z_0/4$. Let $\ell(\phi) = \lambda_d(\rho e^{\mathrm{i}\phi})/\zeta_d$. The critical equation reads $\tilde x/(\rho(p\rho+h)) = S\omega_\beta(1+\eta)$ with $|\eta| \le \frac{1}{10}$, so $(n-1)h/(p\rho) \le 3\zeta_d$, and the ϕ-derivatives of ℓ up to order four are bounded. Also $-\ell''(0) = \rho^2\lambda_d''(\rho)/\zeta_d = (1+\eta)(2p\rho+h)/(p\rho+h) + O(\rho + 1/S) \in [0.4, 3]$. Remark B.2 with $m = 1$, the amplitude $e^{-\mathrm{i}\phi}$ and ζ_d in place of T gives $\mathcal{A}_d = \Lambda(0)\rho^{-1}e^{\lambda_d(\rho)}(2\pi\zeta_d(-\ell''(0)))^{-1/2}(1+O(1/\zeta_d))$, which proves the claims, since $\Lambda(0)e^{\lambda_d(\rho)} = w_d(\rho)\Lambda(\rho)$.

Consequence. $\mathbf{D} - (1+\varepsilon(0))\mathcal{A}_d$ is the integral of $w_d\Lambda(\varepsilon(z) - \varepsilon(0))$, which is at most $C(\rho/S)\mathcal{A}_d^{\mathrm{abs}}$ in modulus. Finally $\rho \le C/S$ with few visits and $\rho \le 2\sqrt{\tilde x/(\omega_\beta S)}$ with many. □

Lemma E.6 (the facet states discretized) *Uniformly as in Theorem 6.12, $P_N(n) = h^{k-1}\mathbf{D}(1+O(T^2/N))$.*

Proof Step 1. By Lemma 2.3, $P_N(n) = \sum_m u_m$ with $u_m = W(m)h^{M-1}\prod_{l\in F}\binom{n_l-1}{m_l-1}(1-hq_l)^{n_l-m_l}$. Since $M-1 = m_i+(k-2)+\sum_{j\in G}(m_j-1)$, as in Step 1 of the proof of Lemma 5.5 we get $u_m = \kappa_m v_m$ with

$$v_m = W(m)h^{k-2}h^{m_i}\binom{n-1}{m_i-1}p^{n-m_i}\prod_{j\in G}e^{-q_j s_j}\frac{s_j^{m_j-1}}{(m_j-1)!} \ge 0,$$

and with κ_m the factor of that proof with the product over G only. The state i is kept exact. By the coefficient identity with $q' = Q$, $\sum_m v_m = h^{k-1}\mathbf{D}$, which is positive by Lemma E.5. So $P_N(n)/(h^{k-1}\mathbf{D}) = \mathbb{E}_w\kappa_m$ for the probability w proportional to v_m. With $M_G = \sum_{j\in G} m_j$ and $n^G_{\min} = \min_{j\in G} n_j \ge \varepsilon_{\mathcal{K}}N/2$, the argument for (A7) gives $(1 - M_G^2/(2n^G_{\min}))e^{-q^2_{\max}T^2/N} \le \kappa_m \le e^{hq_{\max}M_G}$.

Step 2. Let $\sigma = \mathrm{e}$, and let q' be Q with the rates out of the states of G multiplied by σ. Every letter of G in a word, except possibly the last, is followed by a jump out of it, so $W'(m) \ge \sigma^{M_G-1}W(m)$, and $\mathbb{E}_w\sigma^{M_G-1} \le \mathbf{D}'/\mathbf{D}$ with $\mathbf{D}' = \sum_K b_K[z^K]f_G'^{(z)}(s)$ by the coefficient identity. Here $f_G'^{(z)}$ is built from $\sigma A(z)$, $\mu^{(z)}$, $\mathbf{1}+z\sigma c^-$ and $q_G - z\sigma c^-\circ c^+$. All its coefficients in z are nonnegative, so $\mathbf{D}' \le \rho^{-1}w_d(\rho)f_G'^{(\rho)}(s)$. We claim that $f_G'^{(r)}(S\beta) \le CS^{(k-2)/2}e^{CS}\Lambda(r)$ for real $0 \le r \le z_0$. For $|G| = 1$ this is $f_G'^{(r)} \le \sigma e^{(\sigma-1)r\omega S}\Lambda(r)$, from the closed forms. For $|G| \ge 2$ we use Lemma A.3 for the weights $(\sigma A(r), \mu^{(r)}, \mathbf{1}+r\sigma c^-)$ at the radii $\sigma\theta'(\beta,r)$, which are admissible by Lemma 6.10(a), (e). As in Step 2 of the proof of Lemma 4.1, g scales by σ, so $\bar g = \sigma\theta_1(\beta,r)$ by (A3) for $A(r)$, and $\bar b \le \sigma b_r(\theta'(\beta,r)) \le \sigma B_0$ by Lemma 6.10(b), since $\mathbf{1}+r\sigma c^- \le \sigma e^{(r)}$ and $X_{A(r)}(\theta'(\beta,r)) \ge 0$. So $f_G'^{(r)}(S\beta) \le C\exp(S[\sigma\langle\beta,\theta(\beta,r)\rangle - \langle\beta,d(r)\rangle + (\sigma-1)r\langle\beta,c^-\circ c^+\rangle])$, while $\Lambda(r) = S^{-(k-2)/2}C_G(\beta,r)e^{-S\langle\beta,d(r)-\theta(\beta,r)\rangle}$ by Step 4 of the proof of Theorem 6.8, with $C_G \ge c > 0$ on $\mathcal{K}\times[0,z_0]$. This proves the claim. With Lemma E.5 and $\mathbf{D} \ge \frac12\mathcal{A}_d$, $\mathbf{D}'/\mathbf{D} \le C\zeta_d^{1/2}S^{(k-2)/2}e^{CS} \le e^{C'T}$. So the convex function $\Upsilon(c) = \log\mathbb{E}_w e^{c(M_G-1)}$ has $\Upsilon(0) = 0$ and $\Upsilon(1) \le C'T$.

Step 3. As in Step 4 of the proof of Lemma 5.5, $\mathbb{E}_w e^{hq_{\max}M_G} = 1 + O(T^2/N)$ and $\mathbb{E}_w M_G^2 = O(T^2)$, so $\mathbb{E}_w\kappa_m = 1 + O(T^2/N)$. □

Proof of Theorem 6.12 We prove (a), (b) and (c) in turn. (a) By Lemmas E.6 and E.5, $P_N(n) = h^{k-1}(1+\varepsilon(0))\mathcal{A}_d(1+O(\rho/S+T^2/N))$. Lemma 5.5 for the face G, with N' steps, the same h and the time S, applies since $S^2/N' \le T^2/N$, and gives $P_{N'}(\hat n) = h^{k-2}f_G(s)(1+O(T^2/N))$. For $|G| = 1$ this is the formula for the path $j^{N'}$. Also $f_G(s) = \Lambda(0)(1+\varepsilon(0))$ and $\mathbf{K}^h_\beta(n,S) = h\mathcal{A}_d/\Lambda(0)$. Dividing, $1+\varepsilon(0)$ cancels, and $E = O(\rho/S+T^2/N)$. Taking T_0 large and ε_1 small makes $|E| \le \frac12$.

(b) Fix $x' \in \{hn, h(n-1)\}$ and let ρ and $\mathcal{A}$ be the radius and the integral of the proof of Theorem 6.1(a) for x', so that $\mathcal{A}^{\mathrm{abs}} \le C\mathcal{A}$ and $\mathbf{K}_\beta(x',S) = e^{-q_i x'}\mathcal{A}/\Lambda(0)$. For $n = 1$ and $x' = 0$ both sides are $h[z]\widetilde\Lambda_\beta$. On the circle put $y = h/z - hq_i$. Since $h/\rho \le \max(hS, 4h\omega_\beta S/Z_0)$, we have $|y| \le CT^2/N \le \frac12$, and $w_d = e^{(n-1)y}e^{(n-1)(\log(1+y)-y)}$ with $(n-1)|y|^2 \le 2h\tilde x/\rho^2 + 2hq_i^2\tilde x \le CT^2/N$, since $\tilde x/\rho^2 \le 4\omega_\beta S$. Also $e^{(n-1)y} = e^{x'/z - q_i x'}$ if $x' = \tilde x$, with one more factor $e^{-y} = 1+O(T^2/N)$ if $x' = hn$. So $w_d = e^{x'/z-q_ix'}(1+\eta)$ with $|\eta| \le CT^2/N$, and $|\mathcal{A}_d - e^{-q_ix'}\mathcal{A}| \le C(T^2/N)e^{-q_ix'}\mathcal{A}$. This is (b), and with (a) it gives the last display.

(c) The proofs of Theorem 6.1(b) and (c) compare $\mathbf{K}_\beta$ with $\mathbf{B}_\beta$ at the same (x, S). They show $\mathbf{K}_\beta = \mathbf{B}_\beta(1 + O(1/T + \sqrt{x/T}))$ for $x \le 1$, $\mathbf{K}_\beta = \mathbf{B}_\beta e^{O(\sqrt{x/T}(1+x))}$ for $1 \le x \le \delta T$, and $\mathbf{K}_\beta = \mathbf{B}_\beta(1 + c_1(\beta)z_* I_0(\zeta)/I_1(\zeta) + O(x/T))$ when $\pi_\beta^{\text{in}} = \pi_\beta^{\text{out}}$, where $c_1(\beta)z_* I_0/I_1$ is bounded. We combine these with (b). □

References

[1] D. E. Amos, *Computation of modified Bessel functions and their ratios*, Math. Comp. **28** (1974), 239–251. https://doi.org/10.1090/S0025-5718-1974-0333287-7.

[2] M. Benaïm, M. Bravo and M. Faure, *On the occupation measure of evolution models with vanishing mutations*, preprint, 2026. arXiv:2602.06829.

[3] A. Berman and R. J. Plemmons, *Nonnegative Matrices in the Mathematical Sciences*, SIAM, Philadelphia, 1994. https://doi.org/10.1137/1.9781611971262.

[4] L. Bertini, A. Faggionato and D. Gabrielli, *From level 2.5 to level 2 large deviations for continuous time Markov chains*, Markov Process. Related Fields **20** (2014), 545–562. arXiv:1212.6908.

[5] L. Bertini, A. Faggionato and D. Gabrielli, *Large deviations of the empirical flow for continuous time Markov chains*, Ann. Inst. H. Poincaré Probab. Stat. **51** (2015), 867–900. https://doi.org/10.1214/14-AIHP601.

[6] L. Bertini, D. Gabrielli and C. Landim, *Metastable* Γ*-expansion of finite state Markov chains level two large deviations rate functions*, Ann. Appl. Probab. **34** (2024), 3820–3869. https://doi.org/10.1214/24-AAP2051.

[7] B. Bollobás, *Modern Graph Theory*, Springer, New York, 1998. https://doi.org/10.1007/978-1-4612-0619-4.

[8] F. Bouguet and B. Cloez, *Fluctuations of the empirical measure of freezing Markov chains*, Electron. J. Probab. **23** (2018), no. 2, 1–31. https://doi.org/10.1214/17-EJP130.

[9] D. Brydges, R. van der Hofstad and W. König, *Joint density for the local times of continuous-time Markov chains*, Ann. Probab. **35** (2007), 1307–1332. https://doi.org/10.1214/009171906000001024.

[10] G. Carugno, P. Vivo and F. Coghi, *Graph-combinatorial approach for large deviations of Markov chains*, J. Phys. A **55** (2022), 295001. https://doi.org/10.1088/1751-8121/ac79e6.

[11] V. Čekanavičius and P. Vellaisamy, *Compound Poisson and signed compound Poisson approximations to the Markov binomial law*, Bernoulli **16** (2010), 1114–1136. https://doi.org/10.3150/09-BEJ246.

[12] J. Chen and X. Deng, *Large deviations and related problems for absorbing Markov chains*, Stochastic Process. Appl. **123** (2013), 2398–2418. https://doi.org/10.1016/j.spa.2013.02.014.

[13] P. Collet, S. Martínez and J. San Martín, *Quasi-Stationary Distributions: Markov Chains, Diffusions and Dynamical Systems*, Springer, Heidelberg, 2013. https://doi.org/10.1007/978-3-642-33131-2.

[14] J. N. Darroch and E. Seneta, *On quasi-stationary distributions in absorbing discrete-time finite Markov chains*, J. Appl. Probab. **2** (1965), 88–100. https://doi.org/10.2307/3211876.

[15] J. N. Darroch and E. Seneta, *On quasi-stationary distributions in absorbing continuous-time finite Markov chains*, J. Appl. Probab. **4** (1967), 192–196. https://doi.org/10.2307/3212311.

[16] R. Dawson and I. J. Good, *Exact Markov probabilities from oriented linear graphs*, Ann. Math. Statist. **28** (1957), 946–956. https://doi.org/10.1214/aoms/1177706795.

[17] Z. Dietz and S. Sethuraman, *Large deviations for a class of nonhomogeneous Markov chains*, Ann. Appl. Probab. **15** (2005), 421–486. https://doi.org/10.1214/105051604000000990.

[18] Z. Dietz and S. Sethuraman, *Occupation laws for some time-nonhomogeneous Markov chains*, Electron. J. Probab. **12** (2007), no. 23, 661–683. https://doi.org/10.1214/EJP.v12-413.

[19] R. L. Dobrushin, *Limit theorems for a Markov chain of two states*, Izv. Akad. Nauk SSSR Ser. Mat. **17** (1953), 291–330 (in Russian). English translation in Select. Transl. Math. Statist. Probab. **1** (1961), 97–134.

[20] M. D. Donsker and S. R. S. Varadhan, *On a variational formula for the principal eigenvalue for operators with maximum principle*, Proc. Nat. Acad. Sci. U.S.A. **72** (1975), 780–783. https://doi.org/10.1073/pnas.72.3.780.

[21] P. Dupuis and Y. Liu, *On the large deviation rate function for the empirical measures of reversible jump Markov processes*, Ann. Probab. **43** (2015), 1121–1156. https://doi.org/10.1214/13-AOP883.

[22] K. R. Gabriel, *The distribution of the number of successes in a sequence of dependent trials*, Biometrika **46** (1959), 454–460. https://doi.org/10.1093/biomet/46.3-4.454.

[23] I. J. Good, *The frequency count of a Markov chain and the transition to continuous time*, Ann. Math. Statist. **32** (1961), 41–48. https://doi.org/10.1214/aoms/1177705138.

[24] L. A. Goodman, *Exact probabilities and asymptotic relationships for some statistics from m-th order Markov chains*, Ann. Math. Statist. **29** (1958), 476–490. https://doi.org/10.1214/aoms/1177706623.

[25] A. Guillin, B. Nectoux and L. Wu, *Large deviations of the empirical measures of a strong-Feller Markov process inside a subset and quasi-ergodic distribution*, preprint, 2024. arXiv:2411.17216.

[26] L. Hervé and J. Ledoux, *A local limit theorem for densities of the additive component of a finite Markov additive process*, Statist. Probab. Lett. **83** (2013), 2119–2128. https://doi.org/10.1016/j.spl.2013.05.032.

[27] R. Huang, D. Kious, V. Sidoravicius and P. Tarrès, *Explicit formula for the density of local times of Markov jump processes*, Electron. Commun. Probab. **23** (2018), no. 90, 1–7. https://doi.org/10.1214/18-ECP194.

[28] P. Kagey, *Problem 131*, Open Problem Collection, https://peterkagey.com/problems/131/ (accessed October 2026).

[29] S. Kim and C. Landim, Γ-*expansion of the measure-current large deviations rate functional of non-reversible finite-state Markov chains*, Stochastic Process. Appl. **189** (2025), 104733. https://doi.org/10.1016/j.spa.2025.104733.

[30] I. Kontoyiannis and S. P. Meyn, *Spectral theory and limit theorems for geometrically ergodic Markov processes*, Ann. Appl. Probab. **13** (2003), 304–362. https://doi.org/10.1214/aoap/1042765670.

[31] Y. Kovchegov, N. Meredith and E. Nir, *Occupation times and Bessel densities*, Statist. Probab. Lett. **80** (2010), 104–110. https://doi.org/10.1016/j.spl.2009.09.018.

[32] C. Landim, *Metastability from the large deviations point of view: a* Γ*-expansion of the level two large deviations rate functional of non-reversible finite-state Markov chains*, Stochastic Process. Appl. **165** (2023), 275–315. https://doi.org/10.1016/j.spa.2023.09.001.

[33] H. D. Miller, *A convexity property in the theory of random variables defined on a finite Markov chain*, Ann. Math. Statist. **32** (1961), 1260–1270. https://doi.org/10.1214/aoms/1177704865.

[34] P. Ney and E. Nummelin, *Markov additive processes I. Eigenvalue properties and limit theorems*, Ann. Probab. **15** (1987), 561–592. https://doi.org/10.1214/aop/1176992159.

[35] P. Ney and E. Nummelin, *Markov additive processes II. Large deviations*, Ann. Probab. **15** (1987), 593–609. https://doi.org/10.1214/aop/1176992160.

[36] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl and M. A. McClain (eds.), *NIST Digital Library of Mathematical Functions*, Release 1.2.8 of 2026-09-15, https://dlmf.nist.gov/.

[37] P. J. Pedler, *Occupation times for two state Markov chains*, J. Appl. Probab. **8** (1971), 381–390. https://doi.org/10.2307/3211908.

[38] A. Pemmasani, *Uniform Bessel bounds for magnetization crossings in finite Ising chains*, preprint, 2026. arXiv:2610.06935.

[39] A. Pemmasani, *Vertex crossings in a symmetric Markov multinomial model*, preprint, 2026. arXiv:2610.09258v1, version 1 of the arXiv entry of [40].

[40] A. Pemmasani, *Occupation counts of rarely switching Markov chains: stratified local laws, support selection and the boundary layer*, the arXiv version, with Sections 7 and 8 on the complete graph and near a vertex, preprint, 2026. arXiv:2610.09258v2, version 2.

[41] M. Polettini, *BEST statistics of Markovian fluxes: a tale of Eulerian tours and Fermionic ghosts*, J. Phys. A **48** (2015), 365005. https://doi.org/10.1088/1751-8113/48/36/365005.

[42] W. Stadje, *Joint distributions of the numbers of visits for finite-state Markov chains*, J. Multivariate Anal. **70** (1999), 157–176. https://doi.org/10.1006/jmva.1999.1814.

[43] P. Whittle, *Some distribution and moment formulae for the Markov chain*, J. Roy. Statist. Soc. Ser. B **17** (1955), 235–242. https://doi.org/10.1111/j.2517-6161.1955.tb00197.x.

[44] F. Y. Wu, *The Potts model*, Rev. Modern Phys. **54** (1982), 235–268. https://doi.org/10.1103/RevModPhys.54.235.